T2K 6.914×10^{20}-POT Neutrino 4.011×10^{20}-POT Anti-Neutrino 3-Flavour Joint Analysis

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Abstract

We report the results of the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1+6 dataset, using data taken in both neutrino and antineutrino beam mode. The goal is to search for CP violation by setting limits on δ_{CP} and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) (Δm^2_{32} for normal hierarchy and Δm^2_{13} for inverted hierarchy). Though T2K has limited sensitivity to the mass hierarchy given the relatively short baseline, fits under both hypotheses were performed to show which is preferred by data. Furthermore the search for $\bar{\nu}_e$ appearance and the measurement of $\bar{\nu}_\mu$ disappearance are shown as well.

This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] to this dataset. The Run 1-6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 6.914×10^{20} POT in neutrino mode and 4.011×10^{20} POT in antineutrino mode.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino μ-like, 34.284 ± 1.755 (syst) for antineutrino μ-like, 26.594 ± 1.808 (syst) for neutrino e-like, and 3.218 ± 0.238 (syst) for antineutrino e-like single ring events in Super K, given the oscillation parameters of the Asimov data set A in Tab. 12, and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

A joint analysis of all the four selected samples was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

For μ-like samples the templates were built as a function of the reconstructed neutrino energy, while for e-like the analysis templates were built as a function of the reconstructed neutrino energy and lepton angle, in order to improve the separation of different flavour/reaction components and increase the sensitivity to δ_{CP}. The observed single ring events spectra were compared to the expected spectra in order to infer all the 3-flavour oscillation parameters.

As in [10], all the systematic parameters were marginalised [53] in the fit and confidence intervals were quoted for all the oscillation parameters. Since the most important result of this analysis release is the search of CP violation, the Feldman-Cousins method [52] was performed only for the parameter δ_{CP}.

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1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1-6 dataset, using data taken in both neutrino (6.914×10^{20} POT) and antineutrino (4.011×10^{20} POT) mode. This analysis builds on our other analyses presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] to this data set. As in [10], a hybrid frequentist-Bayesian method [53], which consists of marginalizing instead of profiling the nuisance parameters, is used.

The consistency of the data with the expectation is given by comparing the predicted reconstructed neutrino energy and outgoing lepton angle spectra of 1 ring events at Super-K to the observed spectra. The method of calculation of these predicted spectra is presented in Sec. 2. This analysis uses flux tuning based on recent NA61/SHINE/SHINE 2009 data [17][45]. It also uses an indirect extrapolation from ND280 to Super-K, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 [29].

A significant difference between this and the previous analyses is that for e-like samples the analysis templates were built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increase the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates were built as a function of the reconstructed neutrino energy as in the past analyses.

Updated nominal predictions of the reconstructed neutrino energy and lepton angle spectrum of single ring events in Super-K are presented for Run 1-6 in Sec. 2

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [50]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 101 parameters including energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state re-scattering systematics plus the oscillation parameters (nuisance parameters). In this Run 1-6 analysis, all the nuisance parameters are marginalised [53] . The nuisance parameters and their effects on the extrapolated single ring event spectra are described in Sec. 3. The analysis method is presented in detail in Sec. 4.1

Fit validation and coverage studies are presented in Sec. 5.

The main goal of this analysis is to search for CP violation in the contest of the PMNS matrix and confidence intervals were set using the Feldman-Cousins method [52]. Confidence intervals of the other oscillation parameters were set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in Sec. 5.1. The results of the measurement using the Run 1-6 dataset is given in Sec. 6 of this note.

In Sec. 7, 8 and 9 the $\bar{\nu}_e$ appearance analysis is shown while the $\bar{\nu}_\mu$ disappearance analysis is shown in Sec. 10 and 11. Finally, a summary is given in Sec. 12.

2. Predictions of single ring event spectra in Super-K

2.1. Methodology

In this analysis a comparison of the reconstructed energy (E_r) and lepton angle (θ) spectrum of single ring events observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals of the oscillation parameters. For μ-like samples the templates were built as a function of the reconstructed neutrino energy, as in the past analyses, and a unique θ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \beta \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t}. \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and lepton angle bin θ. The construction of the nominal MC templates is discussed in Sec. 2.2. Both flux-tuning weights based on NA61/SHINE hadron-production data (see Sec. 2.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, the lepton angle bin θ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 2.5.

2.2. Construction of nominal Super-K Monte Carlo templates

The nominal Super-K single ring MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying either the single e-like ring selection cuts or the single μ-like ring selection cuts to the official Super-K MC samples. The selection is exactly the same for neutrino and antineutrino beam mode. In total 4 samples are obtained:

- neutrino mode single μ-like ring sample
- antineutrino mode single μ-like ring sample
- neutrino mode single e-like ring sample
- antineutrino mode single e-like ring sample

Details of the MC samples are given in Sec. 2.2.1. The calculation of the normalization of each MC sample is described in Sec. 2.2.2. The official single μ-like ring selection cuts are listed in Sec. 2.2.3. while the official single e-like ring selection cuts are listed in Sec. 2.2.4. The list of MC templates used in this analysis is shown in Sec. 2.2.5, while the true and reconstructed energy as well as lepton angle binning of the templates is discussed in Sec. 2.2.6.

2.2.1. Input Monte Carlo samples

This analysis uses the official 14a Super-K MC samples. These were generated using the 13a JNUBEAM flux, NEUT 5.3.2, skdetsim v13p90 and the 14b Super-K software [33] with minor tunings. The CCQE model was reweighted from the Spectral Function to a Relative Fermi Gas model with a relativistic Random Phase Approximation on an event by event basis. The NC1γ events were scaled up by a factor of two in accordance with improved calculations of single photon emission in NC interactions [34]. Lastly, a simplified reweighting, as a function of the outgoing pion energy, was used to scale the CCCoherent events from the Rein Sehgal model to the Berger-Sehgal model prediction. Six MC samples were generated for both neutrino and antineutrino beam mode running (horn current $\pm$250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample. The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

Sample	Number of events per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
$\bar{\nu}_\mu$	353.789	633988
ν_μ	208.369	634659
$\bar{\nu}_e$	6.811	127013
ν_e	8.941	127262
Oscillated $\bar{\nu}_e$	376.420	126812
Oscillated ν_e	215.099	127278

Table 1: Normalization of input Super-K MC samples [33]

2.2.2. Normalization of Monte Carlo samples

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{2}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section on water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-6 POT (6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode) using the numbers in Tab. 1. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode, by weighting the result respectively by 6.914 POT and 0.4011 POT (the Run 1-6 POT in units of 10^{21} POT).

2.2.3. Single μ-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

The same selection is applied to both neutrino and antineutrino mode samples.

2.2.4. Single e-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single e-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts

- the reconstructed vertex is at least 2 metres away from the ID wall
- the visible energy in the event is above 30 MeV
- there is exactly 1 reconstructed ring in the event
- the reconstructed ring is e-like
- number of decay electrons is zero
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

The same selection is applied to both neutrino and antineutrino mode samples.

2.2.5. List of Monte Carlo templates

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous $\bar{\nu}_\mu$-disappearance and $\bar{\nu}_e$ appearance analyses the NC interaction modes were subdivided into NC1$\pi^\pm$, NC1π^0, NC coherent and NC other. In this analysis the new mode NC1γ, which was previously included in NC other, is used. We have also added 5 CC MEC templates for $\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$, ν_e and oscillated ν_e and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 50 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC MEC,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1γ,
- ν_μ CCQE,
- ν_μ CC MEC,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,

- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC 1γ,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC MEC,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC$1\pi^{\pm}$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC$1\pi^0$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1γ,
- ν_e CCQE,
- ν_e CC MEC,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC$1\pi^{\pm}$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC$1\pi^0$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC 1γ,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC MEC,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC MEC,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC other.

In Tab. 2, 3, 4 and 5 the predicted number of events is shown for all the selected samples and each flavour and reaction mode.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	66.077	0.035	4.291	0.002	0.225	0.001	70.631
CC1pi	16.973	0.017	1.566	0.001	0.067	0.000	18.625
CCcoherent	0.326	0.000	0.080	0.000	0.001	0.000	0.408
MEC	19.038	0.010	0.845	0.000	0.050	0.000	19.944
CCother	7.735	0.008	0.494	0.001	0.002	0.000	8.240
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	117.643	0.297	7.602	0.025	0.345	0.002	125.914

Table 2: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	6.993	0.004	12.792	0.005	0.006	0.014	19.813
CC1pi	2.707	0.003	3.033	0.002	0.002	0.003	5.750
CCcoherent	0.054	0.000	0.169	0.000	0.000	0.000	0.223
MEC	2.503	0.001	2.053	0.001	0.002	0.002	4.562
CCother	1.280	0.001	0.885	0.000	0.000	0.000	2.168
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.508	0.051	19.664	0.032	0.010	0.019	34.284

Table 3: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in tab. 12 is used.

2.2.6. Binning of Monte Carlo templates

Each of the 50 MC templates has 2100 2-dimensional bins (= 84 true energy bins × 25 reconstructed energy bins). The 84 true energy bins are arranged as follows:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV.

An update was done for the binning of the reconstructed variables. The difference with the previous analyses is that for e-like samples the analysis templates were built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.056	2.129	0.001	0.088	15.875	0.099	18.249
CC1pi	0.016	0.348	0.000	0.027	1.688	0.017	2.096
CCcoherent	0.000	0.002	0.000	0.002	0.015	0.002	0.021
MEC	0.008	0.710	0.000	0.022	3.901	0.019	4.659
CCother	0.001	0.045	0.000	0.003	0.029	0.001	0.080
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.469	3.264	0.069	0.145	21.508	0.138	26.594

Table 4: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.375	1.156	2.065
CC1pi	0.002	0.040	0.001	0.060	0.057	0.137	0.298
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.019	0.026
MEC	0.001	0.069	0.000	0.063	0.114	0.185	0.433
CCother	0.001	0.008	0.000	0.004	0.004	0.002	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.141	0.313	0.235	0.478	0.551	1.500	3.218

Table 5: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in tab. 12 is used.

components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increase the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates were built as a function of the reconstructed neutrino energy as in the past analyses.

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 50-MeV bins from 0-1.25 GeV.

which is identical to the past analyses.

For μ-like samples the 73 reconstructed energy bins with constant width are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

The θ binning used for e-like samples is

- 14 bins with constant width from 0° to 140°,
- 1 bin for the range 140° − 180°.

that is the same as used by the $p-\theta$ analysis group [15].

For μ-like samples a unique θ bin, that covers the whole range 0° - 180°, is used. The same reconstructed neutrino energy and lepton angle binning is used for the fit of the single ring reconstructed energy spectra. The predicted e-like 2-dimensional spectra are shown in App. B for each flavour component and compared with the 1-dimensional reconstructed energy spectra. It is clear that the components $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ are better separated by the reconstructed energy Vs θ templates.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis.

2.3. Applying flux tuning based on NA61/SHINE data

In this analysis, the nominal MC templates, which were constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which was tuned on NA61/SHINE hadron production measurements [36][44]. The reweighted MC templates are denoted the *NA61/SHINE-tuned* MC templates in this document. The applied weights were obtained from [36] and are shown as a function of the true neutrino energy in Figs. 1-4. The effect of this flux tune on the predicted spectra events is discussed in Sec. 2.6

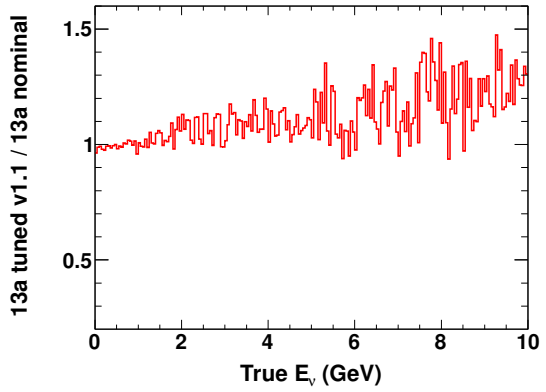


Figure 1: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [36].

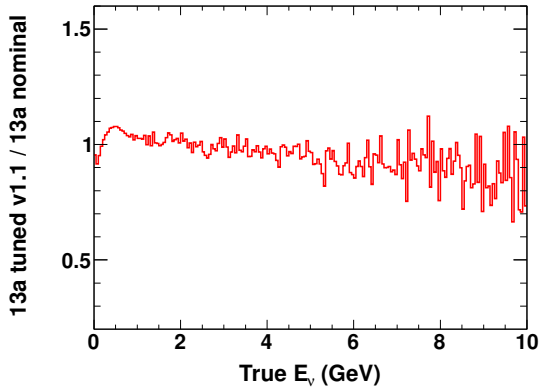


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [36].

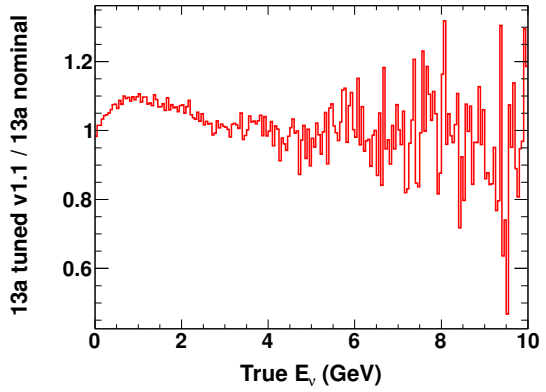


Figure 3: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [36].

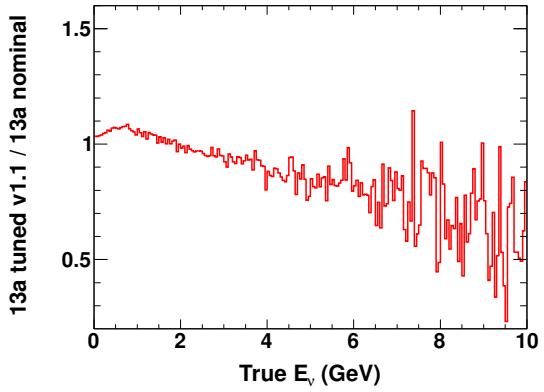


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [36].

2.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples to find the best-fit values of the flux normalisation and cross-section parameters [28]. This analysis uses these best-fit values to reweight the NA61/SHINE-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in Tab. 6, 7, 8 and 9.

The effects of the BANFF tune on the predicted spectra is discussed in Sec. 2.6.

2.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\overset{(-)}{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}. The $\overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_e \to \overset{(-)}{\nu}_e)$, while the oscillated $\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e)$ (the sample was generated assuming 100% of $\overset{(-)}{\nu}_\mu$ transform to $\overset{(-)}{\nu}_e$). The ν_μ templates are weighted with $\mathrm{P}(\nu_\mu \to \nu_\mu)$. The ν_e templates are weighted with $\mathrm{P}(\nu_e \to \nu_e)$ and the oscillated $\nu_\mu \to \nu_e$ templates with $\mathrm{P}(\nu_\mu \to \nu_e)$. All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code were first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [67], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\overset{(-)}{\nu}_e$ and $\overset{(-)}{\nu}_\mu$ can yield $\overset{(-)}{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ Super-K MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

2.6. Predictions of nominal and tuned-MC spectra

In this section, we present the expected spectra in Super-K for various oscillation scenarios. Unless explicitly stated otherwise, the oscillation parameters are set to the values shown in Tab. 12 and the normal mass hierarchy is assumed. All plots were generated for an integrated exposure of 6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode.

Predicted single ring Super-K spectra are shown in Fig. 5. By comparing neutrino and antineutrino mode e-like plots it is clear that E_{reco} Vs θ distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events. More details on the advantage of the 2-dimensional templates are given in App. B.

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like Super-K sample. A Super-K MC study is required to determine whether the Super-K efficiency assumptions made there are valid.

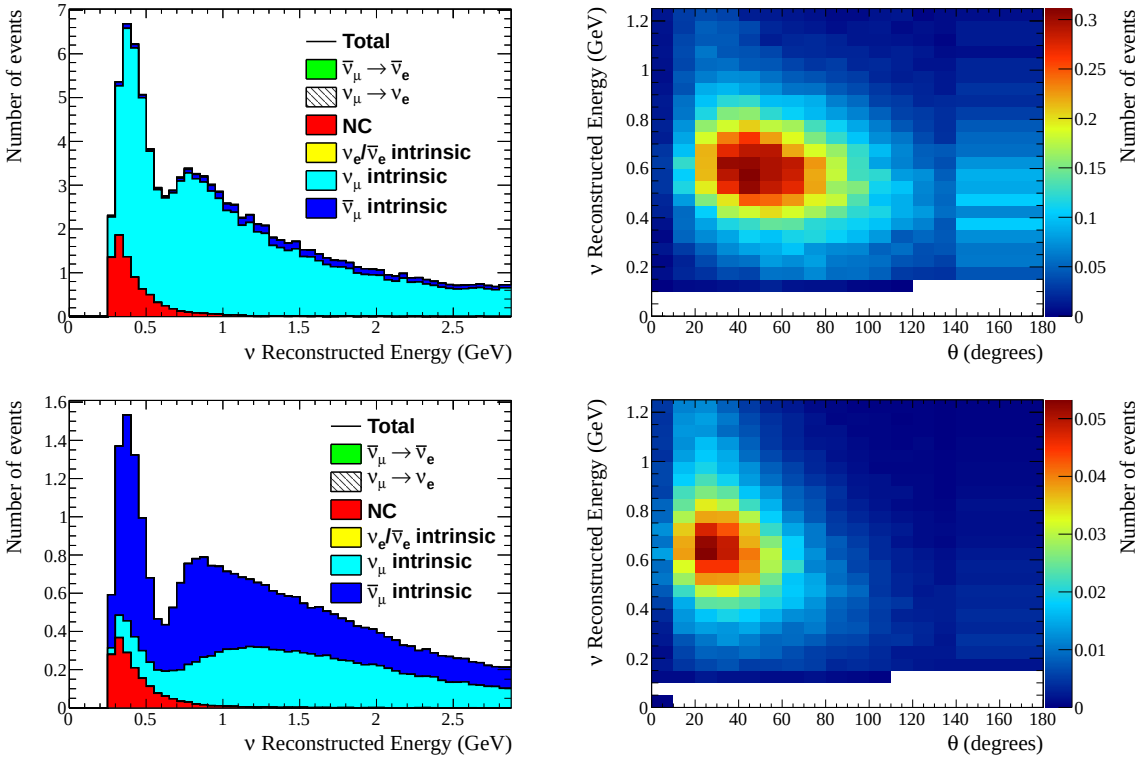


Figure 5: Predicted spectra of neutrino mode muon-like (top left) and electron-like (top right), antineutrino mode muon-like (bottom left) and electron-like (bottom right) events. Muon-like distributions are function of the reconstructed neutrino energy, while the electron-like distributions are function of both the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The muon-like distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in Tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

3.1. Input systematic parameters

This analysis considers 101 systematic parameters which can be grouped into 4 categories:

1. **Flux and cross-section parameters** (65 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to some cross section systematics they remain unconstrained (detailed later). Parameters fit are 11 ν_μ, 5 $\bar{\nu}_\mu$, 7 ν_e and 2 $\bar{\nu}_e$ neutrino flux parameters, 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ antineutrino flux parameters and 15 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples as described in Ref [28]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 6, 7, 8 and 9. The best-fit values are used to reweight the templates as described in Sec. 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis.

The effects of these correlated uncertainties are evaluated using the Cholesky method. The covariance matrix is "Cholesky decomposed" into the product of a lower triangular matrix (all elements above the diagonal are zero) and its transpose (in which all elements below the diagonal are zero). Random tweaks of each parameter are independently drawn from a Gaussian distribution with mean zero and standard deviation 1, and these random numbers are put into a vector. This vector is then multiplied by the lower triangular matrix to give a vector of tweaked uncertainties that have the correct correlations between them. The BANFF parameters are then varied by these tweaked uncertainties, with all parameters being varied simultaneously. This process may be repeated as many times as necessary.

Some of the cross-section parameters have a complicated, often non-linear, effects on true and reconstructed neutrino energy bins; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for ^{16}O
- Binding energy for ^{16}O
- Nucleon to Δ transition axial form factor (C_5^A)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of theta, true and reconstructed energy, and the binning is given in Ref. [35].

The effects of the flux and cross-section normalization uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalization for a limited range of true energy). The cross section parameters which simply normalise the overall spectrum component are:

- MEC, applied to all the MEC events
- MEC-$\bar{\nu}$, applied to all the antineutrino MEC events
- CC coherent for ^{16}O normalisation
- NC coherent normalisation
- NC other normalisation
- NC 1γ

Parameter	Description	Best fit	1σ pre/postfit fractional error
0	FHC ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.128	0.099 / 0.064
1	FHC ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.156	0.103 / 0.061
2	FHC ν_μ flux normalisation, E = 0.5 - 0.6 GeV	1.148	0.096 / 0.051
3	FHC ν_μ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.087 / 0.043
4	FHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.104	0.113 / 0.046
5	FHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.100	0.092 / 0.045
6	FHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.127	0.070 / 0.044
7	FHC ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.124	0.074 / 0.048
8	FHC ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.121	0.087 / 0.049
9	FHC ν_μ flux normalisation, E = 5.0 - 7.0 GeV	1.075	0.098 / 0.053
10	FHC ν_μ flux normalisation, E = 7.0 - 30.0 GeV	1.064	0.114 / 0.065
11	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV	1.100	0.103 / 0.081
12	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.112	0.079 / 0.048
13	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.111	0.084 / 0.060
14	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.116	0.086 / 0.070
15	FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV	1.162	0.086 / 0.069
16	FHC ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.134	0.090 / 0.052
17	FHC ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.135	0.090 / 0.049
18	FHC ν_e flux normalisation, E = 0.7 - 0.8 GeV	1.135	0.086 / 0.047
19	FHC ν_e flux normalisation, E = 0.8 - 1.5 GeV	1.119	0.081 / 0.043
20	FHC ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.115	0.079 / 0.046
21	FHC ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.111	0.084 / 0.050
22	FHC ν_e flux normalisation, E = 4.0 - 30.0 GeV	1.118	0.094 / 0.067
23	FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV	1.121	0.074 / 0.057
24	FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV	1.153	0.128 / 0.117

Table 6: Summary of neutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

- CC ν_e normalisation
- CC $\bar{\nu}_e$ normalisation

Also the uncertainties due to "second class currents" are implemented as CC $\bar{\nu}_e$ and CC ν_e parameters with a 2% anti-correlated constraint between them. The BANFF parameters are applied to the MC templates as specified in Tab. 43. The normalisation parameter of the NC1γ component was not used in the previous analyses, since it was included in NC other.

The predicted number of events in antineutrino e-like sample is now smaller compared to the $\bar{\nu}_e$ appearance analysis [10], 3.218 instead of 3.729. The differences between the current and previous oscillation analyses are mainly due to:

- rescaling of CC coherent and NC1γ components as described in 2.2.1
- new value for $\sin^2\theta_{13}$ corresponding to the PDG 2015 release, as shown in Tab. 12
- a new MEC parameter applied only to $\bar{\nu}$, which BANFF best-fit is 0.578 and reduces the expected total number of $\bar{\nu}_e$ events, while in the past analysis a single systematic parameter was applied to all the MEC events and its BANFF best-fit was about 1.031.

Parameter	Description	Best fit	1σ fractional error
25	RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.098	0.094 / 0.072
26	RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.121	0.079 / 0.052
27	RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.130	0.077 / 0.048
28	RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.155	0.081 / 0.054
29	RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.111	0.080 / 0.055
30	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	1.118	0.104 / 0.071
31	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	1.127	0.102 / 0.060
32	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	1.117	0.096 / 0.052
33	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	1.121	0.085 / 0.044
34	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.155	0.125 / 0.066
35	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.132	0.105 / 0.057
36	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.139	0.080 / 0.053
37	RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	1.141	0.074 / 0.054
38	RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	1.151	0.094 / 0.071
39	RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	1.133	0.093 / 0.070
40	RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	1.082	0.130 / 0.110
41	RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.117	0.069 / 0.051
42	RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.112	0.085 / 0.071
43	RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	1.126	0.095 / 0.058
44	RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	1.127	0.091 / 0.051
45	RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	1.133	0.091 / 0.052
46	RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	1.132	0.084 / 0.046
47	RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	1.125	0.080 / 0.056
48	RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	1.119	0.089 / 0.071
49	RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	1.166	0.156 / 0.141

Table 7: Summary of antineutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Parameter	Description	Best fit	1σ pre/postfit fractional error
50	Meson exchange current normalisation for ^{16}O	1.546	0.352 / 0.343
51	C_5^A nucleon to Δ transition axial form factor	0.789	0.119 / 0.062
52	Scale of isospin 1/2 nonresonant background	1.042	0.154 / 0.134
53	CCQE axial-mass scaling factor	0.931	0.058 / 0.028
54	Resonance-production axial-mass scaling factor	0.889	0.158 / 0.041
55	BANFF; Second current class axial	1.000	1.000 / 1.000
56	BANFF; Second current class vector	1.000	1.000 / 1.000
57	Fermi momentum for ^{16}O	1.041	0.055 / 0.105
58	CC other shape	-0.022	0.400 / 0.208
59	Binding energy for ^{16}O	0.882	0.333 / 0.282
60	CC coherent for ^{16}O normalisation	0.858	0.300 / 0.228
61	NC coherent normalisation	0.931	0.300 / 0.298
62	NC other normalisation	1.000	0.300 / 0.300
63	CC ν_e normalisation	1.000	0.028 / 0.028
64	NC 1γ normalisation	1.000	1.000 / 1.000
65	CC $\bar{\nu}_e$ normalisation	1.000	0.028 / 0.028
66	Antineutrino Meson exchange current normalisation for ^{16}O	0.578	1.000 / 0.177

Table 8: Summary of cross section systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Parameter	Description	1σ fractional error
0	SKDet + FSI/SI 0; E_{reco} 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
1	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.017
2	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
3	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu)	0.077
4	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu)	1.004
5	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu)	0.598
6	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re)	0.100
7	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re)	0.030
8	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re)	0.033
9	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.225
10	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.166
11	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.028
12	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.063
13	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.040
14	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.061
15	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re)	0.488
16	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re)	0.194
17	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re)	0.590
18	SKDet + FSI/SI 0; Ereco 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.018
19	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.016
20	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$CCQE (1Rmu); RHC	0.016
21	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu); RHC	0.093
22	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu); RHC	1.004
23	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu); RHC	0.597
24	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re); RHC	0.060
25	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re); RHC	0.027
26	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re); RHC	0.056
27	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.244
28	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.123
29	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	0.904
30	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.044
31	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.032
32	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.068
33	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re); RHC	0.482
34	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re); RHC	0.171
35	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re); RHC	0.538

Table 9: Summary of SK detector + FSI +SI + PN systematics included in the *VALOR* joint fit analysis.

Parameter	Sample	Category	Energy range (GeV)
0 | 18	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
1 | 19	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
2 | 20	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
3 | 21	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$E_{reco} < 0.35$
4 | 22	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.35 < E_{reco} < 0.80$
5 | 23	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.80 < E_{reco} < 1.25$
6 | 24	electron-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
7 | 25	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
8 | 26	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
9 | 27	electron-like	all NC	$E_{reco} < 0.35$
10 | 28	electron-like	all NC	$0.35 < E_{reco} < 0.80$
11 | 29	electron-like	all NC	$0.80 < E_{reco} < 1.25$
12 | 30	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$E_{reco} < 0.40$
13 | 31	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$0.40 < E_{reco} < 1.10$
14 | 32	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$1.10 < E_{reco} < 30.0$
15 | 33	muon-like	all $\nu_\mu/\bar{\nu}_\mu$ CCnQE	$E_{reco} < 30.0$
16 | 34	muon-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 30.0$
17 | 35	muon-like	all NC	$E_{reco} < 30.0$

Table 10: Binning of Super-K selection efficiency systematic parameters in covariance matrices for both electron- and muon-like samples for each category and reconstructed neutrino energy range. The same parametrization is used for both neutrino and antineutrino mode samples. The parameter number is shown for both neutrino (0-17) and antineutrino (18-35) mode.

2. **Super-K efficiencies & effects of intranuclear and secondary re-interactions** (36 parameters).

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Each Monte Carlo control sample is fitted to the data using the efficiency as a fit parameter, and the uncertainties in the efficiencies are estimated by adding the covariance matrix containing the differences between the best-fit and nominal efficiencies to the covariance matrix containing the statistical errors in the fitted efficiencies [30] [31].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [20]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [20]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the Super-K detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [23]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single e-like or μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [19]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [20]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the Super-K detector efficiency and FSI+SI systematics together, and add their covari-

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gauss	0.085 ± 0.005
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only	uniform	$[0; 0.4]$
$\sin^2 2\theta_{12}$	gauss	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	uniform	$[2; 3] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gauss	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	fixed	NH or IH

Table 11: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [64]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [64], while all the other oscillation parameter values correspond to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH.

ance matrices linearly. Each of these covariance matrices have 36 correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in Tab. **??**. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **Super-K energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [32]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 43 and is common to all the super-K samples.

4. **Oscillation parameters and mass hierarchy** (7 parameters).

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters are marginalised except the mass hierarchy which is fixed to either normal or inverted hierarchy. The treatment of the oscillation parameters in the analysis is shown in Tab. 11 while the nominal values are shown in Tab. 12. For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", or not, denoted as "T2K only". In the first case a gaussian prior given by the combined measurement of reactor experiments is used, while in the latter case a flat prior which covers a large range of values is adopted. With respect to the older global fit [63], the value of $\sin^2\theta_{13}$ has been reduced of about 10% and the error reduced as well. Since an excess of events was seen in the T2K Run 1-4 neutrino mode e-like sample and not much data are collected in neutrino mode during the Run 5+6, this change is expected to have an impact on the search for CP violation, making the confidence intervals tighter.

Parameter(s)	Asimov A	Asimov B
$\sin^2\theta_{23}$	0.528	0.45
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	0.085 (0.0217)	0.085 (0.0217)
$\sin^2 2\theta_{12}$	0.846	0.846
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601	0
Mass Hierarchy	Normal	Normal

Table 12: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [64], while all the other oscillation parameter values correspond to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH. The only difference between the Asimov data sets A and B is given by δ_{CP}, set to 0, and $\sin^2\theta_{23}$ which corresponds to the opposite octant.

3.2. Evaluation of the effects of systematic parameters on the predicted single e-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 3.1 on the predicted number of single ring events and the predicted spectra in Super-K. All results are generated for an integrated exposure of 6.914×10^{20} POT for neutrino and 4.011×10^{20} POT for antineutrino mode and the BANFF-tuned MC templates are used.

In Tab. 13, 14, 15 and 16 the effect of each single systematic parameter obtained by varying its value by $\pm1\sigma$ is shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
All syst	120.349	125.91	153.13	21.616 %	103.868	-17.508 %
$f^{banff}_{0;t,r}$	120.349	125.914	126.429	0.40884 %	125.399	-0.40884 %
$f^{banff}_{1;t,r}$	120.349	125.914	126.409	0.39338 %	125.418	-0.39338 %
$f^{banff}_{2;t,r}$	120.349	125.914	126.033	0.094558 %	125.795	-0.094558 %
$f^{banff}_{3;t,r}$	120.349	125.914	126.012	0.078072 %	125.815	-0.078072 %
$f^{banff}_{4;t,r}$	120.349	125.914	126.834	0.73082 %	124.994	-0.73082 %
$f^{banff}_{5;t,r}$	120.349	125.914	126.771	0.68095 %	125.056	-0.68095 %
$f^{banff}_{6;t,r}$	120.349	125.914	126.733	0.65073 %	125.094	-0.65073 %
$f^{banff}_{7;t,r}$	120.349	125.914	126.452	0.42768 %	125.375	-0.42768 %
$f^{banff}_{8;t,r}$	120.349	125.914	126.471	0.44277 %	125.356	-0.44277 %
$f^{banff}_{9;t,r}$	120.349	125.914	126.126	0.16892 %	125.701	-0.16892 %
$f^{banff}_{10;t,r}$	120.349	125.914	125.96	0.036964 %	125.867	-0.036964 %
$f^{banff}_{11;t,r}$	120.349	125.914	125.941	0.021242 %	125.887	-0.021242 %
$f^{banff}_{12;t,r}$	120.349	125.914	125.934	0.015671 %	125.894	-0.015671 %
$f^{banff}_{13;t,r}$	120.349	125.914	126.002	0.069974 %	125.826	-0.069974 %
$f^{banff}_{14;t,r}$	120.349	125.914	126.08	0.13209 %	125.747	-0.13209 %
$f^{banff}_{15;t,r}$	120.349	125.914	126.063	0.1185 %	125.765	-0.1185 %
$f^{banff}_{16;t,r}$	120.349	125.914	125.914	0.00034323 %	125.913	-0.00034323 %
$f^{banff}_{17;t,r}$	120.349	125.914	125.915	0.00058236 %	125.913	-0.00058236 %
$f^{banff}_{18;t,r}$	120.349	125.914	125.914	0.00041762 %	125.913	-0.00041762 %
$f^{banff}_{19;t,r}$	120.349	125.914	125.916	0.0021262 %	125.911	-0.0021262 %
$f^{banff}_{20;t,r}$	120.349	125.914	125.917	0.0028071 %	125.91	-0.0028071 %
$f^{banff}_{21;t,r}$	120.349	125.914	125.917	0.0024707 %	125.911	-0.0024707 %
$f^{banff}_{22;t,r}$	120.349	125.914	125.916	0.0015978 %	125.912	-0.0015978 %
$f^{banff}_{23;t,r}$	120.349	125.914	125.914	0.00041569 %	125.913	-0.00041569 %
$f^{banff}_{24;t,r}$	120.349	125.914	125.915	0.0011942 %	125.912	-0.0011942 %
$f^{banff}_{Norm_{MEC}}$	120.349	125.914	130.333	3.5095 %	121.495	-3.5095 %
$f^{banff}_{C_5^A}$	120.349	125.914	127.621	1.3562 %	124.311	-1.2726 %
$f^{banff}_{Bg_{RES}}$	120.349	125.914	126.985	0.85109 %	124.971	-0.74846 %
$f^{banff}_{M_A^{QE}}$	120.349	125.914	127.832	1.5237 %	123.975	-1.5395 %
$f^{banff}_{M_A^{RES}}$	120.349	125.914	127.008	0.86933 %	124.837	-0.85554 %
$f^{banff}_{p_f}$	120.349	125.914	123.763	-1.7078 %	127.899	1.5763 %
$f^{banff}_{Shape_{CCoth}}$	120.349	125.914	126.435	0.41379 %	125.393	-0.41379 %
$f^{banff}_{E_B}$	120.349	125.914	125.899	-0.01212 %	125.93	0.012715 %
$f^{banff}_{Norm_{CCcoh}}$	120.349	125.914	126.022	0.086181 %	125.805	-0.086181 %
$f^{banff}_{Norm_{NCcoh}}$	120.349	125.914	125.919	0.0044519 %	125.908	-0.0044519 %
$f^{banff}_{Norm_{NCother}}$	120.349	125.914	126.901	0.78393 %	124.927	-0.78393 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	120.349	125.914	125.926	0.0093354 %	125.902	-0.0093354 %
$f^{banff}_{Norm_{NC1\gamma}}$	120.349	125.914	125.918	0.0029615 %	125.91	-0.0029615 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	120.349	125.914	125.914	0.00011837 %	125.914	-0.00011837 %
$f^{banff}_{Norm_{MECBar}}$	120.349	125.914	126.172	0.20522 %	125.655	-0.20522 %
$f^{SK}_{E;r}$	120.349	125.914	125.914	-4.5145e-14 %	125.914	-7.9003e-14 %
$f^{SK+FSI}_{0;t,r}$	120.349	125.914	126.074	0.1271 %	125.754	-0.1271 %
$f^{SK+FSI}_{1;t,r}$	120.349	125.914	126.556	0.5104 %	125.271	-0.5104 %
$f^{SK+FSI}_{2;t,r}$	120.349	125.914	126.675	0.60427 %	125.153	-0.60427 %
$f^{SK+FSI}_{3;t,r}$	120.349	125.914	128.014	1.6677 %	123.814	-1.6677 %
$f^{SK+FSI}_{4;t,r}$	120.349	125.914	126.336	0.33551 %	125.491	-0.33551 %
$f^{SK+FSI}_{5;t,r}$	120.349	125.914	130.738	3.8311 %	121.09	-3.8311 %

Table 13: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 6.914×10^{20} POT for neutrino mode.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	24.2563	26.5935	26.6217	0.10583 %	26.5654	-0.10583 %
$f^{banff}_{1;t,r}$	24.2563	26.5935	26.7186	0.47018 %	26.4685	-0.47018 %
$f^{banff}_{2;t,r}$	24.2563	26.5935	26.8418	0.93345 %	26.3453	-0.93345 %
$f^{banff}_{3;t,r}$	24.2563	26.5935	26.8365	0.91353 %	26.3506	-0.91353 %
$f^{banff}_{4;t,r}$	24.2563	26.5935	26.8768	1.0652 %	26.3102	-1.0652 %
$f^{banff}_{5;t,r}$	24.2563	26.5935	26.6316	0.14302 %	26.5555	-0.14302 %
$f^{banff}_{6;t,r}$	24.2563	26.5935	26.6004	0.025826 %	26.5867	-0.025826 %
$f^{banff}_{7;t,r}$	24.2563	26.5935	26.5978	0.015909 %	26.5893	-0.015909 %
$f^{banff}_{8;t,r}$	24.2563	26.5935	26.5992	0.021436 %	26.5878	-0.021436 %
$f^{banff}_{9;t,r}$	24.2563	26.5935	26.5962	0.010031 %	26.5909	-0.010031 %
$f^{banff}_{10;t,r}$	24.2563	26.5935	26.595	0.0054078 %	26.5921	-0.0054078 %
$f^{banff}_{11;t,r}$	24.2563	26.5935	26.5975	0.014836 %	26.5896	-0.014836 %
$f^{banff}_{12;t,r}$	24.2563	26.5935	26.5963	0.010428 %	26.5907	-0.010428 %
$f^{banff}_{13;t,r}$	24.2563	26.5935	26.596	0.0093637 %	26.591	-0.0093637 %
$f^{banff}_{14;t,r}$	24.2563	26.5935	26.595	0.00542 %	26.5921	-0.00542 %
$f^{banff}_{15;t,r}$	24.2563	26.5935	26.5948	0.0046444 %	26.5923	-0.0046444 %
$f^{banff}_{16;t,r}$	24.2563	26.5935	26.6185	0.093913 %	26.5685	-0.093913 %
$f^{banff}_{17;t,r}$	24.2563	26.5935	26.628	0.12954 %	26.5591	-0.12954 %
$f^{banff}_{18;t,r}$	24.2563	26.5935	26.6109	0.065228 %	26.5762	-0.065228 %
$f^{banff}_{19;t,r}$	24.2563	26.5935	26.6464	0.19869 %	26.5407	-0.19869 %
$f^{banff}_{20;t,r}$	24.2563	26.5935	26.598	0.016936 %	26.589	-0.016936 %
$f^{banff}_{21;t,r}$	24.2563	26.5935	26.594	0.0017641 %	26.5931	-0.0017641 %
$f^{banff}_{22;t,r}$	24.2563	26.5935	26.5937	0.00071333 %	26.5933	-0.00071333 %
$f^{banff}_{23;t,r}$	24.2563	26.5935	26.6008	0.027288 %	26.5863	-0.027288 %
$f^{banff}_{24;t,r}$	24.2563	26.5935	26.5937	0.00079831 %	26.5933	-0.00079831 %
$f^{banff}_{Norm_{MEC}}$	24.2563	26.5935	27.6259	3.882 %	25.5612	-3.882 %
$f^{banff}_{C^A_5}$	24.2563	26.5935	26.8014	0.78157 %	26.3972	-0.73812 %
$f^{banff}_{Bg_{RES}}$	24.2563	26.5935	26.6753	0.30745 %	26.5216	-0.27037 %
$f^{banff}_{M^{QE}_A}$	24.2563	26.5935	27.0363	1.6648 %	26.1369	-1.7172 %
$f^{banff}_{M^{RES}_A}$	24.2563	26.5935	26.6925	0.37237 %	26.4929	-0.37839 %
$f^{banff}_{p_f}$	24.2563	26.5935	25.9799	-2.3076 %	27.1501	2.093 %
$f^{banff}_{Shape_{CCoth}}$	24.2563	26.5935	26.6063	0.048094 %	26.5807	-0.048083 %
$f^{banff}_{E_B}$	24.2563	26.5935	26.5533	-0.15134 %	26.626	0.12201 %
$f^{banff}_{Norm_{CCcoh}}$	24.2563	26.5935	26.5991	0.021135 %	26.5879	-0.021135 %
$f^{banff}_{Norm_{NCcoh}}$	24.2563	26.5935	26.6567	0.23764 %	26.5303	-0.23764 %
$f^{banff}_{Norm_{NCother}}$	24.2563	26.5935	26.6352	0.15691 %	26.5518	-0.15691 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	24.2563	26.5935	27.2933	2.6315 %	25.8937	-2.6315 %
$f^{banff}_{Norm_{NC1\gamma}}$	24.2563	26.5935	27.0873	1.8568 %	26.0997	-1.8568 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	24.2563	26.5935	26.6014	0.029817 %	26.5856	-0.029817 %
$f^{banff}_{Norm_{MECBar}}$	24.2563	26.5935	26.6061	0.047159 %	26.581	-0.047159 %
$f^{SK}_{E;r}$	24.2563	26.5935	26.5935	1.3359e-14 %	26.5935	-1.3359e-14 %
$f^{SK+FSI}_{6;t,r}$	24.2563	26.5935	26.8445	0.94359 %	26.3426	-0.94359 %
$f^{SK+FSI}_{7;t,r}$	24.2563	26.5935	27.0622	1.7624 %	26.1248	-1.7624 %
$f^{SK+FSI}_{8;t,r}$	24.2563	26.5935	26.7121	0.44605 %	26.4749	-0.44605 %
$f^{SK+FSI}_{9;t,r}$	24.2563	26.5935	26.6525	0.22178 %	26.5345	-0.22178 %
$f^{SK+FSI}_{10;t,r}$	24.2563	26.5935	26.6244	0.116 %	26.5627	-0.116 %
$f^{SK+FSI}_{11;t,r}$	24.2563	26.5935	26.6021	0.0321 %	26.585	-0.0321 %
$f^{SK+FSI}_{12;t,r}$	24.2563	26.5935	26.6184	0.093385 %	26.5687	-0.093385 %
$f^{SK+FSI}_{13;t,r}$	24.2563	26.5935	26.6657	0.27133 %	26.5214	-0.27133 %
$f^{SK+FSI}_{14;t,r}$	24.2563	26.5935	26.6662	0.27323 %	26.5209	-0.27323 %
$f^{SK+FSI}_{15;t,r}$	24.2563	26.5935	26.8604	1.0036 %	26.3266	-1.0036 %
$f^{SK+FSI}_{16;t,r}$	24.2563	26.5935	26.7358	0.53512 %	26.4512	-0.53512 %
$f^{SK+FSI}_{17;t,r}$	24.2563	26.5935	26.7166	0.46264 %	26.4705	-0.46264 %

Table 14: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 6.914×10^{20} POT for neutrino mode.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
All syst	33.354	34.284	42.342	23.505 %	28.0412	-18.209 %
$f^{banff}_{0;t,r}RHC$	33.354	34.2841	34.3212	0.1084 %	34.2469	-0.1084 %
$f^{banff}_{1;t,r}RHC$	33.354	34.2841	34.3316	0.13851 %	34.2366	-0.13851 %
$f^{banff}_{2;t,r}RHC$	33.354	34.2841	34.4224	0.4034 %	34.1458	-0.4034 %
$f^{banff}_{3;t,r}RHC$	33.354	34.2841	34.5137	0.66983 %	34.0544	-0.66983 %
$f^{banff}_{4;t,r}RHC$	33.354	34.2841	34.5214	0.69216 %	34.0468	-0.69216 %
$f^{banff}_{5;t,r}RHC$	33.354	34.2841	34.4278	0.41934 %	34.1403	-0.41934 %
$f^{banff}_{6;t,r}RHC$	33.354	34.2841	34.3655	0.23759 %	34.2026	-0.23759 %
$f^{banff}_{7;t,r}RHC$	33.354	34.2841	34.301	0.049526 %	34.2671	-0.049526 %
$f^{banff}_{8;t,r}RHC$	33.354	34.2841	34.2964	0.036085 %	34.2717	-0.036085 %
$f^{banff}_{9;t,r}RHC$	33.354	34.2841	34.4627	0.52119 %	34.1054	-0.52119 %
$f^{banff}_{10;t,r}RHC$	33.354	34.2841	34.4614	0.51727 %	34.1067	-0.51727 %
$f^{banff}_{11;t,r}RHC$	33.354	34.2841	34.4688	0.53874 %	34.0994	-0.53874 %
$f^{banff}_{12;t,r}RHC$	33.354	34.2841	34.3882	0.30381 %	34.1799	-0.30381 %
$f^{banff}_{13;t,r}RHC$	33.354	34.2841	34.3968	0.32868 %	34.1714	-0.32868 %
$f^{banff}_{14;t,r}RHC$	33.354	34.2841	34.3136	0.086075 %	34.2546	-0.086075 %
$f^{banff}_{15;t,r}RHC$	33.354	34.2841	34.2898	0.016631 %	34.2784	-0.016631 %
$f^{banff}_{16;t,r}RHC$	33.354	34.2841	34.2853	0.0035678 %	34.2828	-0.0035678 %
$f^{banff}_{17;t,r}RHC$	33.354	34.2841	34.2856	0.0045846 %	34.2825	-0.0045846 %
$f^{banff}_{18;t,r}RHC$	33.354	34.2841	34.2841	0.00019776 %	34.284	-0.00019776 %
$f^{banff}_{19;t,r}RHC$	33.354	34.2841	34.2841	0.00023211 %	34.284	-0.00023211 %
$f^{banff}_{20;t,r}RHC$	33.354	34.2841	34.2841	0.00018231 %	34.284	-0.00018231 %
$f^{banff}_{21;t,r}RHC$	33.354	34.2841	34.2843	0.0007317 %	34.2838	-0.0007317 %
$f^{banff}_{22;t,r}RHC$	33.354	34.2841	34.2845	0.0011599 %	34.2837	-0.0011599 %
$f^{banff}_{23;t,r}RHC$	33.354	34.2841	34.2846	0.0015682 %	34.2835	-0.0015682 %
$f^{banff}_{24;t,r}RHC$	33.354	34.2841	34.2846	0.0016844 %	34.2835	-0.0016844 %
$f^{banff}_{Norm_{MEC}}$	33.354	34.2841	35.2949	2.9484 %	33.2732	-2.9484 %
$f^{banff}_{C^A_5}$	33.354	34.2841	34.7665	1.4073 %	33.8496	-1.2673 %
$f^{banff}_{Bg_{RES}}$	33.354	34.2841	34.6114	0.95474 %	33.9962	-0.8396 %
$f^{banff}_{M^{QE}_A}$	33.354	34.2841	34.7782	1.4412 %	33.7964	-1.4226 %
$f^{banff}_{M^{RES}_A}$	33.354	34.2841	34.557	0.79619 %	34.0229	-0.76184 %
$f^{banff}_{p_f}$	33.354	34.2841	33.7369	-1.596 %	34.8188	1.5598 %
$f^{banff}_{Shape_{CCoth}}$	33.354	34.2841	34.4278	0.41939 %	34.1403	-0.41939 %
$f^{banff}_{E_B}$	33.354	34.2841	34.2825	-0.0045403 %	34.2856	0.0043363 %
$f^{banff}_{Norm_{CCcoh}}$	33.354	34.2841	34.3434	0.17311 %	34.2247	-0.17311 %
$f^{banff}_{Norm_{NCcoh}}$	33.354	34.2841	34.2866	0.0073745 %	34.2815	-0.0073745 %
$f^{banff}_{Norm_{NCother}}$	33.354	34.2841	34.5427	0.75445 %	34.0254	-0.75445 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	33.354	34.2841	34.2846	0.0015892 %	34.2835	-0.0015892 %
$f^{banff}_{Norm_{NC1\gamma}}$	33.354	34.2841	34.2844	0.00099164 %	34.2837	-0.00099164 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	33.354	34.2841	34.2848	0.0022231 %	34.2833	-0.0022231 %
$f^{banff}_{Norm_{MECBar}}$	33.354	34.2841	34.9123	1.8323 %	33.6559	-1.8323 %
$f^{SK}_{E;r}$	33.354	34.2841	34.2841	-2.0725e-14 %	34.2841	-2.0725e-14 %
$f^{SK+FSI}_{0;t,r}RHC$	33.354	34.2841	34.3274	0.12641 %	34.2407	-0.12641 %
$f^{SK+FSI}_{1;t,r}RHC$	33.354	34.2841	34.4079	0.36115 %	34.1602	-0.36115 %
$f^{SK+FSI}_{2;t,r}RHC$	33.354	34.2841	34.5168	0.67883 %	34.0513	-0.67883 %
$f^{SK+FSI}_{3;t,r}RHC$	33.354	34.2841	35.0403	2.2058 %	33.5278	-2.2058 %
$f^{SK+FSI}_{4;t,r}RHC$	33.354	34.2841	34.3304	0.13529 %	34.2377	-0.13529 %
$f^{SK+FSI}_{5;t,r}RHC$	33.354	34.2841	35.3403	3.0809 %	33.2278	-3.0809 %

Table 15: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 4.011×10^{20} POT for antineutrino mode.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
All syst	3.11707	3.2183	4.2182	31.07 %	2.48667	-22.734 %
$f^{banff}_{0;t,r}RHC$	3.11707	3.21831	3.23425	0.49537 %	3.20237	-0.49537 %
$f^{banff}_{1;t,r}RHC$	3.11707	3.21831	3.22891	0.32934 %	3.20771	-0.32934 %
$f^{banff}_{2;t,r}RHC$	3.11707	3.21831	3.22364	0.16581 %	3.21297	-0.16581 %
$f^{banff}_{3;t,r}RHC$	3.11707	3.21831	3.22045	0.066577 %	3.21617	-0.066577 %
$f^{banff}_{4;t,r}RHC$	3.11707	3.21831	3.22085	0.078879 %	3.21577	-0.078879 %
$f^{banff}_{5;t,r}RHC$	3.11707	3.21831	3.22043	0.065975 %	3.21619	-0.065975 %
$f^{banff}_{6;t,r}RHC$	3.11707	3.21831	3.22509	0.21083 %	3.21152	-0.21083 %
$f^{banff}_{7;t,r}RHC$	3.11707	3.21831	3.23439	0.49965 %	3.20223	-0.49965 %
$f^{banff}_{8;t,r}RHC$	3.11707	3.21831	3.2365	0.56517 %	3.20012	-0.56517 %
$f^{banff}_{9;t,r}RHC$	3.11707	3.21831	3.25195	1.0453 %	3.18467	-1.0453 %
$f^{banff}_{10;t,r}RHC$	3.11707	3.21831	3.2239	0.17371 %	3.21272	-0.17371 %
$f^{banff}_{11;t,r}RHC$	3.11707	3.21831	3.21966	0.041857 %	3.21696	-0.041857 %
$f^{banff}_{12;t,r}RHC$	3.11707	3.21831	3.21896	0.020249 %	3.21766	-0.020249 %
$f^{banff}_{13;t,r}RHC$	3.11707	3.21831	3.21907	0.023667 %	3.21755	-0.023667 %
$f^{banff}_{14;t,r}RHC$	3.11707	3.21831	3.21857	0.0081707 %	3.21805	-0.0081707 %
$f^{banff}_{15;t,r}RHC$	3.11707	3.21831	3.21855	0.0074537 %	3.21807	-0.0074537 %
$f^{banff}_{16;t,r}RHC$	3.11707	3.21831	3.23255	0.44247 %	3.20407	-0.44247 %
$f^{banff}_{17;t,r}RHC$	3.11707	3.21831	3.21849	0.0055449 %	3.21813	-0.0055449 %
$f^{banff}_{18;t,r}RHC$	3.11707	3.21831	3.22275	0.13805 %	3.21387	-0.13805 %
$f^{banff}_{19;t,r}RHC$	3.11707	3.21831	3.22312	0.14949 %	3.2135	-0.14949 %
$f^{banff}_{20;t,r}RHC$	3.11707	3.21831	3.22095	0.081949 %	3.21567	-0.081949 %
$f^{banff}_{21;t,r}RHC$	3.11707	3.21831	3.22701	0.2705 %	3.2096	-0.2705 %
$f^{banff}_{22;t,r}RHC$	3.11707	3.21831	3.21896	0.020273 %	3.21766	-0.020273 %
$f^{banff}_{23;t,r}RHC$	3.11707	3.21831	3.21839	0.002446 %	3.21823	-0.002446 %
$f^{banff}_{24;t,r}RHC$	3.11707	3.21831	3.21838	0.002219 %	3.21824	-0.002219 %
$f^{banff}_{Norm_{MEC}}$	3.11707	3.21831	3.31425	2.9812 %	3.12237	-2.9812 %
$f^{banff}_{C^A_5}$	3.11707	3.21831	3.25166	1.0364 %	3.19031	-0.86988 %
$f^{banff}_{Bg_{RES}}$	3.11707	3.21831	3.23459	0.5058 %	3.20399	-0.4448 %
$f^{banff}_{M^{QE}_A}$	3.11707	3.21831	3.26276	1.3812 %	3.17411	-1.3732 %
$f^{banff}_{M^{RES}_A}$	3.11707	3.21831	3.22955	0.34945 %	3.20849	-0.30505 %
$f^{banff}_{p_f}$	3.11707	3.21831	3.14236	-2.3599 %	3.29191	2.2869 %
$f^{banff}_{Shape_{CCoth}}$	3.11707	3.21831	3.22119	0.089515 %	3.21543	-0.089518 %
$f^{banff}_{E_B}$	3.11707	3.21831	3.21749	-0.025439 %	3.21891	0.018569 %
$f^{banff}_{Norm_{CCcoh}}$	3.11707	3.21831	3.22518	0.21355 %	3.21144	-0.21355 %
$f^{banff}_{Norm_{NCcoh}}$	3.11707	3.21831	3.2514	1.0282 %	3.18522	-1.0282 %
$f^{banff}_{Norm_{NCother}}$	3.11707	3.21831	3.22886	0.32779 %	3.20776	-0.32779 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	3.11707	3.21831	3.24258	0.75412 %	3.19404	-0.75412 %
$f^{banff}_{Norm_{NC1\gamma}}$	3.11707	3.21831	3.33932	3.76 %	3.0973	-3.76 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	3.11707	3.21831	3.27411	1.734 %	3.1625	-1.734 %
$f^{banff}_{Norm_{MECBar}}$	3.11707	3.21831	3.2943	2.3612 %	3.14232	-2.3612 %
$f^{SK}_{E;r}$	3.11707	3.21831	3.21831	-1.3799e-14 %	3.21831	-4.1397e-14 %
$f^{SK+FSI}_{6;t,r}RHC$	3.11707	3.21831	3.22805	0.30255 %	3.20857	-0.30255 %
$f^{SK+FSI}_{7;t,r}RHC$	3.11707	3.21831	3.25713	1.2064 %	3.17948	-1.2064 %
$f^{SK+FSI}_{8;t,r}RHC$	3.11707	3.21831	3.24365	0.78746 %	3.19297	-0.78746 %
$f^{SK+FSI}_{9;t,r}RHC$	3.11707	3.21831	3.22378	0.16988 %	3.21284	-0.16988 %
$f^{SK+FSI}_{10;t,r}RHC$	3.11707	3.21831	3.2219	0.11164 %	3.21472	-0.11164 %
$f^{SK+FSI}_{11;t,r}RHC$	3.11707	3.21831	3.22052	0.068625 %	3.2161	-0.068625 %
$f^{SK+FSI}_{12;t,r}RHC$	3.11707	3.21831	3.22172	0.10612 %	3.21489	-0.10612 %
$f^{SK+FSI}_{13;t,r}RHC$	3.11707	3.21831	3.23071	0.38524 %	3.20591	-0.38524 %
$f^{SK+FSI}_{14;t,r}RHC$	3.11707	3.21831	3.23982	0.66827 %	3.1968	-0.66827 %
$f^{SK+FSI}_{15;t,r}RHC$	3.11707	3.21831	3.28217	1.9844 %	3.15444	-1.9844 %
$f^{SK+FSI}_{16;t,r}RHC$	3.11707	3.21831	3.25076	1.0083 %	3.18586	-1.0083 %
$f^{SK+FSI}_{17;t,r}RHC$	3.11707	3.21831	3.2479	0.91957 %	3.18871	-0.91957 %

Table 16: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 4.011×10^{20} POT for antineutrino mode.

The effects of all the combined systematic uncertainties on the reconstructed energy spectrum of single ring events is shown in Fig. 6 and 7 for the typical oscillation scenario with parameters of Asimov datasets A. In App. D the same plots are shown for different oscillation scenarios. These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm 1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 17, 18, 19 and 20 with the typical parameter values given in Tab. 12. For the measurement of δ_{CP} it important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples, which is the consequence of correlations between systematic parameters. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. On the other hand if correlations exist but are not taken into account, the measurement of δ_{CP} could be biased. Depending on the value of δ_{CP}, correlations or anti-correlations between parameters are more conservative. If true $\delta_{CP} = \pm\pi/2$ the most conservative case is given by anti-correlation in the e-like samples, since the oscillation asymmetry could be hidden by the systematic uncertainties. If true $\delta_{CP} \sim 0$ uncorrelated parameters reduce the sensitivity to δ_{CP}. The effect of the systematic parameters on the ratio between neutrino and antineutrino mode events are shown in Tab. 21. The most critical systematic parameters are σ_{ν_e} and $\sigma_{\bar{\nu}_e}$. In fact they are partially anti-correlated in this analysis and are applied respectively to ν_e and $\bar{\nu}_e$ events.

In order to reduce the CPU time, the toys MC The toys MC were produced using 1-dimensional templates E_{reco} e-like templates, instead of the 2-dimensional E_{reco} Vs θ, used in the analysis. However it has been checked that using 1- or 2- dimensional templates does not make any difference in the drawing of the error envelopes and the effect of the systematic uncertainties on the total number of events.

In Figs. 7 the error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters, are shown.

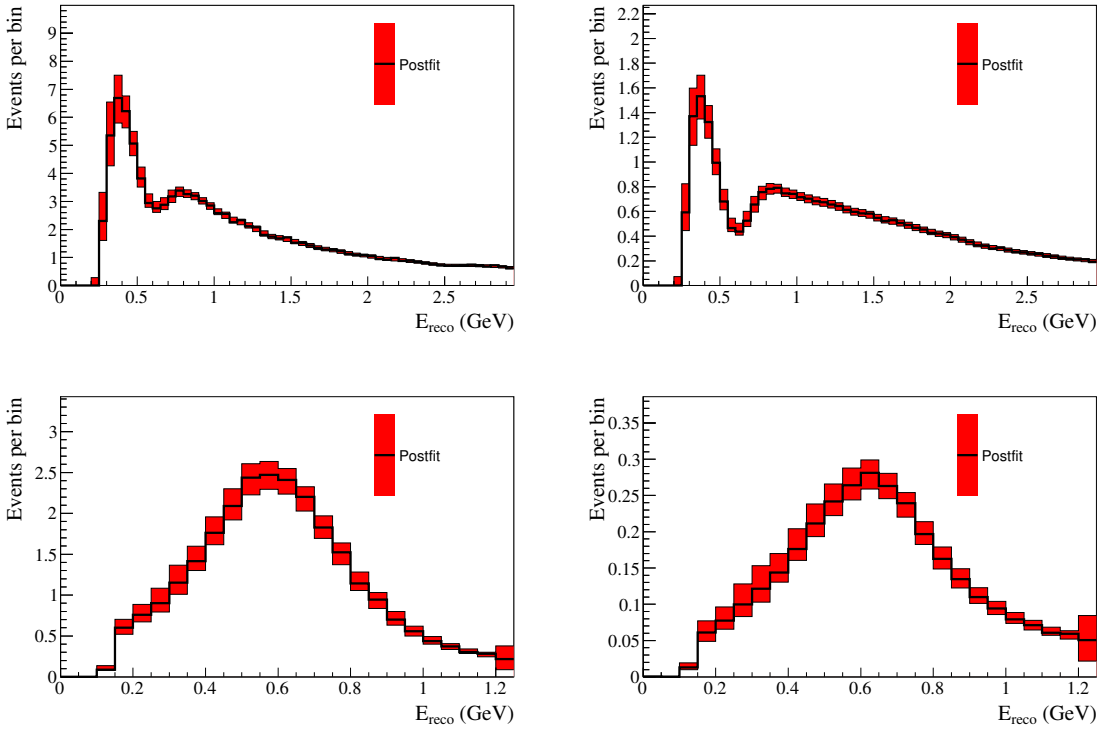


Figure 6: Error envelopes for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

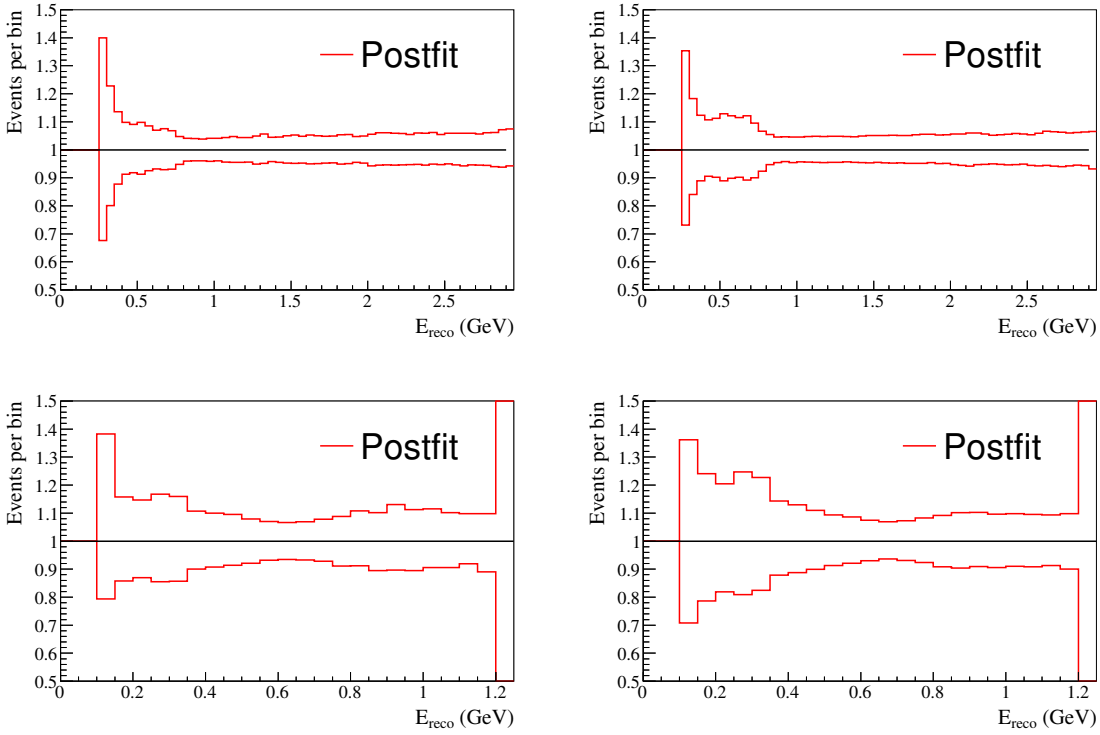


Figure 7: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In Fig. 8, 9, 10, 11, 12 the error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.

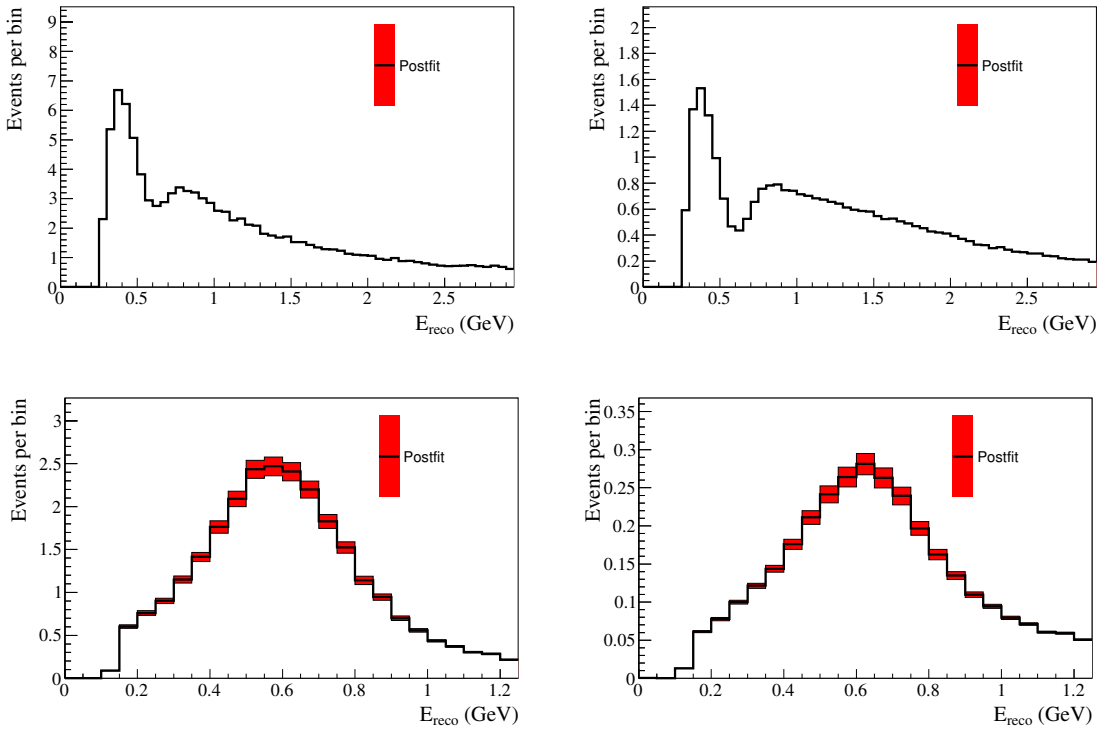


Figure 8: Error envelopes for the energy spectrum as a result of the oscillation systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

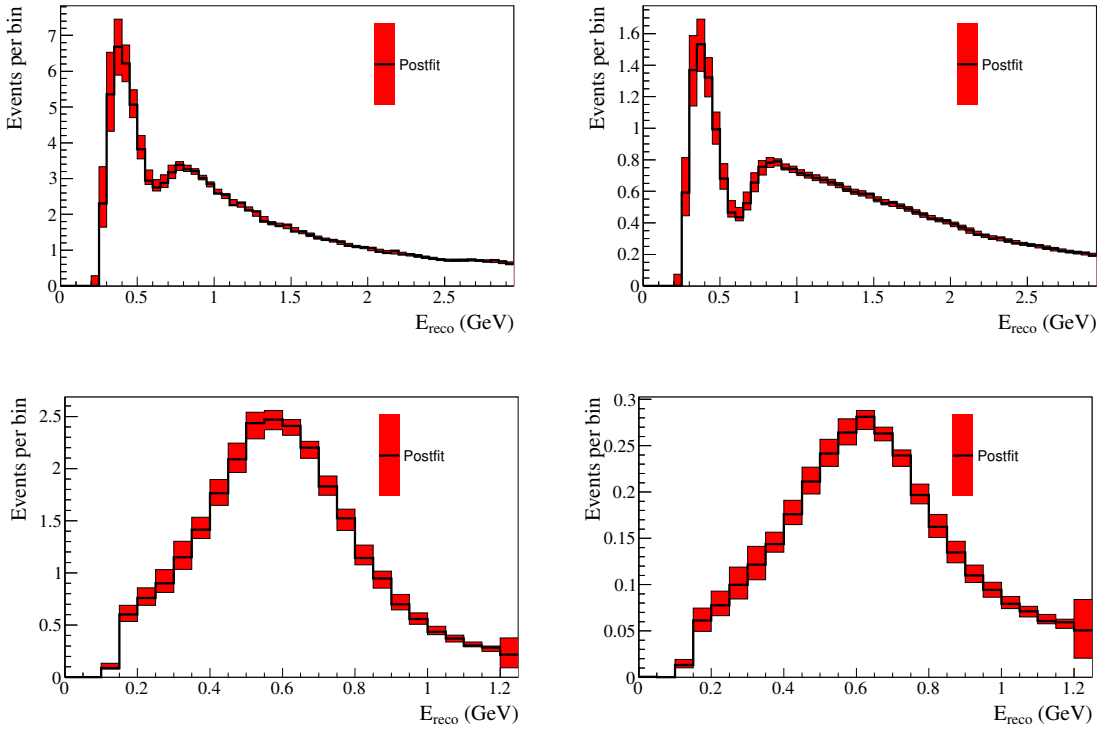


Figure 9: Error envelopes for the energy spectrum as a result of the super-K + FSI + SI + PN systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

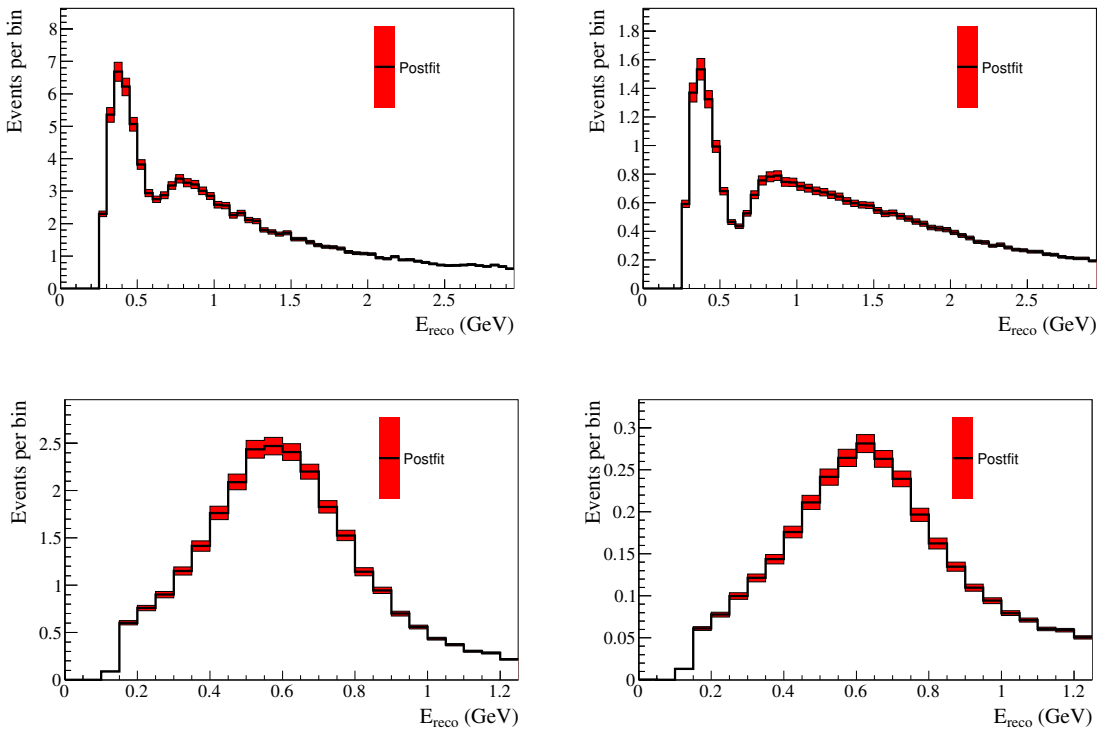


Figure 10: Error envelopes for the energy spectrum as a result of the flux systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

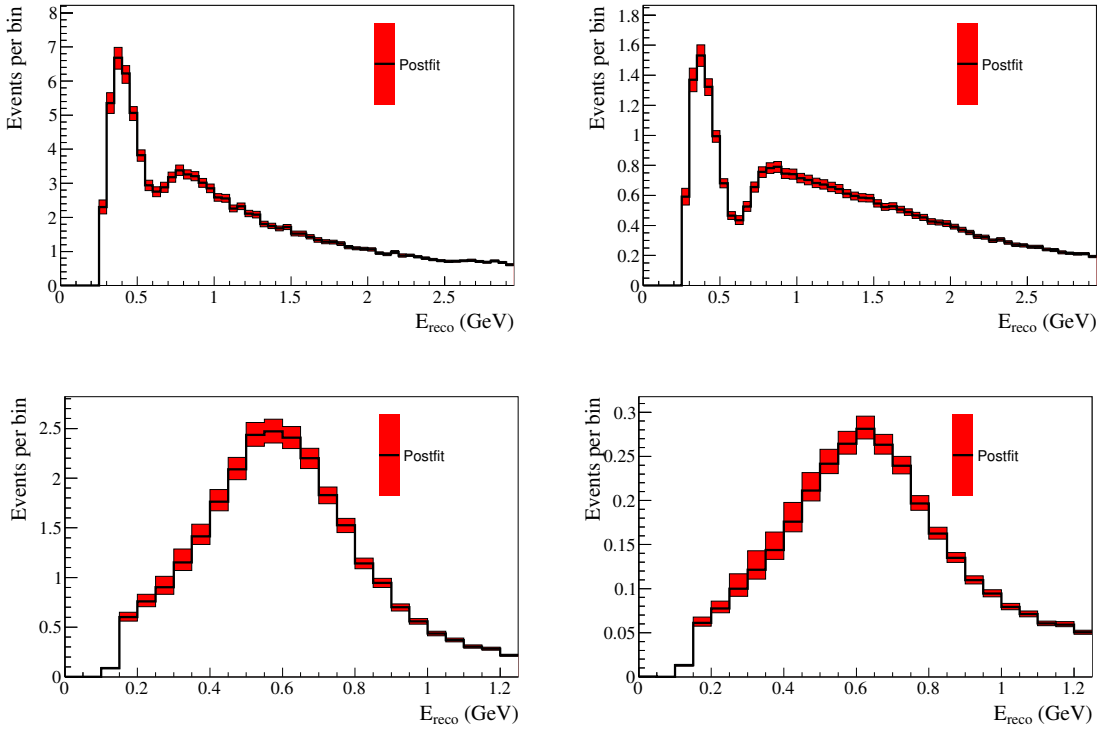


Figure 11: Error envelopes for the energy spectrum as a result of the cross section systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

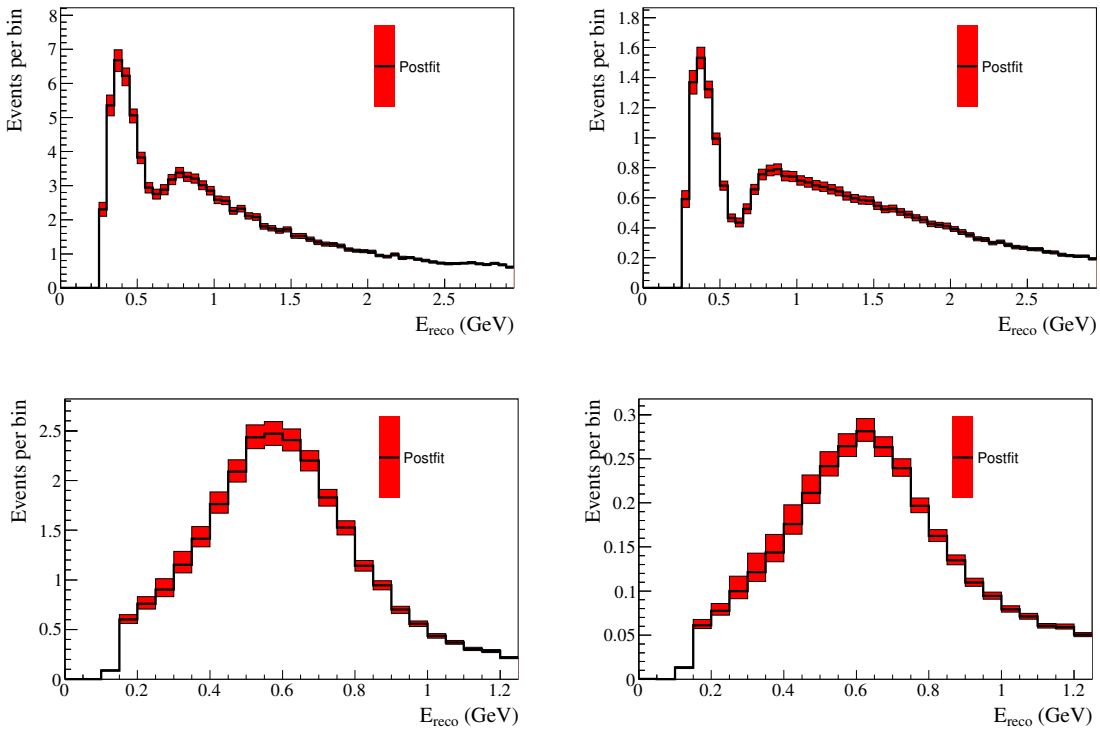


Figure 12: Error envelopes for the energy spectrum as a result of the flux plus cross section systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In Fig. 13, 14, 15, 16, 17 the ratio error envelopes of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.

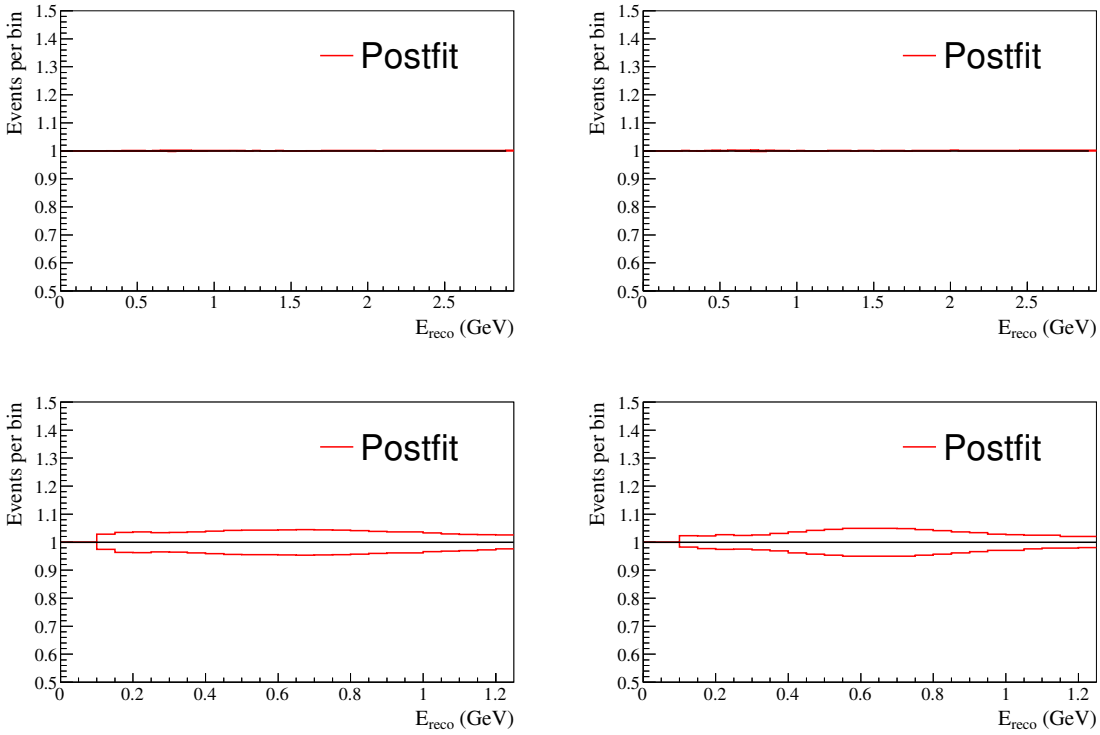


Figure 13: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the oscillation systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

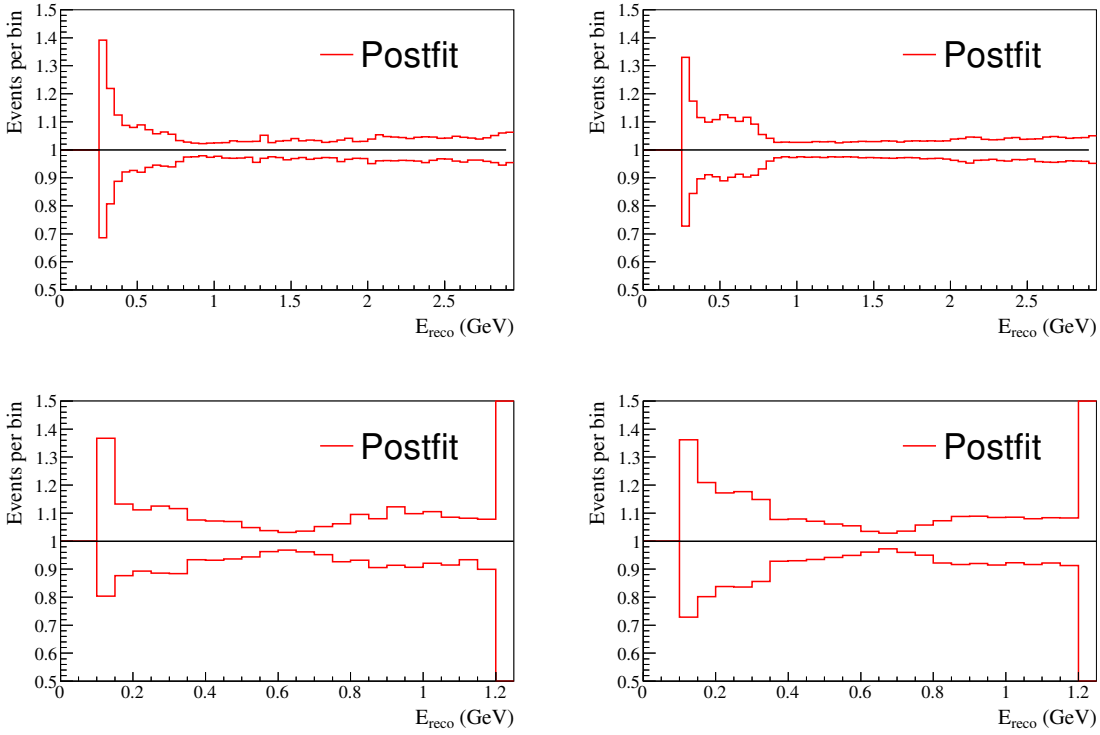


Figure 14: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of super-K + FSI + SI + PN systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

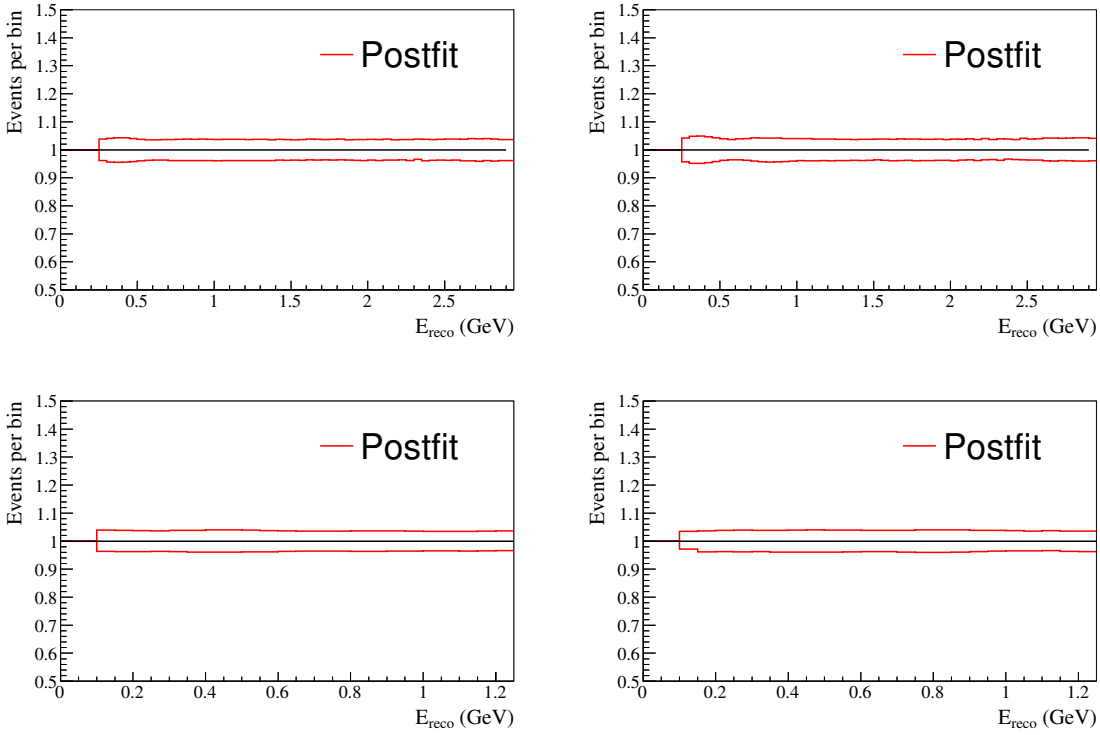


Figure 15: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

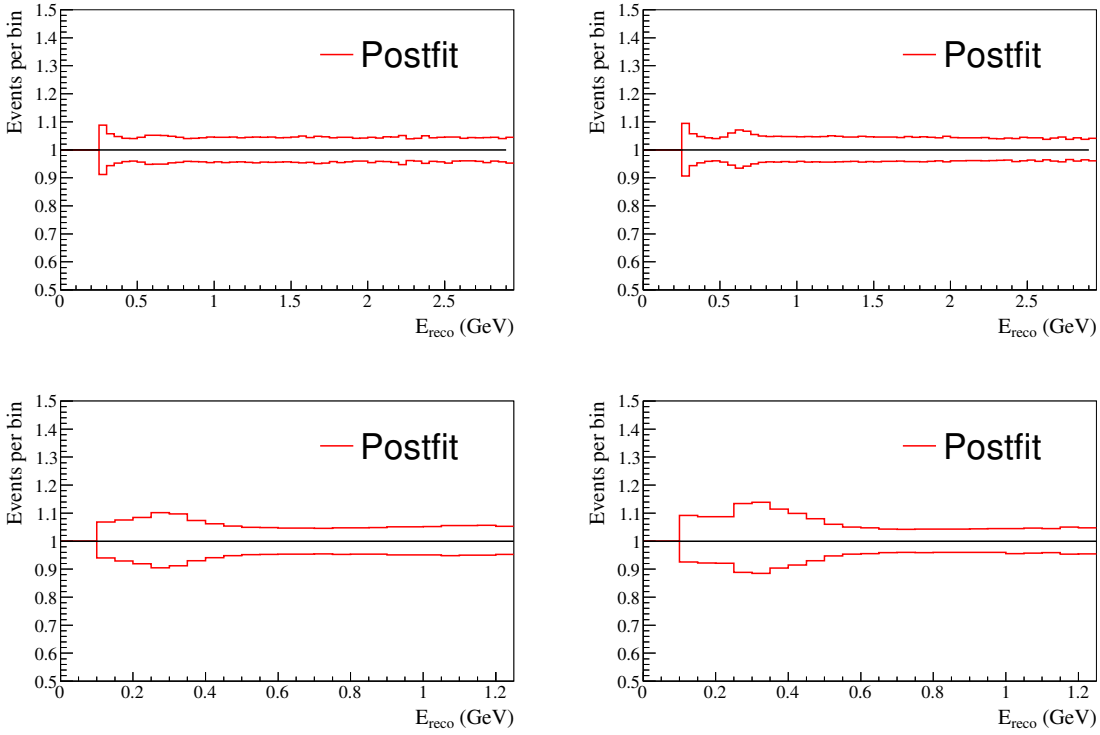


Figure 16: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the cross section systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

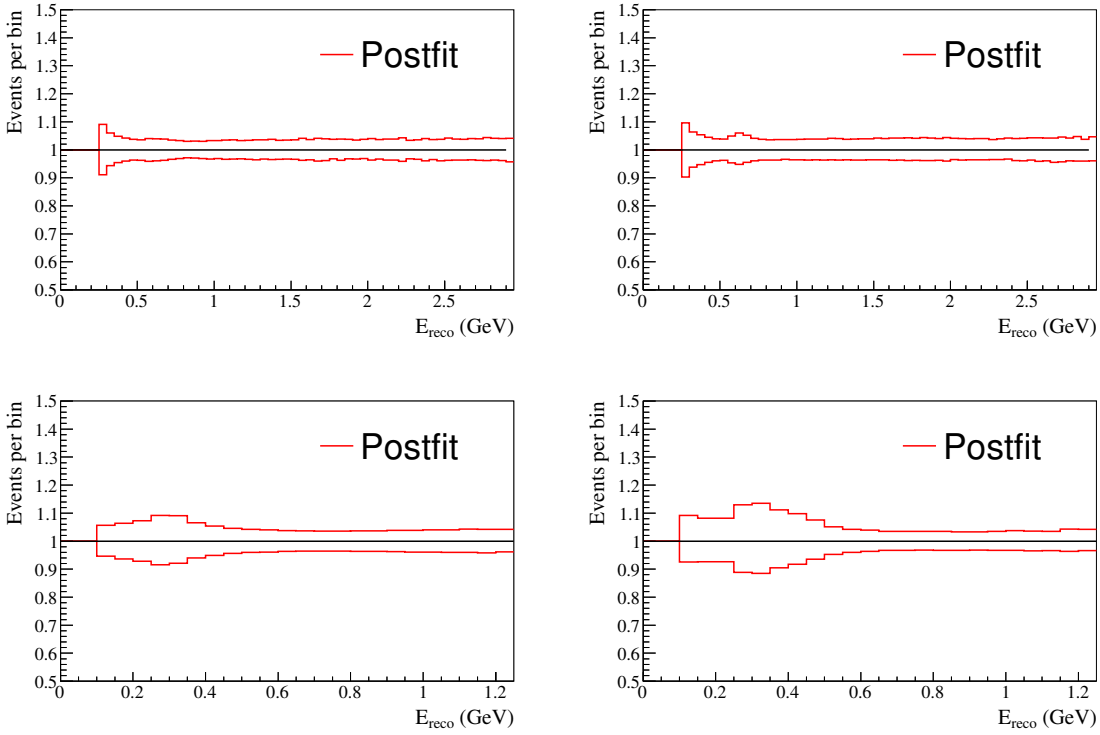


Figure 17: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of the flux + cross section systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set A shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.15%
Flux	3.59%
XSec Corr Tot	4.02%
MEC (corr)	3.54%
MECbar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.10%
Flux+XSec	2.92%
Flux+XSec+SKDet+FSI+SI	5.09%
Oscillations	0.03%
All	5.13%
All (pre-fit)	11.9%

Table 17: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.50%
Flux	3.67%
XSec Corr Tot	4.25%
MEC (corr)	3.85%
MECbar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.47%
XSec nue/numu	2.60%
XSec Tot	5.20%
Flux+XSec	4.13%
Flux+XSec+SKDet+FSI+SI	5.53%
Oscillations	4.16%
All	6.80%
All (pre-fit)	12.6%

Table 18: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.66%
XSec Corr Tot	4.08%
MEC (corr)	2.96%
MECbar (corr)	1.82%
NC other (uncorr)	0.75%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.14%
Flux+XSec	3.36%
Flux+XSec+SKDet+FSI+SI	5.18%
Oscillations	0.03%
All	5.12%
All (pre-fit)	12.7%

Table 19: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.97%
Flux	3.78%
XSec Corr Tot	4.30%
MEC (corr)	3.00%
MECbar (corr)	2.36%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.93%
XSec nue/numu	1.51%
XSec Tot	5.43%
Flux+XSec	4.61%
Flux+XSec+SKDet+FSI+SI	6.21%
Oscillations	4.00%
All	7.41%
All (pre-fit)	14.3%

Table 20: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	μ-like RMS/Mean	e-like RMS/Mean
SKDet	0.007	0.016
FSI+SI	0.026	0.035
SKDet+FSI+SI	0.026	0.038
Flux	0.017	0.017
XSec NC other (uncorr)	0.000	0.002
XSec NC 1γ (uncorr)	0.000	0.015
XSec ν_e / ν_μ (uncorr)	0.000	0.031
XSec Tot	0.017	0.042
Flux+XSec	0.018	0.041
Flux+XSec+SDet+FSI+SI	0.032	0.057
Oscillations	0.000	0.008
All	0.032	0.058

Table 21: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

4. Analysis procedure

4.1. Fit method

The goal of this analysis is to measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, $|\Delta m^2_{32}|$ and search for CP violation by setting limits on δ_{CP} for a given hypothesis and mass hierarchy, which is fixed to either normal or inverted. All the systematic as well as oscillation parameters, which are nuisance, are marginalised. The treatment of the oscillation parameters is shown in Tab. 12. The parameters of interest will be denoted with the vector $\vec{\theta}$, since more than 1 parameter at a time can be measured. $\vec{\theta}$ can include the combination of all the oscillation parameters listed above.

The *VALOR* analysis uses a binned likelihood-ratio method where the estimation of $\vec{\theta}$ is obtained by comparing the observed and predicted Super-K E_{reco} Vs θ spectra for 1-ring events.

Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the Super-K reconstructed energy spectrum prediction (e.g. parameters that control the beam, neutrino interaction simulations, detector response and oscillations). The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{3}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The total likelihood is given by the product of the likelihood terms of each selected sampled,

$$\lambda(\vec{\theta};\mathbf{a}) = \prod_{s=1}^{N_{samples}}\lambda_s(\vec{\theta};\mathbf{a}) \tag{4}$$

where $N_{samples}$ is the number of selected samples used in the analysis or in by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all the super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \tag{5}$$

The advantage of the likelihood ratio method over the extended maximum-likelihood method is that, in the large-sample limit, the quantity $-2\ln\lambda(\vec{\theta};\mathbf{a})$ in Eq. 3 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

4.2. Marginalization

In the previous analyses, except [10], a fully frequentist approach, in which all nuisance parameters were profiled, was used. Now a hybrid Bayesian-frequentist method is used. The nuisance parameters are marginalised over while the parameters of interest are minimized with respect to the likelihood distribution. The marginalization of the nuisance parameters give a so-called marginal likelihood that depends on only on a vector of θ dimensions where θ is the number of parameters of interest. $\vec{\theta}$, is written as:

$$\lambda_{marg}(\vec{\theta}) = \int_A \lambda(\vec{\theta};\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i) \tag{6}$$

where $\mathbf{a}$ is the set of nuisance parameter, $\pi(\mathbf{a})$ is the prior of the nuisance parameters. n is the number of toys MC performed for the marginalization and $\mathbf{a_i}$ is the single set of nuisance parameters extracted in the "i-th" toy MC. Many sets of $\mathbf{a}$ ($n = 10k$) should be extracted in order to effectively sample the space of the

nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement.

We derive the marginal log-likelihood, which we will denoted as χ^2 from now on:

$$\chi^2_{marg}(\vec{\theta}) = -2\ln\lambda_{marg}(\vec{\theta}) = -2\ln\left(\frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i)\right) \tag{7}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood. During the marginalization process, random values of the nuisance parameters are drawn according to the prior constraints which come from fits to near detector data and control samples at Super-K. Therefore, each throw is already weighted based on the prior information, and as a result the nuisance parameter values are not included in the calculation of the likelihood.

For this reason a modified version of Eq. 3, which doesn't include the penalty term, is used.

Sometimes the free parameters, which are not marginalised, are constrained with a penalty term. In this case a modified version of Eq. 7 is used:

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + (\vec{\theta}-\vec{\theta_0})^T V^{-1}(\vec{\theta}-\vec{\theta_0}) \tag{8}$$

where $\vec{\theta_0}$ is the nominal value of the parameters of interest, $\vec{\theta}$ is its fitted value and V is the covariance matrix which relates all the free parameters. If the constrained parameter θ_{constr} is not correlated with the other parameters of interest, V is reduced to the uncertainty $\Delta\theta^{constr}$, and Eq. 8 becomes

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + \frac{(\theta^{constr}-\theta_0^{constr})^2}{(\Delta\theta^{constr})^2} \tag{9}$$

This is the case of $\sin^2\theta_{13}$ when it is measured, i.e. it is parameter of interest, but at the same time it is constrained with the measurement of the reactor experiments. When a penalty term is applied to a parameter of interest it will be specified.

The toys MC used for the marginalization of the nuisance parameters are performed exactly in the same way as described in sec. 3.1, taking into account the correlations, but without statistical fluctuations. First n distributions of E_{reco} Vs θ for e-like and E_{reco} only for μ-like samples are produced using the extracted systematic parameters for 1×10^{20} POT so that they may be scaled appropriately when comparing to datasets of differing POT. Finally they are stored in output files which can be used for whatever needed fit. A fit on a single dataset is performed as following:

- all the n toy MC distributions are read from the files,
- each distribution is normalised to the number of POT of the data set,
- for each toy distribution Eq. 3 is calculated and $\lambda(\vec{\theta},\mathbf{a}) = e^{-(-2\ln\lambda(\vec{\theta},\mathbf{a}))/2}$ is obtained,
- the χ^2 value is finally obtained, using Eq. 7, for a given set of parameter values of interest, $\vec{\theta}$.

This procedure is followed for each point in the $\vec{\theta}$ grid, where the binning is so fine that it does not impact on the precision of the measurement. Finally the best fit is found by looking for the $\vec{\theta}$ grid point with the minimum χ^2_{bf} value. This point corresponds to the best-fit parameters $\vec{\theta}_{bf}$. The minimization can be also performed by using a discrete descent gradient algorithm implemented in the VALOR framework that allows the minimum χ^2 to be found without the need of calculating the full χ^2 distribution across all grid points. This method is employed when a large number of fits is required, such as the calculation of critical values for Feldman-Cousins contours, as it drastically reduces the amount of CPU time required. A validation of the minimization algorithm will be shown in sec. 5 for the analyses where it is used.

4.3. Best-fit spectra with marginalization

When profiling, the best fit spectra are defined by the MC prediction with the largest likelihood.

However, since the nuisance parameters are marginalised, there does not exist a single set which corresponds to the best fit. Therefore a new method has been developed to represent the best fit spectra which is consistent with oscillation results.

At the end of the fit the nuisance parameters have an effective value which corresponds to the weighted average over all the toys MC where the weights are the likelihood values of each single toy given by Eq. 3. This becomes clearer when considering Eq. 6.

The following procedure is used. From each toy MC i, used to marginalise the nuisance parameters, a different spectrum, with $N^j(\vec{\theta}_{bf};\mathbf{a})$ number of events in each bin j, is produced as well as a likelihood $\lambda(\vec{\theta}_{bf},\mathbf{a_i})$ value, calculated using Eq. 6, where $\vec{\theta}_{bf}$ is the best-fit value of the free parameters vector and $\mathbf{a_i}$ is the vector of nuisance parameters corresponding to the toy i. The best-fit number of events in the bin j of the spectrum is calculated as

$$N^j_{marg} = \frac{\sum_{i=1}^{n} \mathcal{L}(\vec{\theta}_{bf};\vec{f}_i)\cdot N^j(\vec{\theta}_{bf};\vec{f}_i)}{\sum_{i=1}^{n}\mathcal{L}(\vec{\theta}_{bf};\vec{f}_i)} \tag{10}$$

where n is the number of toys MC.

One can easily find a relation between the profiled and marginalised best-fit spectra. In the limit of infinite number of toys MC, i.e. the full nuisance parameters space is scanned, the likelihood can be written as

$$\lambda_{marg}(\vec{\theta}_{bf}) = \frac{1}{n}\sum_{i=1,\ i\neq bf}^{n}\Big[\lambda(\vec{\theta}_{bf};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\Big] \tag{11}$$

where $\lambda_{prof} = \lambda(\vec{\theta}_{bf};\mathbf{a_{bf}})$ is the profiled best-fit likelihood. If we denote as $N^j_{prof} = N^j_i(\vec{\theta}_{bf};\mathbf{a_{bf}})$ the number of events in the bin j obtained with profiling, Eq. 10 can be written as

$$N^j_{marg} = \frac{\sum_{i=1,\ i\neq bf}^{n}\mathcal{L}(\vec{\theta}_{bf};\mathbf{a_i})\cdot\mathbf{N^j}(\tilde{\theta}_{\mathbf{bf}};\mathbf{a_i}) + \lambda_{\mathbf{prof}}\cdot\mathbf{N^j_{prof}}}{\sum_{i=1}^{n}\lambda(\vec{\theta}_{bf};\mathbf{a_i})} \tag{12}$$

4.4. Confidence intervals

The full χ^2 distribution is then shifted with respect to $\chi^2(\vec{\theta}_{bf})$:

$$\Delta\chi^2(\vec{\theta}) = \chi^2(\vec{\theta}) - \chi^2(\vec{\theta}_{bf}) \tag{13}$$

Confidence intervals are set on $\Delta\chi^2$. If

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{14}$$

where $\Delta\chi^2_{crit}$ is the critical value for a certain confidence level X%(CL), that value of $\vec{\theta}$ is excluded at the X% CL. In the gaussian approximation constant values of $\Delta\chi^2_{crit}$ can be used. It will be called "constant $\Delta\chi^2$ method".

However in some cases the gaussian approximation is not reliable and the "Feldman-Cousins method" [52] must be used. For each point of the $\vec{\theta}$ grid a different critical value $\Delta\chi^2(\vec{\theta})_{crit}$ is calculated. The procedure consists of the following steps:

1. many toys MC (e.g. 10k) are performed assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties;
2. for each toy first $\chi^2(\vec{\theta}_{fix})$ is obtained by minimizing $\chi^2(\vec{\theta})$ where the oscillation parameters $\vec{\theta}$ are kept fixed to the values corresponding to the grid point and the nuisance parameters are marginalised. Then $\chi^2(\vec{\theta})$ is minimized, leaving the oscillation parameters $\vec{\theta}$ as free and $\chi^2(\vec{\theta}_{bf})$ is obtained.
3. $\Delta\chi^2(\vec{\theta})$ is calculated using Eq. 7 or 9.
4. The steps 2 and 3 are repeated for each toy MC and a distribution of $\Delta\chi^2(\vec{\theta})$ ($f(\Delta\chi^2)$) is obtained.

5. The critical value for $X\%$ CL, $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_{-\infty}^{\Delta\chi^2_{crit}} f(\Delta\chi^2) d(\Delta\chi^2) = X\% \tag{15}$$

6. This procedure is repeated for all the points of the $\vec{\theta}$ grid and several critical values are computed

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition of Eq. 14.

Several analyses are performed and confidence intervals are computed for:

- 1D δ_{CP} with reactor constraint: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D δ_{CP} T2K only: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the uniform prior, shown in Tab. 22, is used.
- 1D $\sin^2\theta_{13}$ T2K only: the parameter of interest is $\sin^2\theta_{13}$ and all the other oscillation parameters are marginalised;
- 1D $|\Delta m^2_{32}|$ with reactor constraint: the parameter of interest is $|\Delta m^2_{32}|$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 1D $\sin^2\theta_{23}$ with reactor constraint: the parameter of interest is $\sin^2\theta_{23}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in Tab. 22, is used.
- 2D $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is a free parameter, the gaussian prior of reactor measurements shown in Tab. 22, is used, through a penalty term as in Eq. 9.
- 2D $\sin^2\theta_{13}$ Vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ Vs $|\Delta m^2_{32}|$ with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ while all the other oscillation parameters are marginalised;

The 1D χ^2 confidence intervals with "constant $\Delta\chi^2$ method" are use to set the $\pm1\sigma$ ranges.

Each point of the $\vec{\theta}$ grid corresponds to the center of the histogram bins, i.e. the χ^2 values are calculated at each histogram bin center. The ranges and binning used for analyses listed above are shown in Tab. 22. All the 1-dimensional analyses, where the precision on the best-fit point is important and must be quoted, are performed with a fine binning. Since 2-dimensional analyses are performed only to set contours with "constant $\Delta\chi^2$ method" and a fit with a very high precision is not needed to find the best-fit point, courser binning are used. Furthermore a very fine 2-dimensional binning would require much more CPU time and storage space. In the future $\bar{\nu}/\nu$ joint analysis release when the "Feldman-Cousins method" will be needed also for 2-dimensional fits, a finer binning will be used.

4.5. Goodness of fit

The goodness-of-fit tests are done by minimising the expression in Eq. 7 for δ_{CP} as free parameter with reactor constraint. The choice of this binning followed a decision by the oscillation analysis group that p-values should be calculated using the rate and the spectra, and that a very coarse binning scheme should be used to ensure that there are sufficient numbers of events in each bin; this scheme has 5 bins from 0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV. 1000 toy experiments are generated for each mass hierarchy, and

Parameter	Min	Max	Step	Bins
	1-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/200$	201
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 200	201
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 200	201
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	2×10^{-3}	$10^{-3}/200$	201
	2-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/50$	51
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	81
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 81	81
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 81	81
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	2×10^{-3}	$10^{-3}/21$	21

Table 22: Grid limits and spacing used for all the 1-dimensional and 2-dimensional marginalization fits and confidence regions. The minimum and maximum values correspond to the center of the χ^2 histogram first and last bin.

both systematic variations and statistical fluctuations are applied to them. All 101 systematic parameters are included in the fits of those datasets.

TODO: DESCRIBE HOW THE P-VALUE IS CALCULATED ???

5. Validation of the $\bar{\nu}/\nu$ joint analysis

The validation of the fit is performed first by comparing the confidence intervals, obtained with Asimov data set corresponding to the oscillation parameters in tab. 12 as well as the Run 1-4 data, with the other group analyses. Then the coverage study is performed for 1-dimensional δ_{CP} with reactor constraint fits, in order to verify if the "constant $\Delta\chi^2$ method" is reliable or the "Feldman-Cousins method" is needed. Pull studies will be shown for 1-dimensional fits of the following oscillation parameters: δ_{CP}, $\sin^2\theta_{13}$, $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$, for which $\pm1\sigma$ ranges will be quoted by using the "constant $\Delta\chi^2$ method". Since the nuisance parameters are marginalised in the fit, pull studies for these parameters do not make sense and will not be shown.

5.1. Sensitivity studies

In Fig. 18 the sensitivity to δ_{CP} with reactor constraint and T2K only is shown for the Asimov data sets A and B.

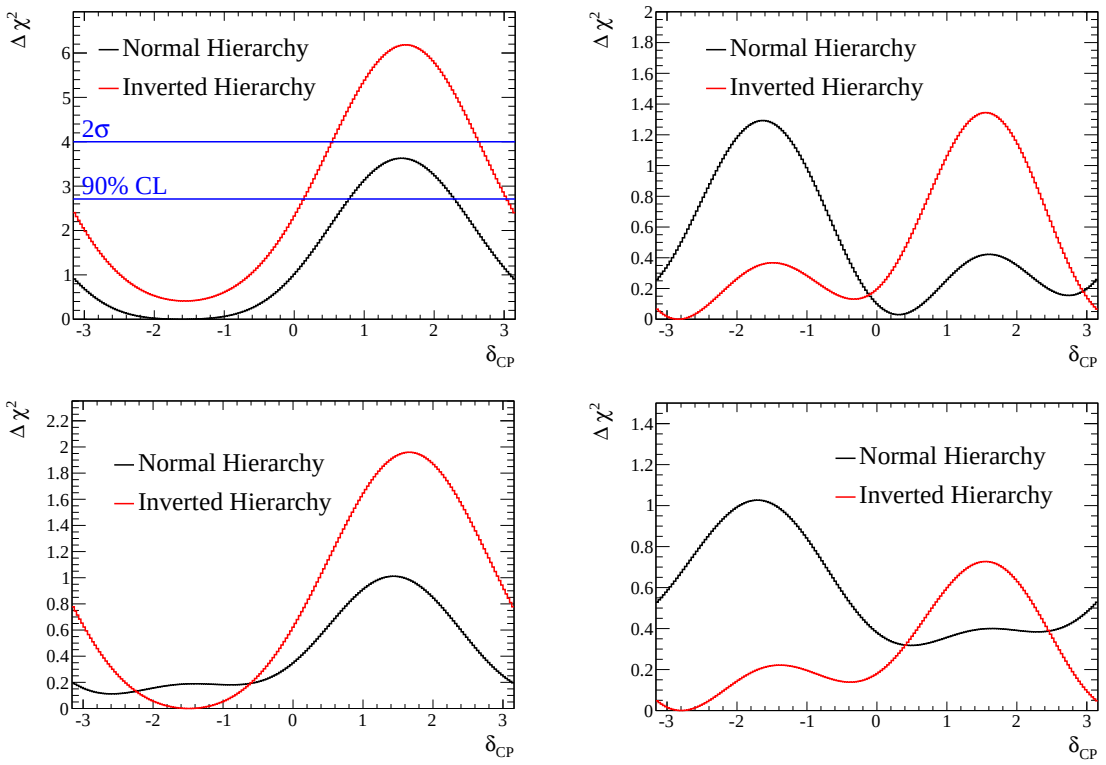


Figure 18: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} with reactor constraint (top) and T2K only data (bottom) for the Asimov data sets A (left) and B (right) of Tab. 12 corresponding to the Run 1-6 statistics is compared to the result obtained by the other analysis groups. The mass hierarchy is minimized: both the normal and inverted hierarchy χ^2 distributions are shifted to the same global best-fit χ^2 value, which is taken as the minimum between normal and inverted hierarchy.

Also the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint and T2K only data, and $\sin^2\theta_{23}$ Vs $|m^2_{32}|$ with reactor constraint are shown in Fig. 19 and 20.

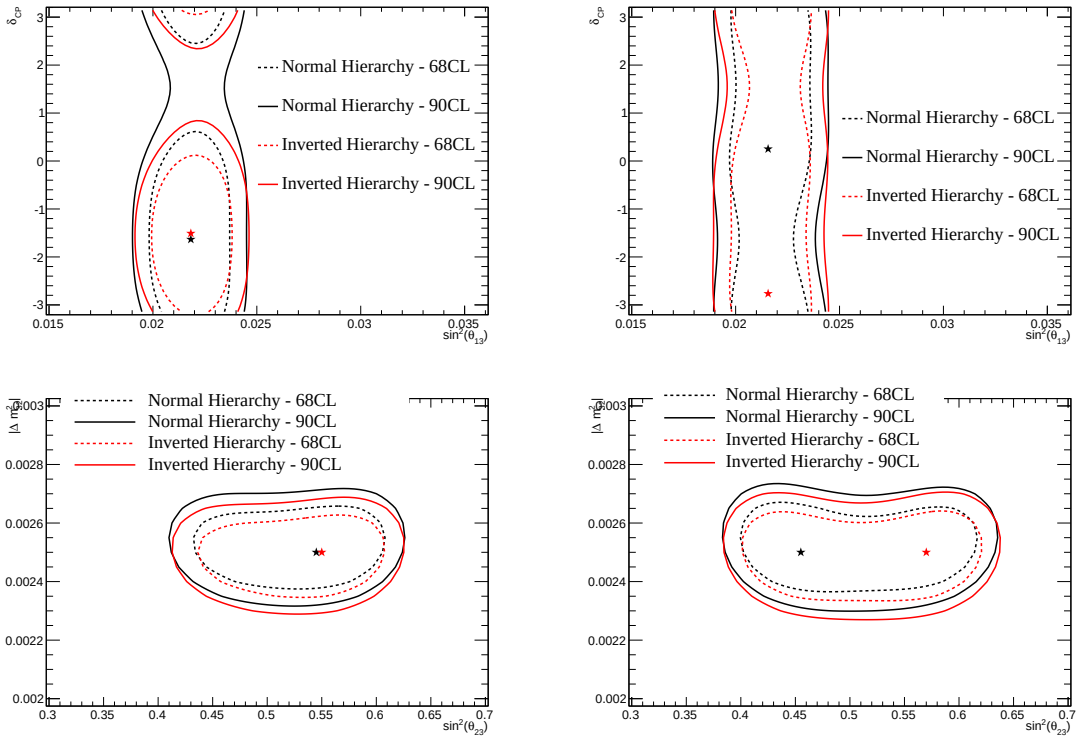


Figure 19: The contours at 68% and 90% CL of $\sin^2\theta_{13}$ Vs δ_{CP} (top) and $\sin^2\theta_{23}$ Vs $|\Delta m^2_{32}|$ (bottom) with reactor constraint for the Asimov data sets A (left) and B (right) of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.

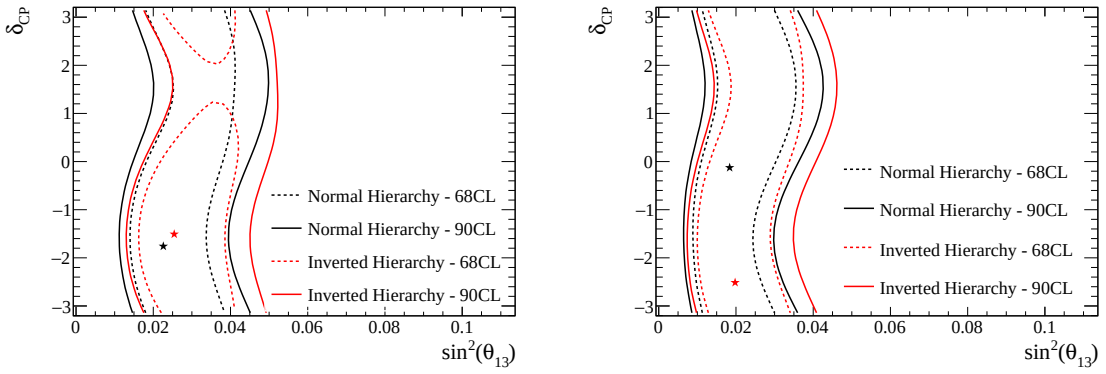


Figure 20: The contours at 68% and 90% CL of $\sin^2\theta_{13}$ Vs δ_{CP} (top) without reactor constraint for the Asimov data sets A (left) and B (right) of tab. 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.

5.2. Coverage study of δ_{CP}

The fit validation is performed also by making coverage studies for the fit of δ_{CP} with reactor constraint, which is the main result of this release of the $\bar{\nu}/\nu$ joint fit, in order to understand if the "Feldman-Cousins method" is needed. The coverage is defined as the number of times the true value δ_{CP}^{true}, used to generate the toys MC, is contained in the confidence interval defined by eq. 14:

$$\Delta\chi^2(\delta_{CP}) = \chi^2(\delta_{CP}^{true}) - \chi^2(\delta_{CP}^{bf}) < \Delta\chi^2_{crit} \tag{16}$$

where δ_{CP}^{bf} is the best-fit value. Adopting the "constant $\Delta\chi^2$ method", the critical value $\Delta\chi^2_{crit}$ at a certain $X\%$ confidence level is used. Many toys MC are performed (10k) and for each one eq. 5.2 is tested. The fit has a good coverage if the number of toys MC, for which eq. 5.2 is satisfied (N^X_{toys}), is equal or close to $X\%$ within the statistical uncertainty of the number of toys MC. If the $N^X_{toys} > X\%$, then the fit has over-coverage, that means the confidence intervals should be tighter than what obtained with the "constant $\Delta\chi^2$ method", while if $N^X_{toys} < X\%$, then the fit has under-coverage and excludes a wider range of values that the in the correct case. In Fig. 21 the coverage study of δ_{CP} 1-dimensional fit with reactor constraint is shown for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$.

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Figure 21: COVERAGE STUDY DCP W/ REACTORS Coverage study of δ_{CP} with reactor constraint for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$ assuming normal hierarchy. All the other oscillation parameters are marginalised and the mass hierarchy is kept fixed to normal in the fit. The statistics corresponds to the Run 1-6 data set.

TODO: ADD COMMENT ON COVERAGE STUDY

5.3. Pull study of oscillation parameters

Pull studies are performed as well for 1-dimensional fits of the oscillation parameters, since 1σ error ranges will be provided for each one independently. All the other oscillation parameters are marginalised. Since for the current low statistics combined with the non-gaussian behaviour they are not expected to be gaussian

with mean 0 and pull width 1, but to have large deviations, for most of the parameters. In particular this is true for δ_{CP}, which is cyclic and the values $\pm\pi/2$ behave like a boundary.

The performance of the fitter was evaluated using ensembles of 5k toy MC experiments, and these were generated with statistical fluctuations and variations of all 101 systematic parameters. Toy ensembles were generated for normal hierarchy, which was kept fixed in the fit, and were fitted using the implemented minimization algorithm as described in sec. 4.2.

Asymmetric errors were calculated by choosing the $\pm 1\sigma$ error as the range of free parameter values that satisfy eq. 5.2 with $\Delta\chi^2 = 1$, i.e. the critical value of the 1σ CL ("constant $\Delta\chi^2$ method").

For a free parameter θ the pull is defined as:

$$\begin{aligned} \text{if}(\theta_{bf} > \theta_{true})\ \text{pull} &= \frac{\theta_{bf} - \theta_{true}}{\sigma^{-}_{bf}} \\ \text{if}(\theta_{bf} < \theta_{true})\ \text{pull} &= \frac{\theta_{bf} - \theta_{true}}{\sigma^{+}_{bf}} \end{aligned} \tag{17}$$

where σ^{+}_{bf} is the positive best-fit error and σ^{-}_{bf} is the negative best-fit error of θ.

The pull study was performed for the oscillation parameters δ_{CP} with reactor constraint, $\sin^2\theta_{13}$ with T2K only data, $\sin^2\theta_{23}$ with reactor constraint and $|\Delta m^2_{32}|$ with reactor constraint for the Asimov data set of Tab. 12 and is shown in Fig. 22.

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Figure 22: PULLS FOR EACH OSCILLATION PARAMETER ASIMOV DATA SET Pull distributions of δ_{CP} with reactor constraint, $\sin^2\theta_{13}$ with T2K only data, $\sin^2\theta_{23}$ with reactor constraint and $|\Delta m^2_{32}|$ with reactor constraint assuming normal hierarchy for the Asimov data set of Tab. 12. In each fit all the other oscillation parameters are marginalised and the mass hierarchy is kept fixed to normal in the fit. The statistics corresponds to the Run 1-6 data set. The pulls are defined as in Eq. 5.3.

TODO: ADD COMMENT ON PULL STUDY

6. Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-6 dataset

The results of the Run 1-6 data set fit for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode is shown in this chapter. In Fig. 23 the predicted spectra of μ-like and e-like samples are compared to the observed data. In Tab. 23 the expected and observed number of events in Run 1-6 data set are shown. The predicted number of events are shown for the oscillation parameters in tab. 12 of Asimov data sets A and B. All the predicted and observed events are in good agreement. A small excess of events with respect to the expectation is observed in the neutrino mode e-like samples, though it is still within 1σ of the statistical uncertainty. This trend is mostly inherited from the Run 1-4 data set, since only 2 neutrino mode e-like events are observed in Run 5+6. Furthermore the latest global fit of the reactor experiments gives a lower value of $\sin^2\theta_{13}$ with a much smaller uncertainty (see Sec. 3.1 and Tab. 12). It is interesting to see that the number of events of antineutrino mode e-like events is in perfect agreement with the assumption of $\delta_{CP} = -1.601$ the best-fit value of the Run 1-4 data set. This means that, for what concerns the event rate, we expect to see agreement between neutrino and antineutrino mode data in the fit of δ_{CP}.

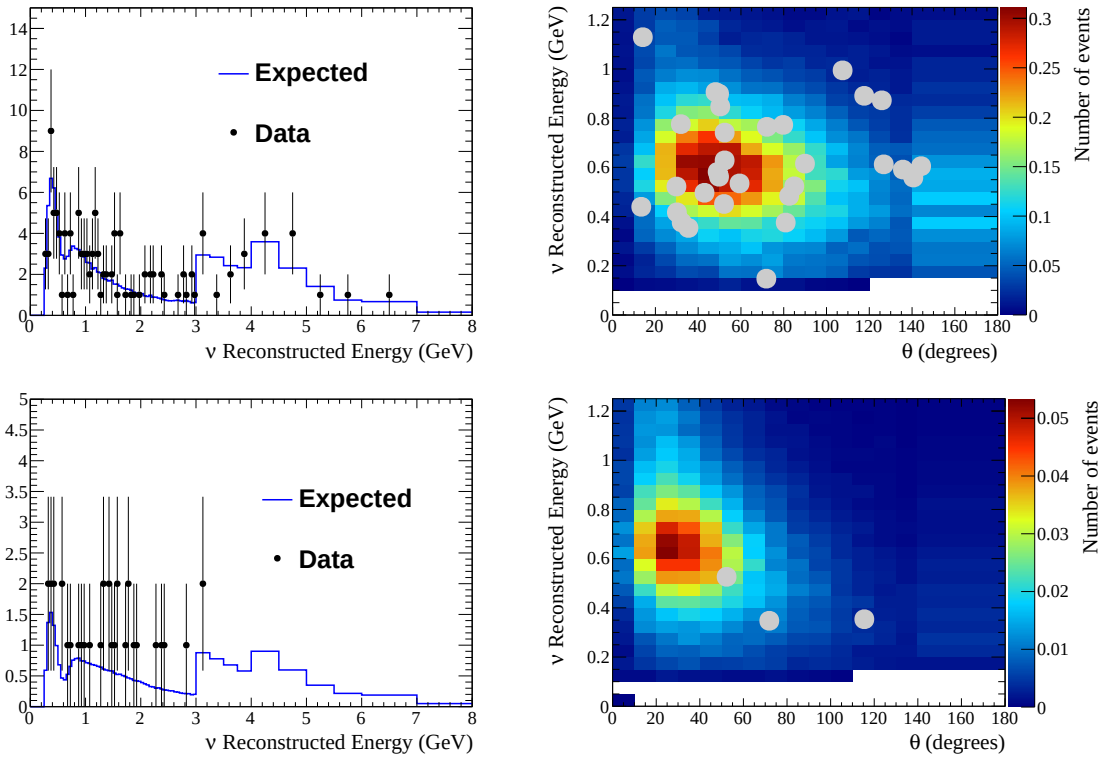


Figure 23: The predicted spectra of neutrino mode muon-like (top left) and electron-like (top right), antineutrino mode muon-like (bottom left) and electron-like (bottom right) events are compared to the Run 1-6 dataset (dots), which corresponds to an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode. The expected distribution corresponds to the oscillation parameters of the Asimov data set A shown in table 12. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9.

Beam mode	Sample	Exp. Asimov A	Exp. Asimov B	Exp. Not Osc	Observed
neutrino	μ-like	125.914	130.464	484.737	124
neutrino	e-like	26.594	19.954	5.698	31
antineutrino	μ-like	34.284	35.032	99.02	34
antineutrino	e-like	3.218	3.355	1.253	3

Table 23: The observed and expected number of events in Run 1-6 data set for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in Tab. 12.

6.1. 1-dimensional χ^2 distributions and best-fit

The same 1-dimensional χ^2 distributions as in sec. 5.1 are produced and shown in Fig. 24. Looking at the χ^2 distribution of $\sin^2\theta_{13}$ with T2K only data we can see that the reactor measurements are in agreement with the T2K data at about 1σ level if $\delta_{CP} \sim -\pi/2$.

In Fig. 25 the χ^2 distributions as a function of δ_{CP} with T2K only data is obtained by running the analysis using either only neutrino mode or antineutrino mode samples for the Run 1-6 data set, in order to check the compatibility of neutrino and antineutrino mode data. The result is mainly driven by the excess of the observed number of events in e-like neutrino mode sample compared to what is expected. Though the significance given by the antineutrino mode samples is still low, it is interesting to note that also the antineutrino data prefer a value of δ_{CP} close to $-\pi/2$.

In Fig. 26 the confidence intervals obtained with the "Feldman-Cousins method" are shown for δ_{CP} with reactor constraint. Different critical values are calculated independently for normal and inverted hierarchy.

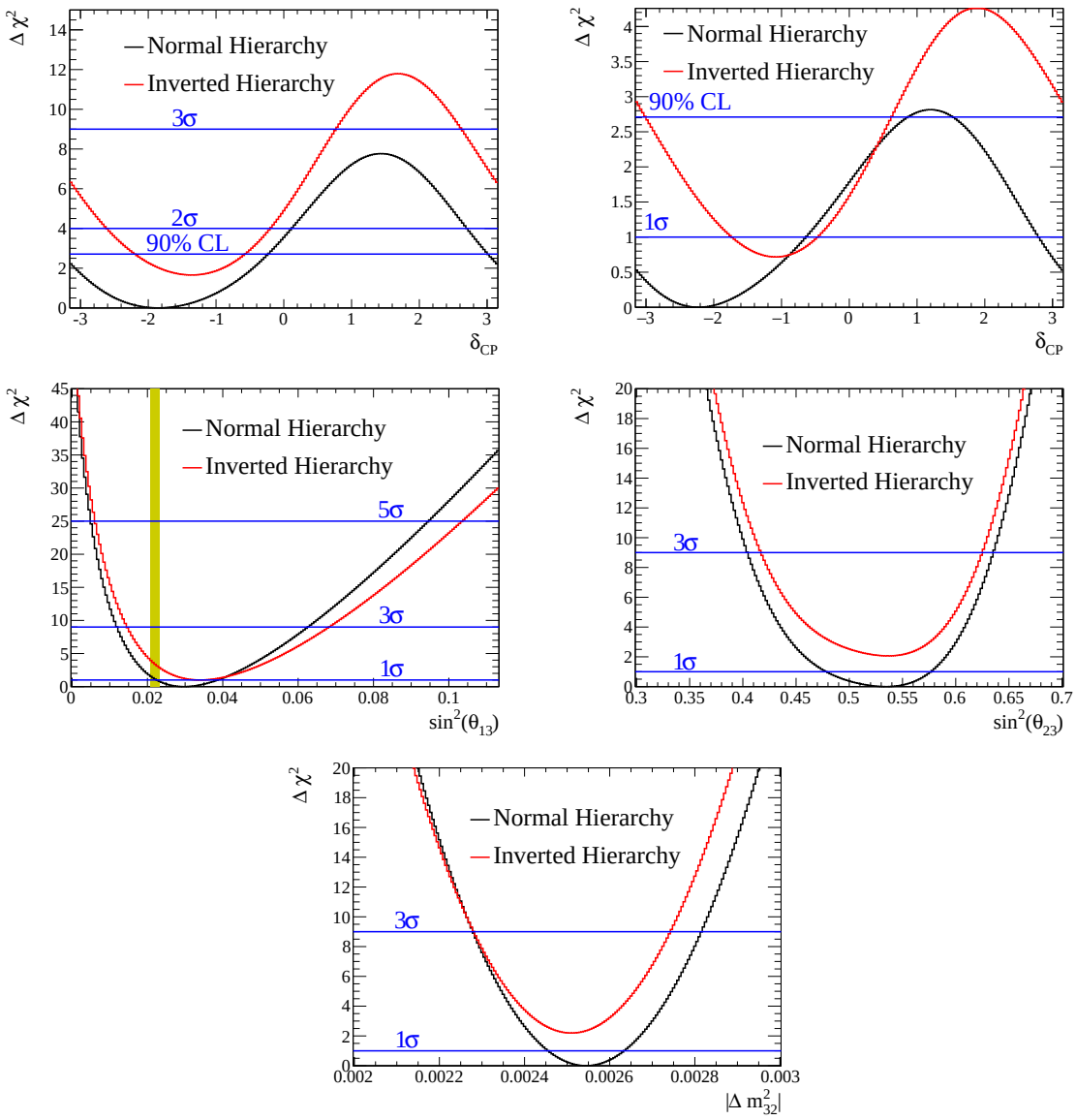


Figure 24: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of δ_{CP} with reactor constraint, δ_{CP} with T2K data only, $\sin^2\theta_{13}$ with T2K data only, $\sin^2\theta_{23}$ with reactor constraint and Δm^2_{32} (Δm^2_{13}) with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [64], shown by the yellow bar.

In Tab. 24 the best-fit values as well as the 1σ ranges for each oscillation parameter are shown, for both normal and inverted hierarchy. The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional χ^2 distributions shown in Fig. 24, where one oscillation parameter is free and all the other oscillation as well as the systematic parameters are marginalised. The mass hierarchy is minimized as well: for both normal and inverted hierarchy the 1σ ranges are calculated with respect to the same global minimum.

The best-fit spectra are computed as described in sec. 4.3 and are compared to the unoscillated nominal spectra. For e-like samples, where 2-dimensional templates are used, the projection onto the reconstructed neutrino energy axis is shown.

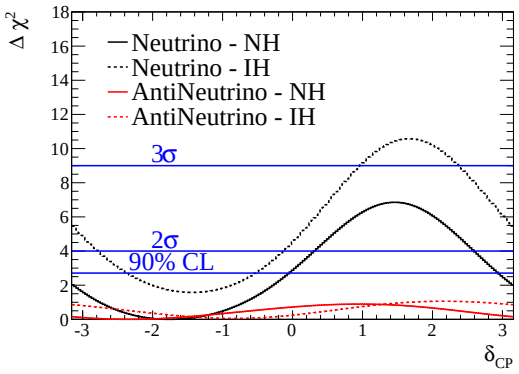


Figure 25: Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of δ_{CP} with reactor constraint, δ_{CP} with T2K data only, $\sin^2\theta_{13}$ with T2K data only, $\sin^2\theta_{23}$ with reactor constraint and Δm^2_{32} (Δm^2_{13}) with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 22. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [64], shown by the yellow bar.

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Figure 26: MEASURED FELDMAN-COUSINS DELTACP W/ REACTORS Measured $\Delta\chi^2$ distributions with the full run 1-6 data set as a function of δ_{CP} with reactor constraint, δ_{CP} with T2K data only, $\sin^2\theta_{13}$ with T2K data only, $\sin^2\theta_{23}$ with reactor constraint and Δm^2_{32} (Δm^2_{13}) with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [64], shown by the yellow bar.

6.2. 2-dimensional confidence intervals

The same 2-dimensional confidence intervals as in sec. 5.1 are produced. In Fig. 28 the corresponding 2-dimensional contours produced with the Run 1-6 data set are shown. As in 1-dimensional analysis results, looking at contours of $\sin^2\theta_{13}$ Vs δ_{CP} with T2K only data it is clear that the reactor measurements are in agreement with the T2K data at 1σ level if $\delta_{CP} \sim -\pi/2$.

6.3. Goodness of fit

The goodness of fit study is performed as described in sec. 4.5. All the oscillation as well as the systematic parameters are marginalised except δ_{CP}, which is fixed to its best-fit value, since in the analysis it is minimized. The result corresponds to T2K only data.

TODO: ADD COMMENTS ABOUT GOF AND P-VALUE RESULT

	Normal Hierarchy			Inverted Hierarchy		
Parameter	Best-fit	χ^2_{min}	$\pm1\sigma$ interval	Best-fit	χ^2_{min}	$\pm1\sigma$ interval
δ_{CP} (T2K only)	XXX	XXX	[XXX; XXX] & [XXX; XXX]	XXX	XXX	[XXX; XXX]
δ_{CP} (reactors)	XXX	XXX	[XXX; XXX]	XXX	XXX	-
$\sin^2\theta_{13}$	XXX	XXX	[XXX; XXX]	XXX	XXX	-
$\sin^2\theta_{23}$	XXX	XXX	[XXX; XXX]	XXX	XXX	-
Δm^2_{32} or Δm^2_{13} (10^{-3} eV$^2/c^4$)	XX	XXX	[XXX; XXX]	XXX	XXX	-

Table 24: TO UPDATE

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Figure 27: TO UPDATE MEASURED BEST-FIT ERECO SPECTRA COMPARED TO DATA AND PREDICTED UNOSCILLATED SPECTRA

6.4. Discussion of the results

TODO: ADD COMMENTS ABOUT RESULTS

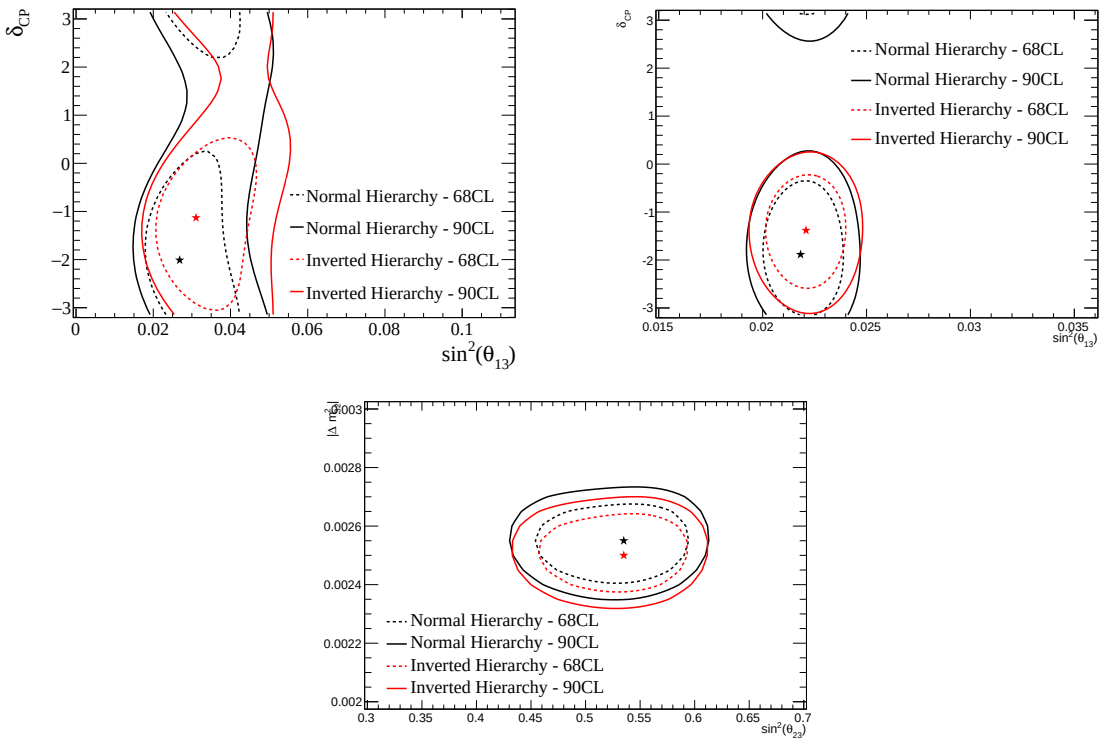


Figure 28: Measured confidence intervals with the full run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. From top left to bottom right the following combinations of oscillation parameters are shown: $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only, $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint and $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with reactor constraint. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently with respect to the global minimum of each hierarchy (star). All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in tab. 11. The measured contour of $\sin^2\theta_{13}$ Vs δ_{CP} without reactor constraint is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [64], shown by the yellow bar.

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Figure 29: TO UPDATE MEASURED GOODNESS OF FIT

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Figure 30: COMPARISON OF OUR WITH OTHER ANALYSIS MEASURED CONTOURS

771 TODO: ADD COMMENTS ABOUT COMPARISONS

772

7. $\bar{\nu}_e$ appearance analysis

The significance to $\bar{\nu}_e$ appearance was given as a p-value. The marginalisation method is used and all the Super-K samples are analyzed simultaneously. The oscillation parameters priors used for the marginalisation are shown in table 11. Two different test statistics were used: rate-only and rate+shape

7.1. Rate-only analysis method

The total number of observed events (N_{obs}), without using any information from the reconstructed energy shape, was used to quote the rate-only p-value. In order to simultaneously analyze all the Super-K samples a different procedure with respect to the one used in 2015 analysis [10], where toys were performed by using the posterior distributions of the Run 1-4 Bayesian analysis, has been adopted. For a single fit the following procedure was used:

- N_{marg} toy experiments were performed taking into account only systematic uncertainties, but not the statistical fluctuations
- For each toy the likelihood λ_{marg} was calculated using eq. 6 and including only the samples which are not affected by $\bar{\nu}_e$ appearance: neutrino mode muon-like, neutrino mode electron-like and antineutrino mode muon-like (called "other samples")
- Using the likelihood as a probability distribution function, λ_{marg} was then applied as a weight to the total number of observed events (of the toy) in the antineutrino mode electron-like sample
- The procedure is repeated for all the N_{marg} toy experiments.

N_{marg} is the number of marginalization toy experiments, which are needed to marginalise the systematic parameters and must be distinguished from the toy experiments that are used as fake-data sets in order to perform the p-value calculation. Indeed the latter toys are performed by applying both systematic and statistical uncertainties.

It is worth to note that reweighting the number of events with the likelihood computed with the "other samples", is mathematically the same as performing a likelihood ratio fit of the "other samples", taking the posterior distribution of the systematic parameters and using it to extract the systematic parameters in a toy experiment.

The advantage of this method is that correlations between all the samples are taken correctly into account. In the 2015 analysis the oscillation parameters were extracted from the Run 1-4 posterior distributions of the Bayesian analysis, but the flux, cross section and detector systematic parameters priors were unaffected by the Run 1-4 data fit. The difference between the two methods is negligible since the antineutrino mode e-like sample has a much lower statistic compared to the other samples, however this new method allows to take all the correlations into account correctly and to perform the analysis in one single step without making two different fits, one to obtain the oscillation parameters priors and the other to fit the antineutrino mode e-like data.

7.2. Rate+shape analysis method

The rate+shape analysis method is similar to what is described in 4.4, with the difference that the test statistic is defined as:

$$\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1) \tag{18}$$

where β is the discrete parameter of interest that can take only 0 or 1 values. This test statistic simply compare the oscillation with the non-oscillation hypothesis but does not use the best-fit χ^2 value. More details are given in section 7.3. The method is analogue to what was used in [10], with the only difference that now all the Super-K samples are fitted simultaneously and not only the antineutrino mode e-like one.

7.3. P-value

There are different test statistics which can be used to quote a p-value, as described in [57] . As proposed by the OA group, for this $\bar{\nu}_e$ appearance search two different p-values are quoted. In order to quote the significance given null $\bar{\nu}_e$ appearance, the parameter of interest is set to $\beta = 0$ and $n = 10k$ toys are performed. The systematic parameters are randomized taking into account the correlations as described in section 3.1. Then the E_{reco} Vs θ distribution is fluctuated with Poisson. Then the following p-values are calculated:

- p-value with rate-only analysis: the number of events is used as test statistics, without using the shape information of the reconstructed energy distribution. The distribution $f(N_{SK})$ of the total number of Super-K events in a toy (N_{SK}) is obtained from the performed toys. Then the number of events is measured from the data set (N_{SK}^{obs}) and the 1-sided (right tail) p-value is computed in the following way:

$$\text{p-value (right tail)} = \int_{N_{SK}^{obs}}^{+\infty} f(N_{SK}) dN_{SK} = \frac{\sum_{itoy=1}^{n} w(N_{SK} \geq N_{SK}^{obs})^{itoy}}{n} \tag{19}$$

  where $w(N_{SK} \geq N_{SK}^{obs})$ is 1 if $N_{SK} \geq N_{SK}^{obs}$ otherwise 0. The distribution $f(N_{tot}^{toy})$ is expected to be discrete since it follows the Poisson distribution.

  In an analogous way the left tail 1-sided p-value is computed as following:

$$\text{p-value (left tail)} = \int_{-\infty}^{N_{SK}^{obs}} f(N_{SK}) dN_{SK} = \frac{\sum_{itoy=1}^{n} w(N_{SK} \leq N_{SK}^{obs})^{itoy}}{n} \tag{20}$$

- p-value with rate+shape analysis: the shape information of the reconstructed energy distribution is used. In this case the test statistic is used to compare the case of $\beta = 0$ with $\beta = 1$:

$$\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1) \tag{21}$$

  where $\chi^2(\beta = 0)$ and $\chi^2(\beta = 1)$ are computed using eq. (7). It is worth to note that the parameter β is not fitted but fixed either to 0 or 1. The 1-sided p-value (right tail) is calculated as:

$$\text{p-value (right tail)} = \int_{\Delta\chi^2_{obs}}^{+\infty} f(\Delta\chi^2) d\Delta\chi^2 = \frac{\sum_{itoy=1}^{n} w(\Delta\chi^2 \geq \Delta\chi^2_{obs})^{itoy}}{n} \tag{22}$$

  where $w(\Delta\chi^2 \geq \Delta\chi^2_{obs})$ is 1 if $\Delta\chi^2 \geq \Delta\chi^2_{obs}$ otherwise 0.

  The 1-sided p-value (left tail) is calculated as:

$$\text{p-value (left tail)} = \int_{-\infty}^{\Delta\chi^2_{obs}} f(\Delta\chi^2) d\Delta\chi^2 = \frac{\sum_{itoy=1}^{n} w(\Delta\chi^2 \leq \Delta\chi^2_{obs})^{itoy}}{n} \tag{23}$$

The p-value will be calculated on the left or right tail depending on the position of the observed test statistic in the expected distribution. For example, if the observed test statistic is on the left tail of the expected distribution of the test statistic, a 1-sided p-value is computed on the left tail. Otherwise a 1-sided p-value on the right tail is computed.

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Figure 31: The distribution of the total number of events from each of the 10k toy experiments. The red line corresponds to the total number of events given by the Asimov dataset.

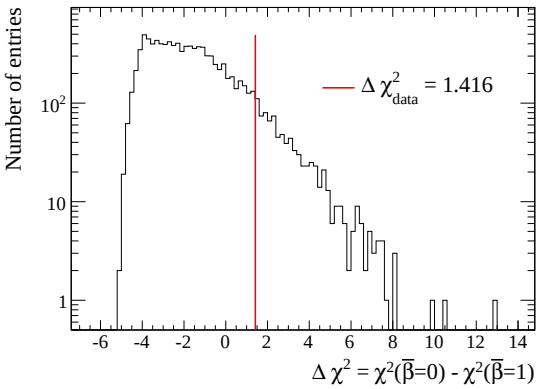


Figure 32: Distribution of the $\Delta\chi^2$ for each of the 10k toy experiments. The red line corresponds to $\Delta\chi^2$ obtained from the fit of the Asimov dataset A. The study is done with Run 1-6 statistic.

8. Sensitivity to $\bar{\nu}_e$ appearance

As discussed within the OA group, since there are not fitted parameters in the analysis, the validation can be done by comparing the rate-only with rate+shape p-values and checking if the corresponding distributions are reasonable. Of course the final validation is given if all the oscillation analyses agree. This is done by performing a sensitivity study to $\bar{\nu}_e$ appearance both with the Asimov dataset, which consists of the most probable dataset that is the nominal expected E_{reco} Vs θ distribution [49], as well as for toy distributions, given a value of β.

The significance to $\bar{\nu}_e$ appearance is quoted using the procedure described in section 7.3. In order to quote the significance to $\bar{\nu}_e$ appearance a 1-sided p-value (right tail) is computed. In addition an mean p-value is also computed by performing 10k toys.

8.1. Asimov rate-only p-value

The expected Asimov rate-only p-value is given by the Asimov dataset A for $\beta = 1$. The corresponding oscillation parameters are shown in table 12, The distribution of the number of superK events used as test statistic is shown in fig. 31. The expected number of events is 3.218 and the quoted p-value is XXX. In table 25 the rate-only p-value is shown for different numbers of observed events.

8.2. Asimov rate+shape p-value

The expected distribution of $\Delta\chi^2$ for all the 10k toy experiments is shown in fig. 32 for Run 1-6 statistic. The $\Delta\chi^2$ from the Asimov dataset is 1.416. The Asimov expected p-value is 0.0858 for the Run 1-6 statistic.



Figure 33: Rate-only (violet) and rate+shape (blue) p-value given by Asimov datasets for several values of β in normal (left) and logarithmic (right) scale. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done simulating the Run 1-6 statistic. 10k toys are used.



Figure 34: Rate-only (violet) and rate+shape (blue) p-value for different number of observed events given by Asimov datasets corresponding to the several values of β shown in fig. 33. Both normal (left) and logarithmic scales (right) are shown. The red line corresponds to the PMNS expectation ($\beta = 1$). The study is done with Run 1-6 statistic. 10k toys are used.

The p-value for the Asimov dataset as a function of β or the total number of events is shown for Run 1-6 statistic in fig. 33 and fig. 34.

8.3. Mean p-value

The mean p-value is computed without using the Asimov dataset, but with toy experiments:

- the distributions shown in fig. 31 and fig. 32 are used respectively for the rate-only and rate+shape p-value computations,
- 10k toys are produced with $\beta = 1$,
- for each toy with $\beta = 1$ the test statistic (i.e. the number of events and $\Delta\chi^2$ given in section 7.3) is computed and a 1-sided p-value (right tail) is quoted using the distributions fig. 31 and fig. 32,
- a distribution of rate-only and rate+shape p-values is obtained and the mean p-value is calculated.

The mean rate-only p-value, given by the mean of the p-value distribution shown in fig. 35, is XXX while the mean rate+shape p-value is 0.170 for Run 1-6 statistic. The mean p-value is larger than the Asimov one, which corresponds to the most probable p-value. It is worth to point out that sometimes it can happen that the rate-only p-value is smaller than the rate+shape one. This can be due to both the discreteness of the rate-only p-value or to the shape information wich can discard the $\bar{\nu}_e$ appearance hypothesis. On average

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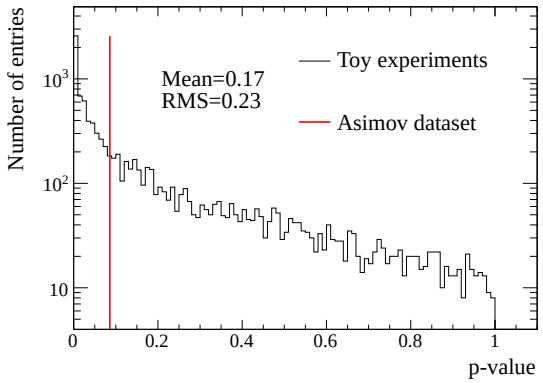


Figure 35: Distribution of rate-only (left) rate+shape (right) p-values obtained by 10k toys with $\beta = 1$. The rate-only p-value is discrete and each filled bin correspond to toys with different number of events. In both distributions the spike at p-value=1 corresponds to toys with no events. The study is performed with Run 1-6 statistic. 10k toys are used.

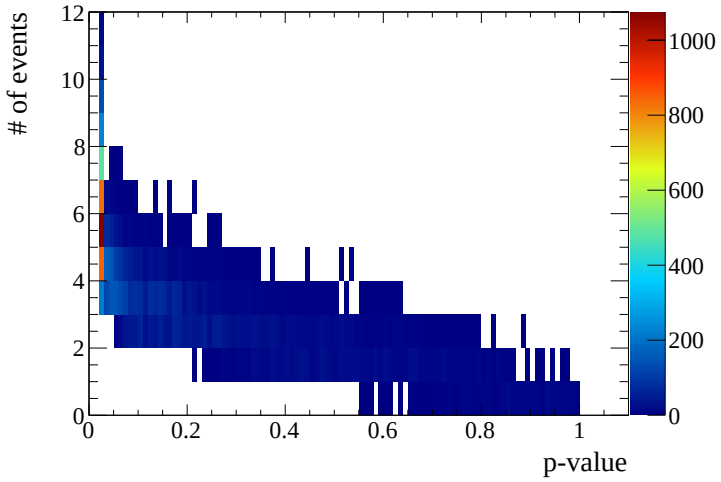


Figure 36: The rate+shape p-value as a function of the number of events for a single toy produced with $\beta = 1$ is shown for the Run 1-6 statistic. 10k toys are produced as described in section 8.3. Each entry corresponds to a different toy.

the rate+shape p-value is smaller than the rate-only one, since it exploits also the shape information, but if a single dataset is used it can happen that, due to the discreteness of test statistic (number of events), the rate-only p-value is smaller than the rate+shape p-value. Indeed a change of the number of events produce a not small jump in the p-value, as shown in the left plot of fig. 35, while the distribution of the rate+shape p-value is smooth and continuos. For this reason it is important to quote both the p-values.

In fig. 36 the rate+shape p-value as a function of the number of events for single toy produced with $\beta = 1$ is shown for the Run 1-6 statistic.

8.4. Summary of expected significance to $\bar{\nu}_e$ appearance

In table 26 the expected rate-only and rate+shape p-values corresponding to Run 1-6 statistic. In table 25 the expected rate-only p-value is shown for different number of observed events for Run 1-6 statistic.

The mean rate-only and rate+shape p-values, obtained considering only p-values smaller than 0.05 (0.10), are shown table 27.

In table 28 mean p-values corresponding to Run 1-6 statistic for those fake datasets with the number of observed events higher than the expected background are shown.

In table 29 the probability to have the rate-only p-value smaller than the rate+shape p-value is shown for Run 1-6 statistic

# of observed events	P-value
$N_{obs} = 2$	XXX
$N_{obs} = 3$	XXX
$\mathbf{N_{obs} = 4}$	**XXX**
$N_{obs} = 5$	XXX
$N_{obs} = 6$	XXX

Table 25: Summary of the expected rate-only p-values for different number of observed events. The simulated number of POT corresponds to Run 1-6. 10k toys are used to get the distribution of the test statistic and the Asimov data is used to compute the p-value. The Asimov dataset for $\beta = 1$ is in bold.

P-value	Asimov	Mean
Rate-only	XXX	XXX
Rate+shape	0.0858	0.170

Table 26: Summary of the expected rate-only and rate+shape p-values for the Run 1-6 statistic. The Asimov p-value is calculated using the Asimov dataset while the mean p-value is calculated making 10k toys and taking the mean of the p-value distribution.

	p-value ≤ 0.10	p-value ≤ 0.05
Fraction of toys (rate-only)	XXX%	XXX%
Fraction of toys (rate+shape)	58.0%	46.5%

Table 27: The probability to have a p-value smaller than 0.05 and 0.10 is shown. 10k toys for the Run 1-6 statistic are produced. Then for each toy the p-value is computed as described in section 8.3.

# obs events > # exp bkg	p-value	Fraction of toys
Rate-only	XXX	XXX%
Rate+shape	0.102	86.9%

Table 28: Summary of the expected rate-only and rate+shape mean p-values for the Run 1-6 statistic. The Asimov p-value is calculated using the Asimov dataset while the mean p-value is calculated making 10k toys and taking the mean of the p-value distribution. Only fake datasets with number of observed events higher than the expected background are considered. The fraction of selected toys is shown as well.

p-value (rate-only) > p-value (rate+shape)	Probability
All toys	85.0%
Toys w/ # obs. events > # exp. bkg	77.4%

Table 29: The probability to have a rate-only p-value larger than the rate+shape p-value is shown. 10k toys for the Run 1-6 statistic. Then for each toy the p-value is computed as described in section 8.3. Both all toys are considered as well as the toys with the number of observed events larger than the number of expected background.

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Figure 37: The distribution of the total number of events from each of the 10k toy experiments. The red line corresponds to the total number of events (left) and $\Delta\chi^2$ (right) given by the Run 5+6 dataset. Data correspond to Run 1-6 statistic. The distribution of the test statistics are obtained by performing 50k toys.

9. Results of the $\bar{\nu}_e$-appearance analysis with the Run 5+6 dataset

9.1. Significance to $\bar{\nu}_e$ appearance

The rate-only and rate+shape p-values for $\bar{\nu}_e$-appearance are performed on the combined Run 1-6 statistic following the same procedure as in section 8. 1-sided p-values are computed on the right tail of the test statistic expected distribution. All the 101 systematic parameters are marginalised. A summary of the observed number of events in Super-K, $\Delta\chi^2$ and p-values is given in table 30.

Observed num. of events	p-value (rate-only)	$\chi^2(\beta = 0)$	$\chi^2(\beta = 1)$	$\Delta\chi^2$	p-value (rate+shape)
X	XXX	XXX	XXX	XXX	XXX

Table 30: Summary of observed number of events, χ^2 for $\beta = 0$ and $\beta = 1$, $\Delta\chi^2$, rate-only and rate+shape p-values from the combined Run 5+6 dataset. The quoted χ^2 values were computed from eq. (7), while $\Delta\chi^2$ from section 7.3 in section 4.1. using the reconstructed energy binning scheme given in section 2.2.6. The p-values are obtained performing 10k toys with $\beta = 0$ in order to get the distribution of the test statistic.

The distribution of the expected test statistics, compared with the one obtained from the Run5-6 dataset, is shown in fig. 37.

In fig. 38 and fig. 39 the distributions of the test statistic, for rate-only and rate+shape analyses, are shown and compared with the data. It is already clear that the current dataset is compatible with both $\beta = 0$ and $\beta = 1$, since the significance for both cases is not large enough.

In order to quantify it, the 1-sided p-value given $\bar{\nu}_e$ appearance was calculated as well. In this case, contrary to the case of null $\bar{\nu}_e$ appearance hypothesis, the 1-sided p-value is computed on the left tail of the expected distribution of the test statistic, where the observed value of $\Delta\chi^2$ is placed. As shown in table 31 the p-value given $\bar{\nu}_e$ appearance is slightly larger than the background only hypothesis. The fit slightly prefers the hypothesis of $\bar{\nu}_e$ appearance, though the significance is not large enough to claim any possible hint yet.

hypothesis	p-value (rate-only)	p-value (rate+shape)
$\beta = 0$	XXX	XXX
$\beta = 1$	XXX	XXX

Table 31: Summary of 1-sided rate-only and rate+shape p-values from the combined Run 5+6 dataset. The 1-sided p-values are obtained performing 50k toys with $\beta = 0$ for background only hypothesis and $\beta = 1$ for $\bar{\nu}_e$ appearance hypothesis, in order to get the distribution of the test statistic. In the first case the p-value given null $\bar{\nu}_e$ appearance is obtained on the right tail of the expected distribution, while for $\beta = 1$ the 1-sided p-value is obtained on the left tail of the expected distribution.





Figure 38: The distribution of the total number of events from each of the 10k toy experiments is shown for both $\beta = 0$ (green) and $\beta = 1$ (red). The black line corresponds to the total number of events observed in the Run 5+6 dataset. Data correspond to Run 1-6 statistic. Both linear (left) and logarithmic (right) scales are shown.





Figure 39: The distribution of the test statistic $\Delta\chi^2$ for each of the 10k toy experiments is shown for both $\beta = 0$ (green) and $\beta = 1$ (red). The black line corresponds to the $\Delta\chi^2$ obtained from the Run 5+6 dataset. Data correspond to Run 1-6 statistic. Both linear (left) and logarithmic (right) scales are shown.

10. $\bar{\nu}_\mu$ disappearance analysis

11. Analysis procedure: $\bar{\nu}_\mu$ dissapearance

11.1. Fit method

The $\bar{\nu}_\mu$ disappearance analysis is performed in an extension of the PMNS framework with additional antineutrino oscillation parameters $\bar{\theta}_{23}$ and $|\Delta\bar{m}^2_{32}|$. In this framework, neutrino oscillation probabilities are calculated with the PMNS 3 flavour hypothesis whereas the antineutrino oscillation probabilities calculated with $\bar{\theta}_{23}$ and $|\Delta\bar{m}^2_{32}|$ along with the standard neutrino oscillation parameters θ_{13}, θ_{12}, Δm^2_{21}, $|\Delta m^2_{32}|$ and mass ordering. Testing the consistency between neutrino and antineutrino disappearance probabilities in this way provides insight into the level of CPT violation in neutrino oscillations while decoupling it from the CP violation which may appear in the appearance channels. Fits are performed using the FHC 1 ring μ-like and RHC 1 ring μ-like samples from the full T2K Run 1-6 dataset.

The measurements of interest are as follows:

- 2D $\sin^2\theta_{23}$ Vs $|\Delta m^2_{32}|$ with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ while all the other oscillation parameters (including $\sin^2(\bar{\theta}_{23})$ and $|\Delta\bar{m}^2_{32}|$) are marginalised; the gaussian prior of reactor measurements shown in Tab. **??**, is used, through a penalty term as in eq. eq. (9).

- 2D $\sin^2(\bar{\theta}_{23})$ Vs $|\Delta\bar{m}^2_{32}|$ with reactor constraint: the parameters of interest are $\sin^2(\bar{\theta}_{23})$ and $|\Delta\bar{m}^2_{32}|$while all the other oscillation parameters including ($\sin^2\theta_{23}$ and $|\Delta m^2_{32}|$) are marginalised; the gaussian prior of reactor measurements shown in Tab. **??**, is used, through a penalty term as in eq. eq. (9).

A comparison of contours from these fits is key to determining the amount of CPT violation.

11.2. Sensitivity studies

All confidence level contours for sensitivity studies were were produced using the constant $\Delta\chi^2$ method described in section 4.4.

In Fig. fig. 40 the sensitivity to $\sin^2(\bar{\theta}_{23})$ Vs $|\Delta\bar{m}^2_{32}|$ with reactor constraint is shown for the Asimov data set and compared with the result obtained by the other analysis groups is shown.

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Figure 40: Shown here are 68% and 90% constant $\Delta\chi^2$ CL contours of $\sin^2(\bar{\theta}_{23})$ Vs $|\Delta\bar{m}^2_{32}|$ with the reactor constraint from the 3 analysis groups from a fit to the Run 1-6 Asimov dataset. Independent contours are shown for each hierarchy.

In Fig. fig. 41 the sensitivity to $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ with reactor constraint is shown for the Asimov data set and compared with the result obtained by the other analysis groups is shown.

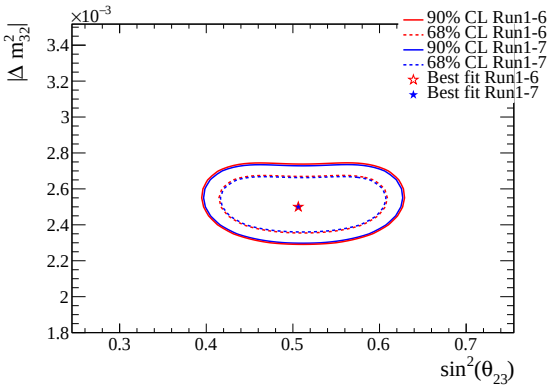
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Figure 41: Shown here are 68% and 90% constant $\Delta\chi^2$ CL contours of $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ with the reactor constraint from the 3 analysis groups from a fit to the Run 1-6 Asimov dataset. Contours were produced independently for each hierarchy hypothesis.

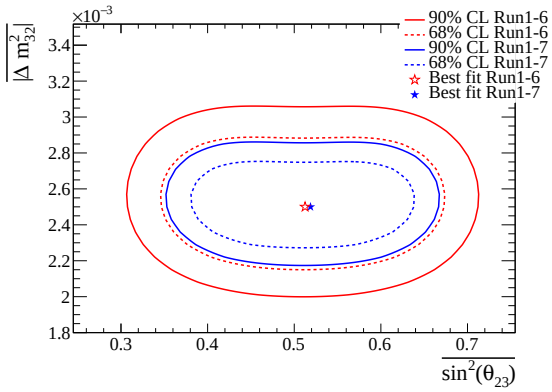
Fig. figs. 42 and 43 show a comparison of the sensitivity to ν and $\bar{\nu}$ parameters $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ from the run 1-6 dataset to the run 1-7 dataset. There is significant improvement in the $\bar{\nu}$ parameter sensitivity in both Asimov fits with the 1-7 dataset 90% CL inside of the run 1-6 dataset 68% CL. Another interesting feature of the Asimov fits is that in some cases, the best fit point moves with the addition of more statistics as in figs. 42b, 42c and 43b. Changes in the best fit point are possible when using a marginal likelihood. As seen perviously employing a marginal likelihood can lead to a best fit point away from the true parameter values because of correlations with parameters that are marginalised. In this case, the parameters marginalised are better constrained in the run 1-7 data fits than the run 1-6 and therefore have a different effect on the best fit point.

This is however a negligible effect in the sensitivity studies as the difference in the $\Delta\chi^2$ between run 1-7 and run 1-6 Asimov fits was found to be insignificant.

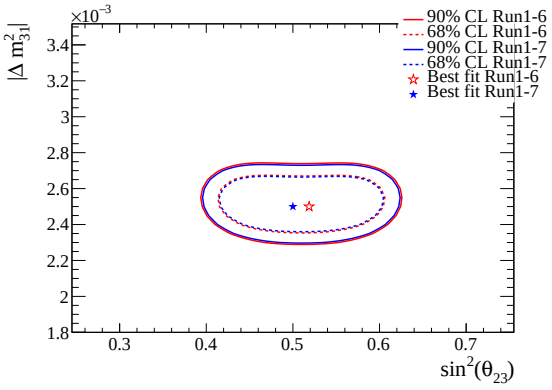
11.3. Results of the $\bar{\nu}_\mu$ disappearance analysis



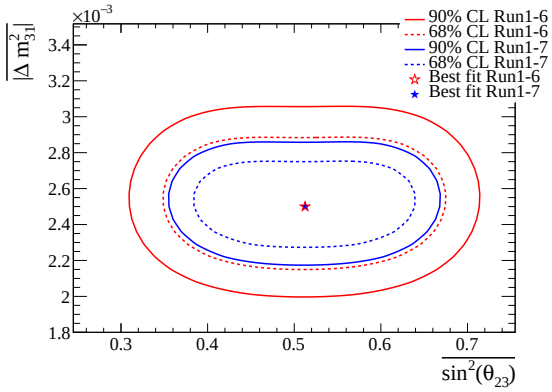
(a) ν oscillation parameter fit for normal hierarchy hypothesis



(b) $\bar{\nu}$ oscillation parameter fit for normal hierarchy hypothesis

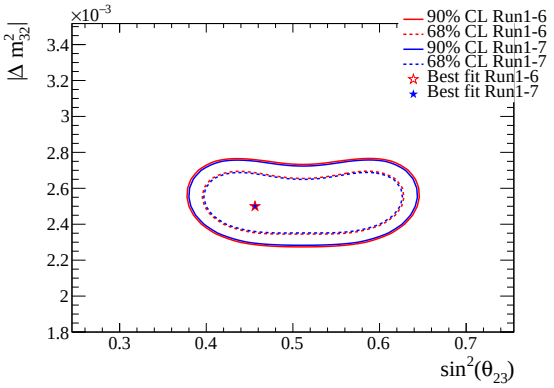


(c) ν oscillation parameter fit for inverted hierarchy hypothesis

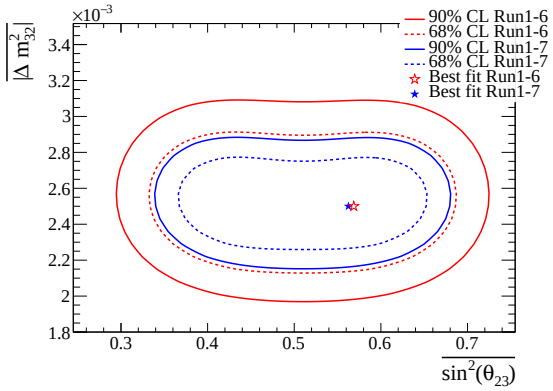


(d) $\bar{\nu}$ oscillation parameter fit for inverted hierarchy hypothesis

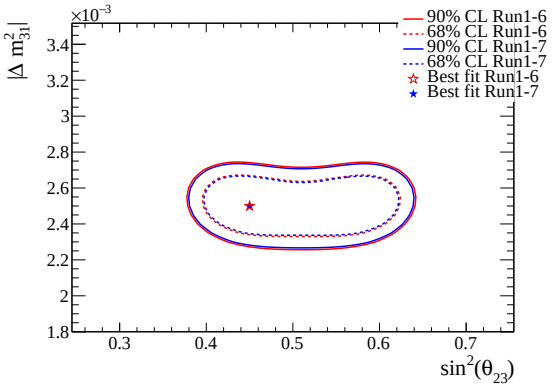
Figure 42: Shown here are 68% and 90% constant $\Delta\chi^2$ CL contours of fits to ν and $\bar{\nu}$ parameters $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ for both the Asimov run 1-6 and Asimov 1-7 datasets for parameter set A in tab. table 12 . Contours were produced independently for each hierarchy hypothesis.



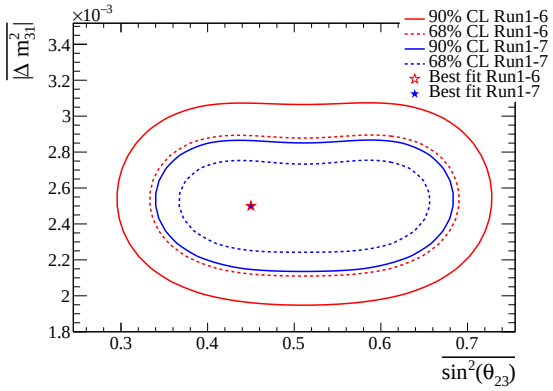
(a) ν oscillation parameter fit for normal hierarchy hypothesis



(b) $\bar{\nu}$ oscillation parameter fit for normal hierarchy hypothesis

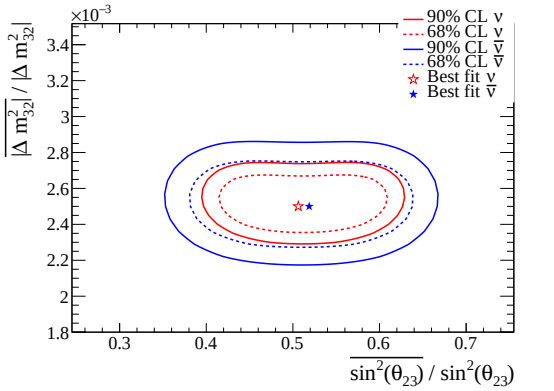


(c) ν oscillation parameter fit for inverted hierarchy hypothesis

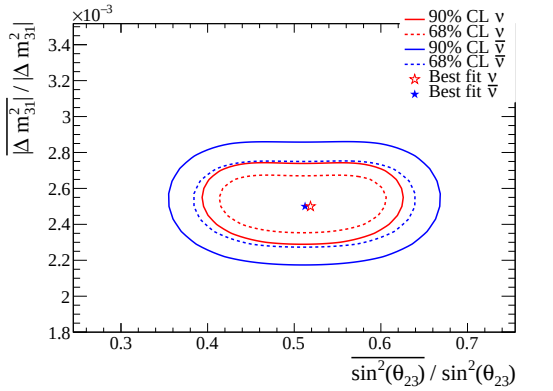


(d) $\bar{\nu}$ oscillation parameter fit for inverted hierarchy hypothesis

Figure 43: Shown here are 68% and 90% constant $\Delta\chi^2$ CL contours of fits to ν and $\bar{\nu}$ parameters $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ for both the Asimov run 1-6 and Asimov 1-7 datasets for parameter set B in tab. table 12 . Contours were produced independently for each hierarchy hypothesis.

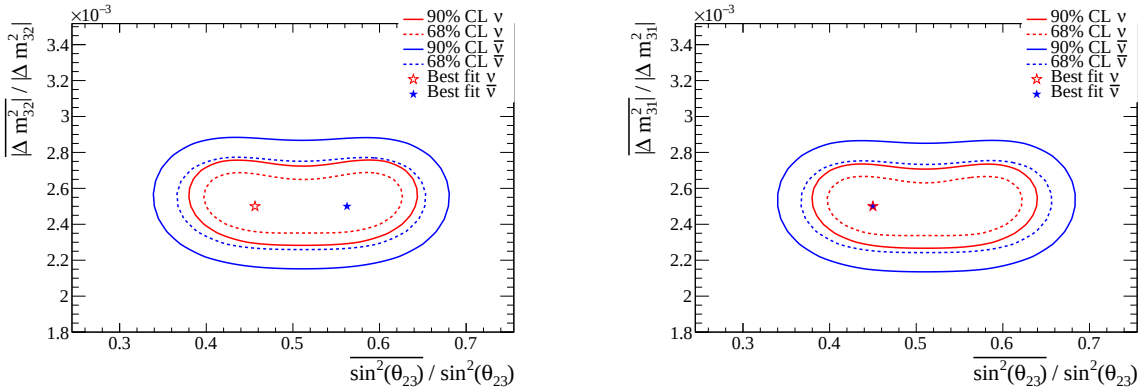


(a) Normal hierarchy



(b) Inverted hierarchy

Figure 44: Shown here are a comparison of the 68% and 90% constant $\Delta\chi^2$ CL contours of fits to ν and $\bar{\nu}$ parameters $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ for the Asimov 1-7 dataset for parameter set A in tab. table 12 . Contours were produced independently for each hierarchy hypothesis.



(a) Normal hierarchy

(b) Inverted hierarchy

Figure 45: Shown here are a comparison of the 68% and 90% constant $\Delta\chi^2$ CL contours of fits to ν and $\bar{\nu}$ parameters $\sin^2(\theta_{23})$ Vs $|\Delta m^2_{32}|$ for the Asimov 1-7 dataset for parameter set B in tab. table 12 . Contours were produced independently for each hierarchy hypothesis.

12. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-6 data set, for an exposure of 6.914×10^{20} POT in neutrino mode and 4.011×10^{20} POT in antineutrino mode. The goal of the analysis is to search for CP violation in the leptonic and put limits on δ_{CP} as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 125.914 ± 6.459 (syst) for neutrino μ-like, 34.284 ± 1.755 (syst) for antineutrino μ-like, 26.594 ± 1.808 (syst) for neutrino e-like, and 3.218 ± 0.238 (syst) for antineutrino e-like single ring events in Super K and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events and reconstructed energy versus lepton angle of single e-like ring events was used to compute the confidence intervals. All 101 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

TODO: ADD COMMENTS ABOUT BEST-FIT OSCILLATION PARAMETERS AND 1SIGMA RANGES

TODO: ADD COMMENTS ABOUT GOODNESS OF FIT

TODO: ADD COMMENTS ABOUT 1D CONTOURS

TODO: ADD COMMENTS ABOUT 2D CONTOURS

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Appendices

A. Predicted number of events

In Tab. 32, 33 and 34, 35 the total number of unoscillated events is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	69.747	0.035	4.335	0.002	0.151	0.001	74.270
CC1pi	17.206	0.017	1.572	0.001	0.049	0.000	18.845
CCcoherent	0.330	0.000	0.080	0.000	0.001	0.000	0.412
MEC	19.725	0.010	0.852	0.000	0.036	0.000	20.623
CCother	7.742	0.008	0.494	0.001	0.002	0.000	8.247
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	122.243	0.297	7.659	0.025	0.238	0.002	130.464

Table 32: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	7.088	0.004	13.306	0.005	0.004	0.016	20.422
CC1pi	2.718	0.003	3.066	0.002	0.002	0.003	5.794
CCcoherent	0.054	0.000	0.173	0.000	0.000	0.000	0.228
MEC	2.525	0.001	2.118	0.001	0.001	0.002	4.649
CCother	1.282	0.001	0.886	0.000	0.000	0.000	2.170
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.637	0.051	20.282	0.032	0.008	0.022	35.032

Table 33: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.062	2.129	0.001	0.088	10.811	0.121	13.211
CC1pi	0.017	0.348	0.000	0.027	1.236	0.020	1.648
CCcoherent	0.000	0.002	0.000	0.002	0.011	0.002	0.017
MEC	0.009	0.710	0.000	0.022	2.750	0.022	3.513
CCother	0.001	0.045	0.000	0.003	0.025	0.001	0.076
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.476	3.264	0.069	0.145	14.833	0.167	19.954

Table 34: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.269	1.376	2.178
CC1pi	0.002	0.040	0.001	0.060	0.045	0.163	0.312
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.022	0.029
MEC	0.001	0.069	0.000	0.063	0.087	0.219	0.439
CCother	0.001	0.008	0.000	0.004	0.003	0.003	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.142	0.313	0.235	0.478	0.404	1.783	3.355

Table 35: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in tab. 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	338.683	0.037	8.672	0.002	0.000	0.000	347.394
CC1pi	39.668	0.018	2.276	0.001	0.000	0.000	41.963
CCcoherent	0.678	0.000	0.136	0.000	0.000	0.000	0.815
MEC	74.638	0.011	1.548	0.000	0.000	0.000	76.198
CCother	8.732	0.009	0.560	0.001	0.000	0.000	9.301
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	469.893	0.301	13.518	0.025	0.000	0.000	483.737

Table 36: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	15.420	0.004	55.525	0.005	0.000	0.000	70.954
CC1pi	4.071	0.003	6.536	0.002	0.000	0.000	10.611
CCcoherent	0.074	0.000	0.612	0.000	0.000	0.000	0.687
MEC	4.721	0.001	7.811	0.001	0.000	0.000	12.534
CCother	1.457	0.001	1.008	0.000	0.000	0.000	2.467
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	26.713	0.052	72.225	0.032	0.000	0.000	99.021

Table 37: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

In Tab. 36, 37 and 38, 39 the total number of unoscillated events is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.413	2.294	0.002	0.094	0.000	0.000	2.803
CC1pi	0.069	0.371	0.001	0.028	0.000	0.000	0.469
CCcoherent	0.002	0.003	0.000	0.002	0.000	0.000	0.006
MEC	0.095	0.759	0.000	0.023	0.000	0.000	0.878
CCother	0.004	0.046	0.000	0.003	0.000	0.000	0.054
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.970	3.504	0.071	0.154	0.000	0.000	5.698

Table 38: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.012	0.203	0.018	0.361	0.000	0.000	0.593
CC1pi	0.004	0.043	0.003	0.064	0.000	0.000	0.114
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.000	0.007
MEC	0.003	0.074	0.003	0.067	0.000	0.000	0.147
CCother	0.001	0.009	0.000	0.004	0.000	0.000	0.015
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.156	0.334	0.255	0.507	0.000	0.000	1.253

Table 39: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

B. Predicted *e*-like spectra

In Fig. 46 each component is shown separately in order to compare the different shapes as a function of E_{reco}. The NC spectrum distribution is quite flat over all the energy range and is quite difficult to separate. The $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events populate the same energy region, around 0.6 GeV, with a very similar shape.

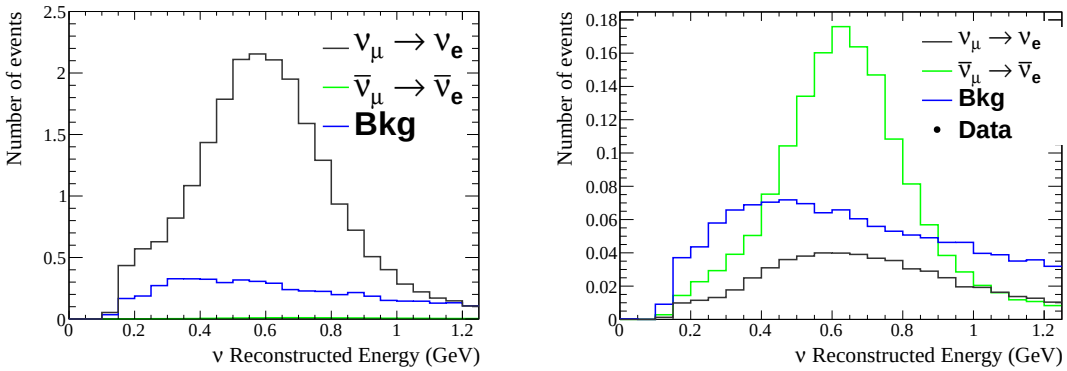


Figure 46: Predicted spectra of reconstructed neutrino energy for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), background, which includes NC, $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

Another kinematical variable is thus needed in order to better separate different flavour / reaction compoonents. In Fig. 50 and 51 the 2-dimensional E_{reco} Vs θ spectra are shown for each flavour component and NC for both neutrino and antineutrino mode *e*-like samples. In particular the shapes of $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ distributions are quite different, thus this templates have a better capability to discriminate signal from background. In general this template parametrization produces quite different shapes for all the flavour and NC components.

In Fig. 49 the projection of the 2-dimensional *e*-like templates onto the E_{reco} and θ axes are shown.

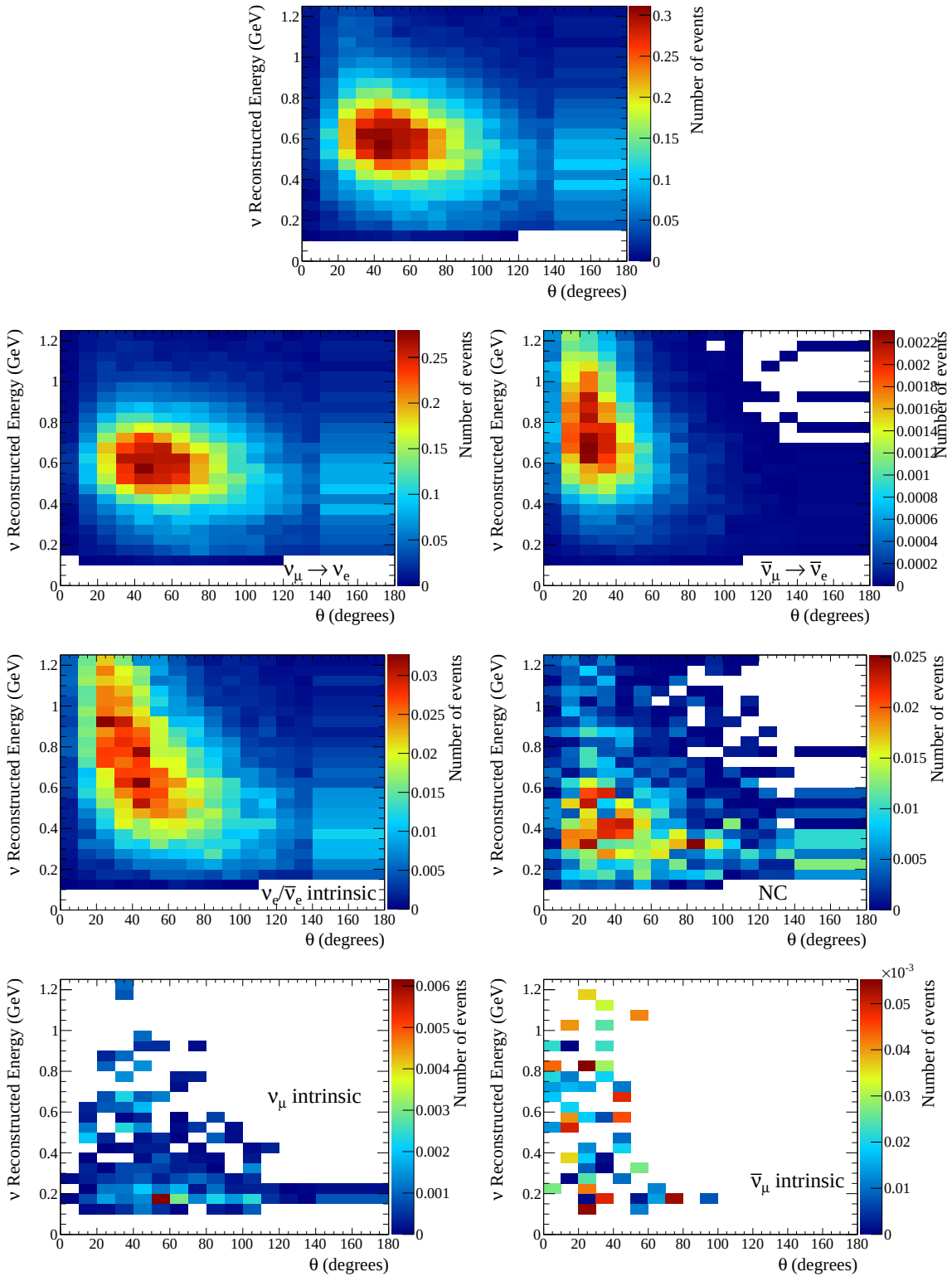


Figure 47: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

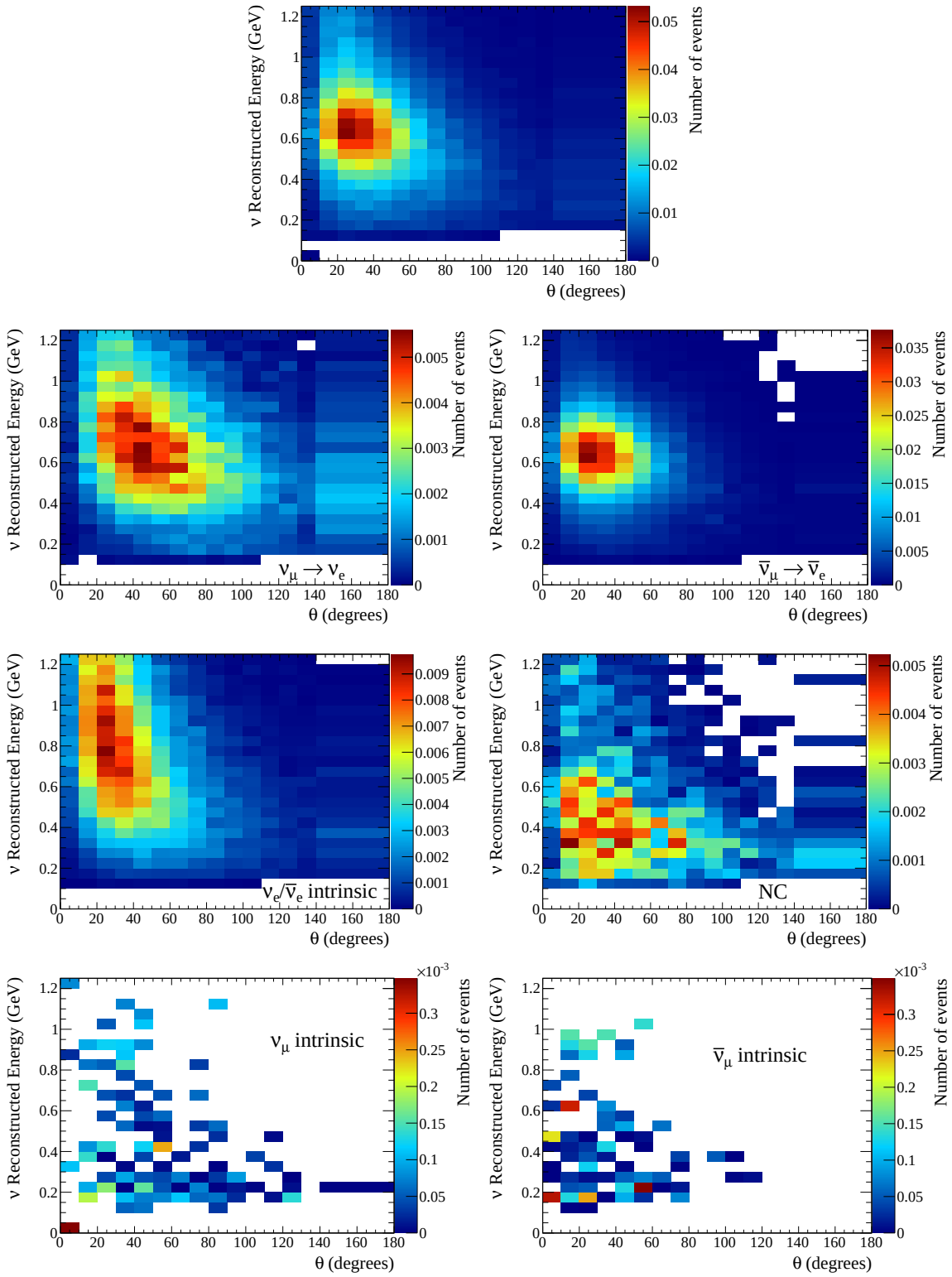


Figure 48: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 4.011×10^{20} POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

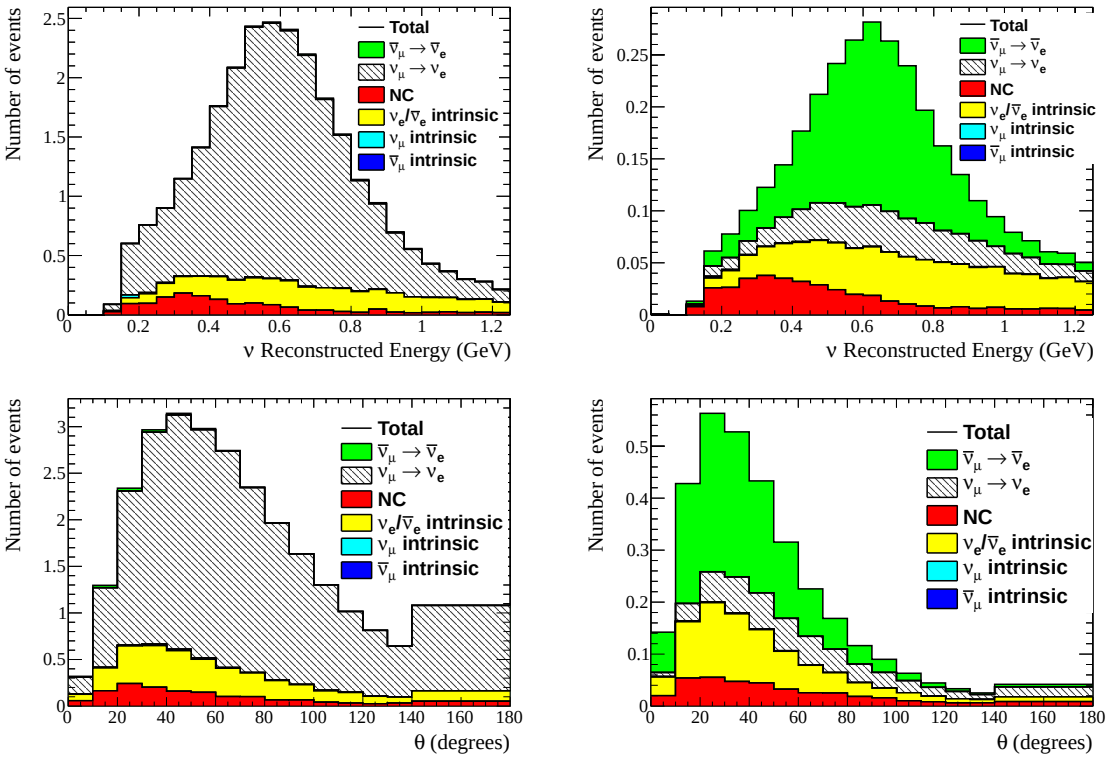


Figure 49: Predicted spectra of the reconstructed neutrino energy (top) and the angle between the neutrino and the outgoing lepton directions (bottom) for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode and of 4.011×10^{20} POT in antineutrino mode. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

C. Observed *e*-like spectra

In Fig. 50 and 51 the predicted *e*-like 2-dimensional spectra are compared to data for both neutrino and antineutrino beam mode.

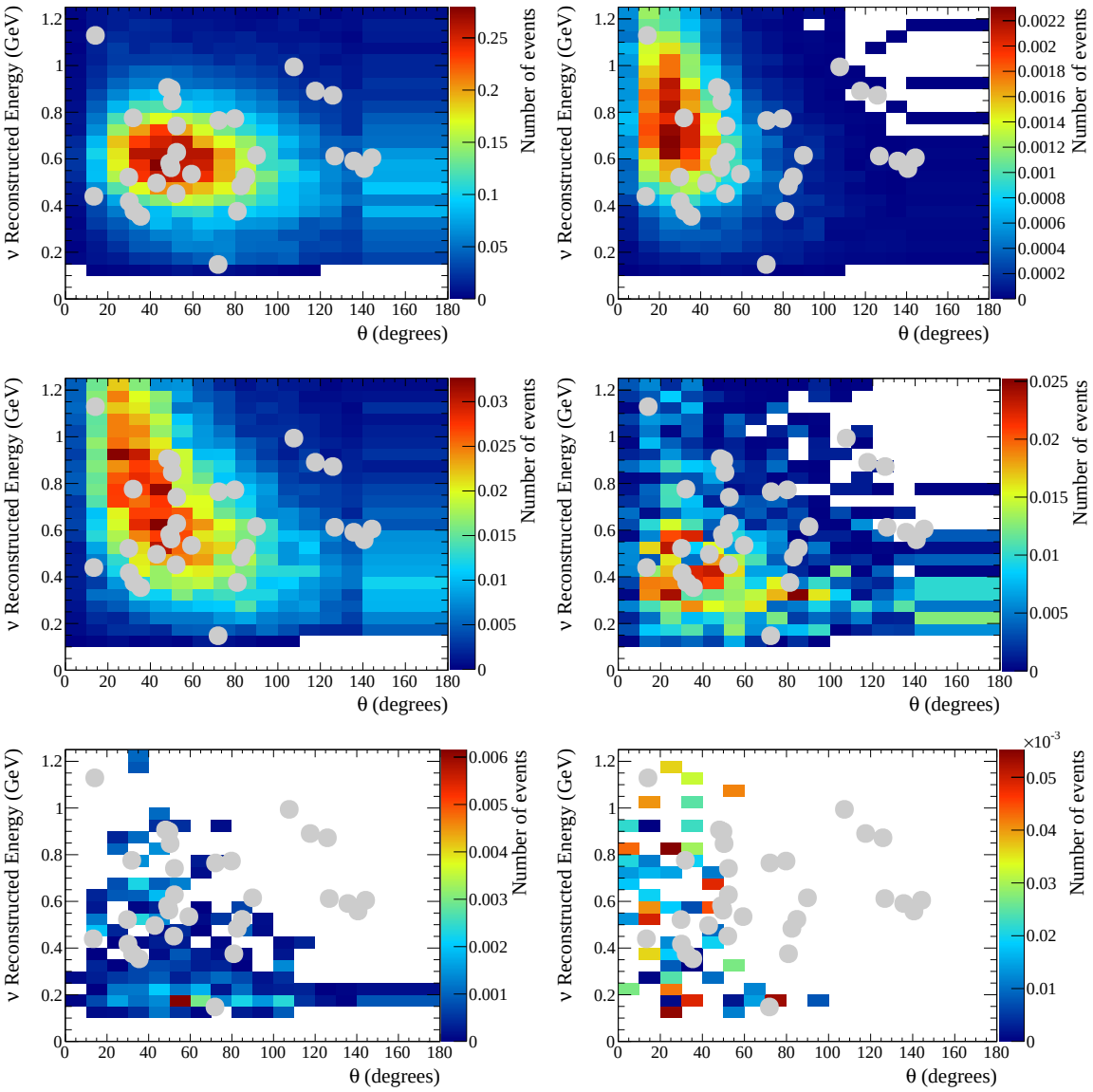


Figure 50: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 6.914×10^{20} POT in neutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

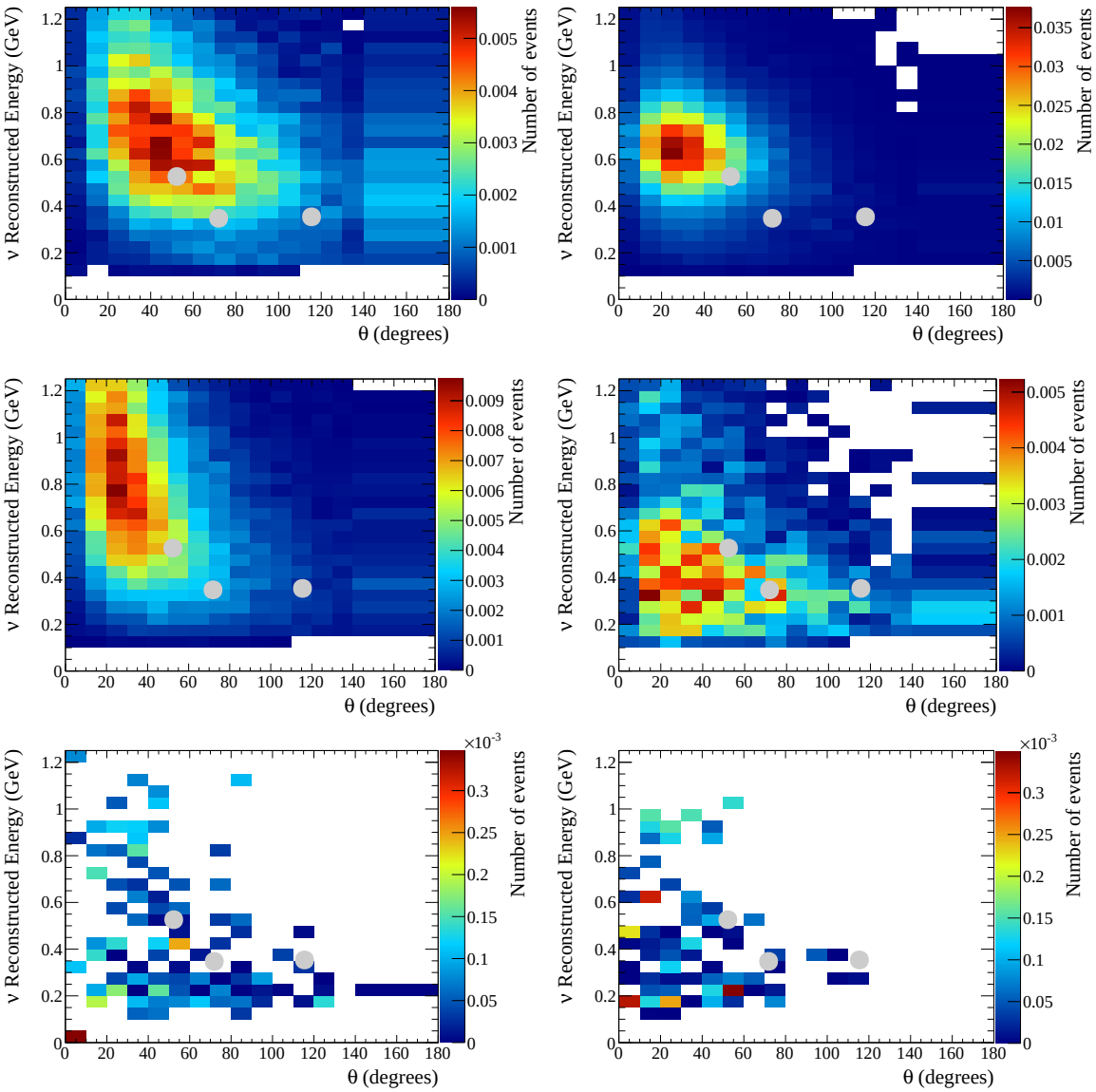


Figure 51: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full run 1-6 data set for an exposure of 4.011×10^{20} POT in antineutrino mode. From top left to bottom right the following components are shown: total, $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8 and 9 and the oscillation parameters corresponding to the Asimov data set A shown in tab. 12.

D. Effect of systematic uncertainties

It tab. 43 it is shown to which reaction and flavour modes each systematic parameter is applied.

$f^{banff}_{13;t,r}$	$f^{banff}_{12;t,r}$	$f^{banff}_{11;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{0;t,r}$	$f^{SK+FSI}_{5;t,r}$	$f^{SK+FSI}_{4;t,r}$	$f^{SK+FSI}_{3;t,r}$	$f^{SK+FSI}_{2;t,r}$	$f^{SK+FSI}_{1;t,r}$	$f^{SK+FSI}_{0;t,r}$	$f^{SK}_{E;r}$	δ_{CP}	Δm^2_{21}	$\sin^2(\theta_{13})$	$\sin^2(\theta_{12})$	
			✓	✓	✓	✓	✓			✓		✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	ν_μ CCQE
			✓	✓	✓	✓	✓			✓		✓	✓			✓				✓	✓	✓	✓	ν_μ CC 1π
			✓	✓	✓	✓	✓			✓		✓	✓			✓				✓	✓	✓	✓	ν_μ CC coherent
			✓	✓	✓	✓	✓			✓		✓	✓				✓	✓	✓	✓	✓	✓	✓	ν_μ CC MEC
			✓	✓	✓	✓	✓			✓		✓	✓			✓				✓	✓	✓	✓	ν_μ CC other
			✓	✓	✓	✓	✓	✓				✓	✓	✓						✓				ν_μ/ν_τ NC $1\pi^{+/-}$
			✓	✓	✓	✓	✓	✓				✓	✓	✓						✓				ν_μ/ν_τ NC $1\pi^0$
			✓	✓	✓	✓	✓	✓				✓	✓	✓						✓				ν_μ/ν_τ NC coherent
			✓	✓	✓	✓	✓	✓				✓	✓	✓						✓				ν_μ/ν_τ NC other
			✓	✓	✓	✓	✓	✓				✓	✓	✓						✓				ν_e NC 1γ
✓	✓	✓															✓	✓	✓	✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CCQE
✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC 1π
✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC coherent
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✓	✓	✓														✓				✓	✓	✓	✓	✓	$\bar{\nu}_\mu$ CC other
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$
✓	✓	✓												✓						✓					$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent
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✓	✓	✓												✓						✓					ν_μ NC 1γ
									✓											✓	✓	✓	✓	✓	ν_e CCQE
									✓											✓	✓	✓	✓	✓	ν_e CC 1π
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								✓												✓					ν_e NC coherent
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			✓	✓	✓	✓	✓		✓						✓					✓	✓	✓	✓	✓	Osc. ν_e CCQE
			✓	✓	✓	✓	✓		✓						✓					✓	✓	✓	✓	✓	Osc. ν_e CC 1π
			✓	✓	✓	✓	✓		✓						✓					✓	✓	✓	✓	✓	Osc. ν_e CC coherent
			✓	✓	✓	✓	✓		✓						✓					✓	✓	✓	✓	✓	Osc. ν_e CC MEC
			✓	✓	✓	✓	✓		✓						✓					✓	✓	✓	✓	✓	Osc. ν_e CC other
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CCQE
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC 1π
															✓					✓	✓	✓	✓	✓	$\bar{\nu}_e$ CC coherent
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								✓												✓					$\bar{\nu}_e$ NC coherent
								✓												✓					$\bar{\nu}_e$ NC other
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✓	✓	✓							✓											✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC coherent
✓	✓	✓							✓											✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC MEC
✓	✓	✓							✓											✓	✓	✓	✓	✓	Osc. $\bar{\nu}_e$ CC other

| | | |

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC MEC	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	ν_e NC 1γ	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC MEC	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	ν_μ NC 1γ	ν_e CCQE	ν_e CC 1π	ν_e CC coherent	ν_e CC MEC	ν_e CC other	ν_e NC $1\pi^{+/-}$	ν_e NC $1\pi^0$	ν_e NC coherent	ν_e NC other	$\bar{\nu}_\mu$ NC 1γ	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC MEC	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC MEC	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	$\bar{\nu}_e$ NC 1γ	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC MEC	Osc. $\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
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$f^{SK+FSI}_{9;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{10;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{11;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
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$f^{SK+FSI}_{15;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{16;t,r}$																																															
$f^{SK+FSI}_{17;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{0;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{1;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{2;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{3;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{4;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{5;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{6;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{7;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\sin^2(\theta_{12})$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\sin^2(\theta_{13})$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Δm^2_{21}	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
δ_{CP}	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{6;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{7;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{8;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{9;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{10;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{11;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{12;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{13;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{14;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{15;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{16;t,r}$						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{17;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{0;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{1;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{2;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{3;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{4;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{5;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{6;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{7;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

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	$\sin^2(\theta_{12})$	$\sin^2(\theta_{13})$	Δm^2_{21}	δ_{CP}	$f^{SK}_{E;r}$	$f^{SK+FSI}_{0;t,r}RHC$	$f^{SK+FSI}_{1;t,r}RHC$	$f^{SK+FSI}_{2;t,r}RHC$	$f^{SK+FSI}_{3;t,r}RHC$	$f^{SK+FSI}_{4;t,r}RHC$	$f^{SK+FSI}_{5;t,r}RHC$	$f^{banff}_{0;t,r}RHC$	$f^{banff}_{1;t,r}RHC$	$f^{banff}_{2;t,r}RHC$	$f^{banff}_{3;t,r}RHC$	$f^{banff}_{4;t,r}RHC$	$f^{banff}_{5;t,r}RHC$	$f^{banff}_{6;t,r}RHC$	$f^{banff}_{7;t,r}RHC$	$f^{banff}_{8;t,r}RHC$	$f^{banff}_{9;t,r}RHC$	$f^{banff}_{10;t,r}RHC$	$f^{banff}_{11;t,r}RHC$	$f^{banff}_{12;t,r}RHC$	$f^{banff}_{13;t,r}RHC$
ν_μ CCQE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC 1π	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC coherent	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC MEC	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ CC other	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC $1\pi^{+/-}$					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC $1\pi^0$					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC coherent					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_μ/ν_τ NC other					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
ν_e NC 1γ					✓						✓														
$\bar{\nu}_\mu$ CCQE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC 1π	✓	✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC coherent	✓	✓	✓	✓	✓			✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC MEC	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC other	✓	✓	✓	✓	✓			✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other					✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ NC 1γ					✓						✓												✓	✓	✓
ν_e CCQE	✓	✓	✓	✓	✓																				
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ν_e CC coherent	✓	✓	✓	✓	✓						✓														
ν_e CC MEC	✓	✓	✓	✓	✓						✓														
ν_e CC other	✓	✓	✓	✓	✓						✓														
ν_e NC $1\pi^{+/-}$					✓						✓														
ν_e NC $1\pi^0$					✓						✓														
ν_e NC coherent					✓						✓														
ν_e NC other					✓						✓														
$\bar{\nu}_\mu$ NC 1γ					✓						✓														
Osc. ν_e CCQE	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC 1π	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC coherent	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC MEC	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
Osc. ν_e CC other	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
$\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓						✓														
$\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓						✓														
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$\bar{\nu}_e$ NC $1\pi^{+/-}$					✓						✓														
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Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓						✓												✓	✓	✓
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓						✓												✓	✓	✓
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓						✓												✓	✓	✓
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓						✓												✓	✓	✓
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓						✓												✓	✓	✓

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	$\sin^2(\theta_{12})$	$\sin^2(\theta_{13})$	Δm^2_{21}	δ_{CP}	$f^{SK}_{E;r}$	$f^{SK+FSI\,RHC}_{6;t,r}$	$f^{SK+FSI\,RHC}_{7;t,r}$	$f^{SK+FSI\,RHC}_{8;t,r}$	$f^{SK+FSI\,RHC}_{9;t,r}$	$f^{SK+FSI\,RHC}_{10;t,r}$	$f^{SK+FSI\,RHC}_{11;t,r}$	$f^{SK+FSI\,RHC}_{12;t,r}$	$f^{SK+FSI\,RHC}_{13;t,r}$	$f^{SK+FSI\,RHC}_{14;t,r}$	$f^{SK+FSI\,RHC}_{15;t,r}$	$f^{SK+FSI\,RHC}_{16;t,r}$	$f^{SK+FSI\,RHC}_{17;t,r}$	$f^{banff\,RHC}_{0;t,r}$	$f^{banff\,RHC}_{1;t,r}$	$f^{banff\,RHC}_{2;t,r}$	$f^{banff\,RHC}_{3;t,r}$	$f^{banff\,RHC}_{4;t,r}$	$f^{banff\,RHC}_{5;t,r}$	$f^{banff\,RHC}_{6;t,r}$	$f^{banff\,RHC}_{7;t,r}$
ν_μ CCQE	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC 1π	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC coherent	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC MEC	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ CC other	✓	✓	✓	✓	✓				✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^{+/-}$					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC $1\pi^0$					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC coherent					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ν_μ/ν_τ NC other					✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
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$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$					✓										✓	✓	✓							
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Osc. ν_e CCQE	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC 1π	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC coherent	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC MEC	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. ν_e CC other	✓	✓	✓	✓	✓	✓	✓	✓										✓	✓	✓	✓	✓	✓	✓
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓										
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓										
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓										
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓										
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓	✓										
$\bar{\nu}_e$ NC $1\pi^{+/-}$					✓																			
$\bar{\nu}_e$ NC $1\pi^0$					✓																			
$\bar{\nu}_e$ NC coherent					✓																			
$\bar{\nu}_e$ NC other					✓																			
$\bar{\nu}_e$ NC 1γ					✓																			
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC MEC	✓	✓	✓	✓	✓	✓	✓	✓																
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓	✓	✓	✓	✓																

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Table 43 – *Continued from previous page*

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Table 43 – *Continued from previous page*

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In tab. 44, 45, 46 and 47 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set B.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	4.02%
Flux	3.56%
XSec Corr Tot	4.03%
MEC (corr)	3.49%
MECbar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.10%
Flux+XSec	2.85%
Flux+XSec+SKDet+FSI+SI	4.96%
Oscillations	0.10%
All	4.96%

Table 44: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.67%
Flux	3.61%
XSec Corr Tot	4.16%
MEC (corr)	3.94%
MECbar (corr)	0.07%
NC other (uncorr)	0.21%
NC 1gamma (uncorr)	1.94%
XSec nue/numu	2.55%
XSec Tot	5.26%
Flux+XSec	4.29%
Flux+XSec+SKDet+FSI+SI	5.68%
Oscillations	4.18%
All	7.19%

Table 45: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B
SKDet+FSI+SI	3.83%
Flux	3.70%
XSec Corr Tot	4.10%
MEC (corr)	2.95%
MECbar (corr)	1.83%
NC other (uncorr)	0.73%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.17%
Flux+XSec	3.31%
Flux+XSec+SKDet+FSI+SI	5.04%
Oscillations	0.06%
All	5.11%

Table 46: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.88%
Flux	3.81%
XSec Corr Tot	4.52%
MEC (corr)	2.92%
MECbar (corr)	2.56%
NC other (uncorr)	0.32%
NC 1gamma (uncorr)	2.81%
XSec nue/numu	1.68%
XSec Tot	5.59%
Flux+XSec	4.62%
Flux+XSec+SKDet+FSI+SI	6.18%
Oscillations	3.59%
All	7.09%

Table 47: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	μ-like RMS/Mean	e-like RMS/Mean
SKDet	0.006	0.012
FSI+SI	0.025	0.036
SKDet+FSI+SI	0.026	0.037
Flux	0.016	0.017
XSec NC other (uncorr)	0.000	0.001
XSec NC 1γ (uncorr)	0.000	0.009
XSec ν_e / ν_μ (uncorr)	0.000	0.033
XSec Tot	0.017	0.044
Flux+XSec	0.018	0.041
Flux+XSec+SDet+FSI+SI	0.031	0.056
Oscillations	0.000	0.006
All	0.032	0.056

Table 48: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set B shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

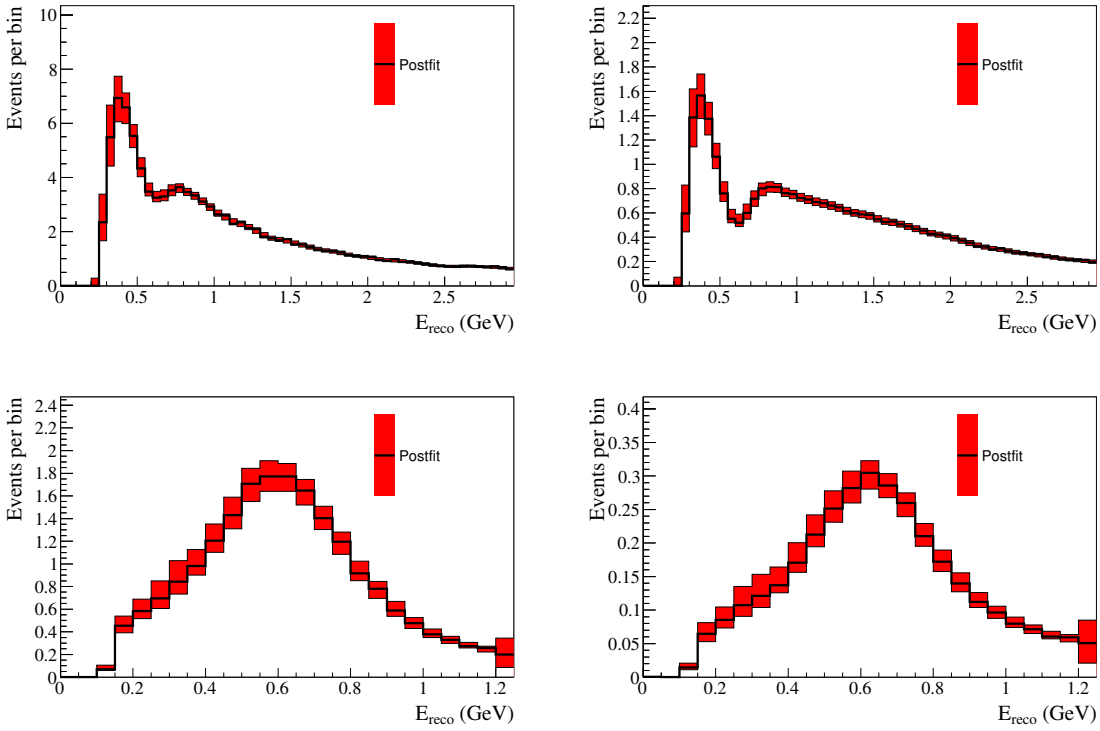


Figure 52: Error envelopes for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

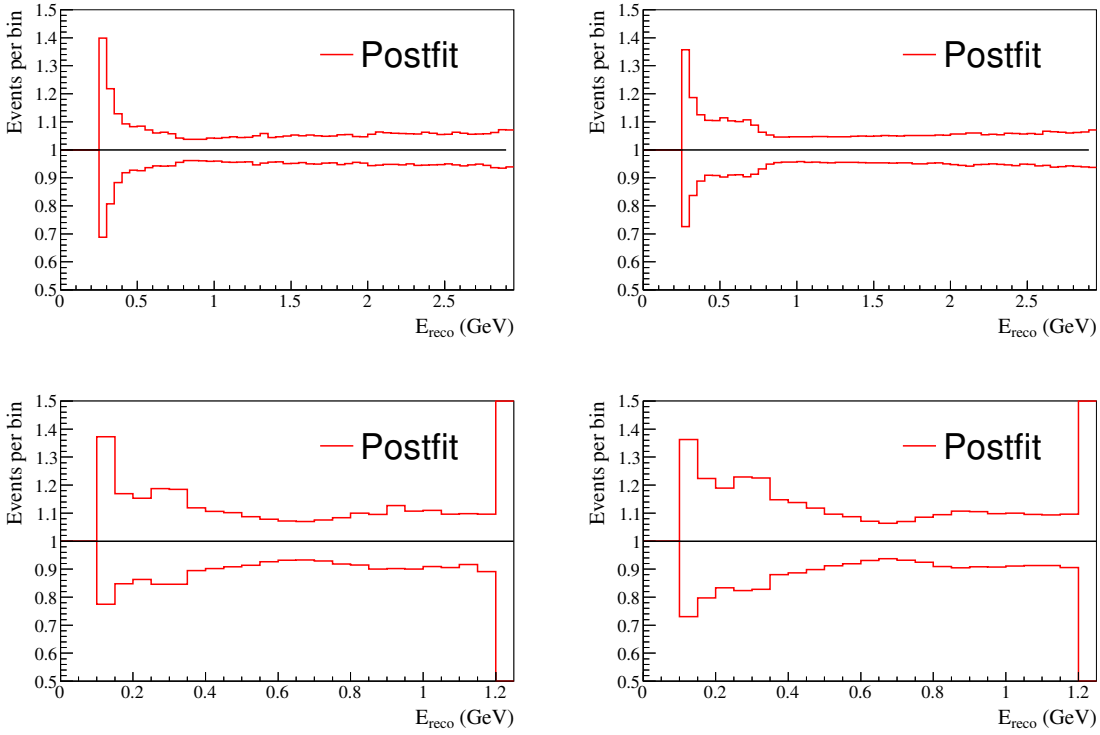


Figure 53: Error envelopes, expressed as a ratio of event count, for the energy spectrum as a result of all the systematic parameters. From top to bottom and left to right the following super-K samples are shown: muon-like neutrino mode, muon-like antineutrino mode, electron-like neutrino mode and electron-like antineutrino mode. 10k toy MC experiments were generated for the Asimov data set B shown in tab. 12 with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. The fractional error in each bin corresponds to the 1σ error obtained from the distribution of the extracted number of events. First the median is found. Then the upper (lower) error corresponds to the number of events for which the 34% of toys is contained in the right (left) side of the median. The continuos line shows the pre-BANFF (blue) and post-BANFF (red) ratios. The distributions correspond to the collected statistics in the full run 1-6 data set. The spectra were generated with the systematic parameters measured with the ND280 data shown in tab. 6, 7, 8, 9 and the oscillation parameters shown in tab. 11. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In tab. 49, 50, 51 and 52 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set A, but without taking into account the correction of the post-fit BANFF systematic parameters as well the additional covariance matrix that take into account the bias in theăflux parameters of the BANFF fit.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.11%
Flux	3.26%
XSec Corr Tot	3.96%
MEC (corr)	3.53%
MECbar (corr)	0.20%
NC other (uncorr)	0.79%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.04%
Flux+XSec	2.75%
Flux+XSec+SKDet+FSI+SI	5.03%
Oscillations	0.03%
All	5.00%

Table 49: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.47%
Flux	3.40%
XSec Corr Tot	4.17%
MEC (corr)	3.95%
MECbar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.49%
XSec nue/numu	2.63%
XSec Tot	5.15%
Flux+XSec	4.08%
Flux+XSec+SKDet+FSI+SI	5.51%
Oscillations	4.19%
All	6.93%

Table 50: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.40%
XSec Corr Tot	4.20%
MEC (corr)	3.00%
MECbar (corr)	1.76%
NC other (uncorr)	0.76%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.27%
Flux+XSec	2.90%
Flux+XSec+SKDet+FSI+SI	4.90%
Oscillations	0.03%
All	4.91%

Table 51: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.53%
XSec Corr Tot	4.44%
MEC (corr)	3.11%
MECbar (corr)	2.24%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.98%
XSec nue/numu	1.52%
XSec Tot	5.57%
Flux+XSec	4.39%
Flux+XSec+SKDet+FSI+SI	6.11%
Oscillations	3.98%
All	7.24%

Table 52: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in tab. 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in tab. 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.